Tromboneless

v0.1.1

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Chapter 1

Tromboneless

This GitHub repository contains the open source hardware and firmware for Tromboneless - an open source device capable of measuring oral input to synthesise the output of a brass trombone using a Raspberry Pi 5, implementing MIDI protocol.

Please refer to our Wiki `wiki` to see component, sequence and state diagrams for the MIDI implementation.

1.1 Table of Contents

- Getting Started
- Dependencies
- Bill of Materials
- Social Media
- Acknowledgements
- Authors & Contributions
- License

1.2 Getting Started:

1. Install Raspberry Pi OS onto your Raspberry Pi.
   For set-up guidance for PiOS, please follow [this guide].(https://www.raspberrypi.com/documentation/computers/getting-started.html#installing-the-operating-system)

1.2.1 Option 1

1. clone this git repository and its submodule dependencies using one line one the command line:

```
git clone --recursive https://github.com/RyanMcB8/Tromboneless.git
```

1.2.2 Option 2

1. clone this git repository using the command:

```
git clone https://github.com/RyanMcB8/Tromboneless.git
```

2. To add the necessary submodules into the workspace, you will need to use the following command line commands:

```
git submodules init
```

```
git submodules update
```

1.3 Dependencies

The Tromboneless App utilises the JUCE framework to create all the widgets necessary for operation.

- Juce
- RTMidi
- libgpiod (specifically for interacting with the GPIO pins on the Pi)
- webkit2gtk-4.1

1.4 Bill of Materials

Below contains the materials required to construct the Tromboneless hardware, with a circuit diagram showing how to configure it.

1.4.1 Controller

Microcontroller	Quantity	Cost (£)
Raspberry Pi 5	1	58.98
Active Cooler for Raspberry Pi 5 (recommended)	1	4.80

1.4.2 Sensors

Sensors	Quantity	Cost (£)
VL53L1X Time of Flight (ToF) Sensor Breakout	1	20.40
CAP1188 8-Key Capacitive Touch Sensor	1	7.70
ADS1115 16-bit ADC	1	6.49

Total Cost: £34.59

1.4.3 Additional Components

Additional Components	Quantity
10.0 k ohm Resistor	1
300.0 ohm Resistor	1
390.0 k ohm Resistor	1
USB MIDI Cable	1
Infrared LED	1
Photodiode	1

1.4.4 Circuit Diagram

Circuit diagram showing how to assemble the Tromboneless :

1.5 Documentation

For documentation of the code used, please refer to the `Documentation` directory where there are both LaTex and HTML versions of the documentation for this project.

1.6 Social Media

- #1 Post on `r/Trombone`
- 28.6k+ Views across `r/Embedded`, `r/Trombone` and `r/linuxaudio`

Follow us for additional updates, including showcase videos and demonstrations:

`Reddit`
`Instagram`

1.7 Authors and Contributions

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1.8 License

1.8.1 ADS1115

- The ADS1115 library was adopted from `Bernd Porr`, which can be sourced `here`.

Chapter 2

Notes on App

2.1 Map of the code

Within the `main.cpp` file is the code responsible for creating main window. This should not need to be modified as the contents of the window are defined elsewhere.

The code within `Layout.cpp` creates the defines the general structure of the window by creating instances of widgets and panels and arranging them to be in the correct position on the screen.

The `Panels.cpp` file is where each panel on the display is created. This sets the backgorund colour of each panel and calls upon the specific widgets which should be in each panel.

The `Widgets.cpp` file hosts the classes for creating more custom widgets that base library of JUCE does not already provide. These include modifications to shapes of the widgets such as the custom calibration sliders and equalizer.

The `tromboneless_data.cpp` file has the variables which interact with the Tromboneless Core functionality and allow for the Core and App to communicate parameters between each other. If there is a need to add more communication channels between the app and core, this is where t should be added. For example, a flag for if a speaker has been connected.

Each of these files has an associated header file, excluding `main.cpp`, which hosts the declarations and custom types which may be used externally. All documentation for the functions are within the header files with more in-depth step by step explanations within the cpp files.

If the files are thought about as a pipe line, the order would be: JUCE --> Widgets --> Panels --> Layout --> main

2.2 Code tutorials specifically for JUCE:

2.2.1 For labels:

This code will display only uppercase characters input.

```
addAndMakeVisible (uppercaseLabel);
uppercaseLabel.setText ("Uppercase:", juce::dontSendNotification);
uppercaseLabel.attachToComponent (&uppercaseText, true);
uppercaseLabel.setColour (juce::Label::textColourId, juce::Colours::orange);
uppercaseLabel.setJustificationType (juce::Justification::right);
addAndMakeVisible (uppercaseText);
uppercaseText.setColour (juce::Label::backgroundColourId, juce::Colours::darkblue);
```

These functions will allow for the text to be editted by the user when the app is open.

```
addAndMakeVisible (inputText);
inputText.setEditable (true);
inputText.setColour (juce::Label::backgroundColourId, juce::Colours::darkblue);
inputText.onTextChange = [this] { uppercaseText.setText (inputText.getText().toUpperCase(),
      juce::dontSendNotification); };
```

When this label is changed by the user, the `onTextChange` sets a lambda function to be excucted whenever there is any change within the label.

2.3 For drop down menus

The drop down menus within the JUCE library are referred to as a `comboBox`.

The lines of code below display an example which first creates an instance of a combo-box called 'styleMenu' and adds optionsusing the `addItem` method. When the choice is changed, the `styleMenuChanged()` function is called upon. The first option within the list is set as the default to be selected.

```
addAndMakeVisible (styleMenu);
styleMenu.addItem ("Plain", 1);
styleMenu.addItem ("Bold", 2);
styleMenu.addItem ("Italic", 3);
styleMenu.onChange = [this] { styleMenuChanged(); };
styleMenu.setSelectedId (1);
setSize (400, 200);
```

Chapter 3

Notes for the changes made to the firmware

Chapter 4

Notes for the Hardware implementation and design

The image above shows a circuit diagram representing the hardware and connections used to create the prototype for the Tromboneless. The chosen peripherals may be changed to suitable alternatives but the code will need to be adapted to have the correct drivers for the new devices.

4.1 Details

4.1.1 Processing

To process and host the program, the Tromboneless used a Raspberry Pi 5 with a linux build which allowed for easy navigation around the directories and remote access for remote testing purposes. As this system was built using C++, this could should be capable of being adapted to work using another main board which supports the standard language. Unfortunately, this would require changing low level drivers and pin defintions within the code base to allow it to work.

One recommendation from the Tromboneless team is to use a cooler on the device to ensure that it doesn't begin to thermal throttle. The Pi was found to crash when building the JUCE submodule without a cooler. The cooler used in development was the `Active Cooler for Raspberry Pi 5` from Raspberry Pi purchased though `The Pi Hut`.

4.1.2 Power

Each of the peripherals is connected directly the Raspberry Pi for both power and ground. Each of the peripherals receives 5V.

4.1.3 Reading data

To read the data from the peripherals, the I2C protocol was utilised to allow for a reduction in pins for communication compared to SPI. To read the data at an appropriate time, each of the peripherals need to send a data ready interrupt to the processor for it to initiate a transfer. This allowed for a reduction in computation for the processor by utilising blocking I/O as opposed to polling for data continuously. This shifted a large amount of computation to the peripherals from the main processor freeing to complete other threads.

4.1.3.1 ADC

To find the amount of pressue the user is exerting into the mouthpiece (blowing), a balloon was enclosed in a tube restricting its width and reducing the light within the tube. The system developed used an IR transmitter and receiver, both of which were diodes. When the balloon expands, the amount of infra-red being reflected increases due to the light bouncing off the base of the balloon. The IR transmitter (an IR LED), was kept on persistently with the receiver having its cathode pin connected to the ADC. The cathode pin also connected to a 10K pull down resister in parallel which created a potential divider allowing the ADC to receive varying voltages as the amount of IR light is varied.

The specific ADC used was the `ADS1115` on a development board from `the Pi hut`.

An alternative to this method is to use a pressure sensor but from testing, the lower end sensors did not have a large enough range to match that which a trombonist would reach during a performance.

4.1.3.2 ToF sensor

To detect the distance the user's hand is from the device, a Time of Flight (ToF) sensor was used. The specific sensor chosen was the `VL53LOX` on a breakout board from `Pimoroni`. By addressing the devices onboard registers, the sensor could be set have the ideal range and sampling frequency for this device.

There are alternative choices for measuring the hand distance such as sonar or linear variable resistors. For this project, LiDAR was the best option for its speed, low interference, accuracy and being non contact.

4.1.3.3 Touch sensor

To measure the amount of pressure the user puts onto the mouthpiece to simulate the change in embouchure or 'buzz' of the trombonists lips, a capacitive touch sensor was developed. This custom sensor was created wrapping a sheet of copper and an electronically insulating material around a curved surface. This created a capacitor. The amount of pressure the user places on the device changes the distance between each of the copper layers which changes the capacitance. This capacitance may be fed directly into the touch sensor board used, the `CAP1188` by Microchip on a breakout board `from Adafruit`.

As the touch sensor and the ToF sensor both have the same I2C address as default, a 390K resister connected to the AD pin of the touch sensor was pulled to ground changing its I2C address from 0x29 to 0x28.

4.1.4 Outputs

4.1.4.1 MIDI

To read the MIDI data which is being output by the Pi, a MIDI compatible device such as a portable synthesiser may be connected to any of the USB-A connectors on the front of the Pi.

4.1.4.2 Video

The 2 mini HDMI ports on the side of the Raspberry Pi may be connected to a monitor (or monitors) to display the video output. When the app is launched, it may be seen in the video output.

Chapter 5

Hierarchical Index

5.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AmplitudeMapper . . . 19
AudioOutput . . . 19
CAP1188 . . . 45
CAP1188Channel . . . 46
CAP1188Printer . . . 46
juce::Component
- CalibrationOnClick . . . 35
  - CalibrateEmbouchure . . . 33
- DropDownMenu . . . 47
- DualRotarySlider . . . 51
  - Barometer . . . 26
- Layout . . . 75
- Panels . . . 114
  - DropDownMenus . . . 50
  - EqualizerPanel . . . 69
  - Sliders . . . 119
- SliderWithLabel . . . 124
  - CalibrationSlider . . . 37
  - verticalMixSlider . . . 142
    - Equalizer . . . 65

CoreWrapper . . . 46
juce::DocumentWindow
- GuiAppApplication::MainWindow . . . 77

Envelope . . . 59
EventHandler . . . 71
I2CBus . . . 74
juce::JUCEApplication
- GuiAppApplication . . . 72

juce::Button::Listener
- CalibrationOnClick . . . 35
- EqualizerPanel . . . 69

juce::Slider::Listener
- CalibrationSlider . . . 37
- DualRotarySlider . . . 51
- Sliders . . . 119

verticalMixSlider 142
juce::LookAndFeel_V4
CalibrationSliderLookAndFeel 40
NeedleLookAndFeel 85
VerticalSliderLookAndFeel 144
MidiCoordinator 79
MidiMessage 81
MidiMessageBuilder 83
Notes 87
Octaves 102
OctavesWithHarmonics 107
TromboneSynth 131
AudioRender 20
PitchMapper::PartialDef 116
PitchMapper 116
PressureSensor 118
RawInputEvent 119
RtMidiSink 119
SynthesiserParameters_t 129
ToFSensor 130
Tromboneless Parameters_t 130

Chapter 6

Class Index

6.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

AmplitudeMapper 19
AudioOutput 19
AudioRender 20
Barometer
A class which can allow for the look and feel of a set of rotary sliders be modified to look like a barometer (pressure gauge) 26
CalibrateEmbouchure
A class which is inherited from the CalibrationOnClick class that adds a dropdown menu next to the button for the user to select which option they wish to calibrate 33
CalibrationOnClick
A class which allows for the additon of a button to the screen without inheriting the button directly. This allows for other widgets to be added to it later without compatibility issues 35
CalibrationSlider
A class that adds more functionality to the slider class from Juce 37
CalibrationSliderLookAndFeel
A class to set the look and feel of a two finger linear slider to look like a trombone where the left most finger is the main body and the right most is the end of the trombones slide 40
CAP1188 45
CAP1188Channel 46
CAP1188Printer 46
CoreWrapper 46
DropDownMenu
A class that creates a drop down menu or 'comboBox' with a label attached to it. This takes up 95% of the available local area. The label is set to have a maxmimum width of 200, but scales as the window size is reduced 47
DropDownMenus
A class which is created to display the drop down menus required by the Tromboneless UI to choose the chosen option for calibration settings 50
DualRotarySlider
A class that adds more functionality to the slider class from Juce 51
Envelope
A class which can allow for the create of an envelope function which outputs a value between 0 and 1 depending on what stage the envelope is in. This uses the ADSR standard in music synthesis where A is the Attack, D is the decay, S is the saturation and R is the rest stage . . . 59
Equalizer
A class which inherits its sliders from `verticalMixSlider` made to act as an equalizer . . 65

EqualizerPanel
A class which may be called to add a panel to the window contains a series of sliders for the equalizer allowing for live editting of the synthesisers sound to ensure it sounds how the Trombonist wishes it to . . . 69
EventHandler . . . 71
GuiAppApplication
Creation of main application class . . . 72
I2CBus . . . 74
Layout
Main class that creates the window . . . 75
GuiAppApplication::MainWindow
This class implements the desktop window that contains an instance of the MainComponent class . . . 77
MidiCoordinator
Class to coordinate Midi message construction based on output from sensor mappers. This class is stateful and responsible for translating high-level sensor data into Midi messages. Utilises MidiMessageBuilder for external synths and directly controls internal synth . . . 79
MidiMessage
Represents a raw MIDI message as a sequence of bytes . . . 81
MidiMessageBuilder
Factory class for constructing common MIDI channel messages . . . 83
NeedleLookAndFeel
A class which may be attached to a rotary slider from the juce framework to change its look and feel. This changes the look of the slider to have an arrow rotate about the centre of the slider pointing to the different values rather than a knob moving in an arc as the basic rotary slider . . . 85
Notes
A class to store the frequency of the notes which may be referenced. The notes are stored in an enum `Notes_t` which allow for the higher level functions to call upon the specific note as opposed to frequency . . . 87
Octaves
Creation of a class to store the frequencies of notes at different octaves. Inherits from `Notes` class.
102
OctavesWithHarmonics
A class which inherits functions and variables from the `Octaves` class allowing for the notes being output to have a set of harmonic frequencies attached to them too . . . 107
Panels
A class which is inherited from Juce:Component reponsible for setting the general style of a panel within the main window . . . 114
PitchMapper::PartialDef . . . 116
PitchMapper
Takes raw input values from the LIDAR and mouthpiece/embouchure sensors and maps them to appropriate values for the musical coordinator to work with . . . 116
PressureSensor
Wrapper around ADS1115rpi library (https://github.com/berndporr/rpi_↩ ads1115) by Bernd Porr that provides an interface for pressure sensing . . . 118
RawInputEvent . . . 119
RtMidiSink . . . 119
Sliders
A class which adds all the linear sliders being used within the calibration section to be within the same
panel . . . 119
SliderWithLabel
A class which inherits attributes from the juce::Component class to create a new component containing a 8 possible label locations around the slider . . . 124
SynthesiserParameters_t
A struct to store any of the synthesier parameters that may be changed within the app. A data changed flag variable should be added to this struct to tell the core to read the new values . . . 129

ToFSensor 130
TrombonelessParameters_t
A struct which stores all the necessary parameters that may be changed on the app side effecting the core functionality. This struct should add a flag which is triggered whenever a parameter has been changed 130
TromboneSynth
A class which may be used to create a custom synth with an envelope attached to it 131
verticalMixSlider
A class that adds more functionality to the slider class from Juce 142
VerticalSliderLookAndFeel
Creation of a class which can be used to change the look of the sliders used within the equalizer for the Tromboneless project 144

Chapter 7

File Index

7.1 File List

Here is a list of all documented files with brief descriptions:

/home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.hpp
This file holds the declarations of all functions and variables needed by the Tromboneless project to change the looks and feel of various different widgets which are being displayed to the end user 155
/home/ryan-mcbride/Documents/Tromboneless/App/inc/Layout.hpp
This file holds the declarations of all functions and variables needed by the Tromboneless project to define the layout of the window which is being displayed to the end user 161
/home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp
A header file to contain the panels used within the Tromboneless user interface window. This will allow for the encapsulation of different areas as opposed to having large amounts of repeat calls in the `Layout.hpp` file 163
/home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
A file which sets the declarations of any custom widgets which have been built off of the juce framework. A large portion of these widgets have been created specifically for the Tromboneless project 166
/home/ryan-mcbride/Documents/Tromboneless/App/src/CustomStyles.cpp
This file holds the defintions of all functions and variables needed by the Tromboneless project to change the looks and feel of various different widgets which are being displayed to the end user 172
/home/ryan-mcbride/Documents/Tromboneless/App/src/Layout.cpp
This file holds the defintions of all functions and variables needed by the Tromboneless project to define the layout of the window which is being displayed to the end user 175
/home/ryan-mcbride/Documents/Tromboneless/App/src/Panels.cpp
This file contains the function defintions for to create the necessary panels that may be used within the Tromboneless user interface 176
/home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp
A file which defines the functions which have been declared within the `Widgets.hpp` file . . 177
/home/ryan-mcbride/Documents/Tromboneless/Core/include/AmplitudeMapper.hpp 178
/home/ryan-mcbride/Documents/Tromboneless/Core/include/audioRender.hpp 178
/home/ryan-mcbride/Documents/Tromboneless/Core/include/CoreWrapper.hpp
A file to allow for the app to view methods attached to the main functioanlity of the core 179
/home/ryan-mcbride/Documents/Tromboneless/Core/include/envelope.hpp
A header file containing all the necessary declarations and references to create a functional envelope class using the ADSR standard 189
/home/ryan-mcbride/Documents/Tromboneless/Core/include/eventHandler.hpp 191
/home/ryan-mcbride/Documents/Tromboneless/Core/include/main.hpp 192
/home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiCoordinator.hpp 192

/home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiMessage.hpp 193
/home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiMessageBuilder.hpp 194
/home/ryan-mcbride/Documents/Tromboneless/Core/include/PitchMapper.hpp 194
/home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp
A header file containing all the necessary declarations and references to allow for the Tromboneless to synthesise the noise a trombone would make when playing the same notes 195
/home/ryan-mcbride/Documents/Tromboneless/Core/include/tromboneSynth.hpp
A file to declare the synthesiser parameters and functions for the trombone specifically using the classes and methods declared in the `synth.hpp` file 199
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/audio_output.hpp 181
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/cap1188_registers.hpp 181
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/cap_sensor.hpp 184
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/i2c_bus.hpp 186
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/Pin_Definitions.h 187
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/pressure_sensor.hpp 187
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/tof_sensor.hpp 188
/home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/USBMidi.hpp 189
/home/ryan-mcbride/Documents/Tromboneless/Core/src/CoreWrapper.cpp
A file which encapsulations the main funcationality to allow for the app to act independently of it 202
/home/ryan-mcbride/Documents/Tromboneless/Core/src/envelope.cpp
Contains the defintions of the functions and variables needed to create an envelope 202
/home/ryan-mcbride/Documents/Tromboneless/Core/src/synth.cpp
C file which contains the defintions for the functions declared in the `synth.hpp` file 203
/home/ryan-mcbride/Documents/Tromboneless/Core/src/tromboneSynth.cpp
A file to define the synthesiser parameters and functions for the trombone specifically using the classes and methods declared in the `synth.hpp` file 204
/home/ryan-mcbride/Documents/Tromboneless/Shared/tromboneless_data.hpp 205

Chapter 8

Class Documentation

8.1 AmplitudeMapper Class Reference

Public Member Functions

- bool **calculateBaseline** (float pressureSample)
- float **getBaseline** ()
- bool **noteEdge** (float pressureSample)

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/AmplitudeMapper.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/AmplitudeMapper.cpp

8.2 AudioOutput Class Reference

Public Member Functions

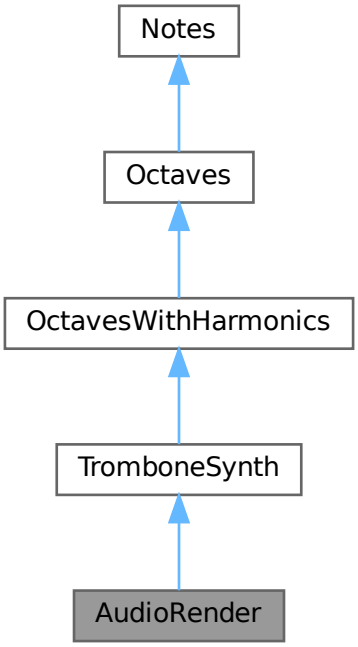
- void **writeSamples** (const int16_t ∗buffer, int frames)

The documentation for this class was generated from the following files:

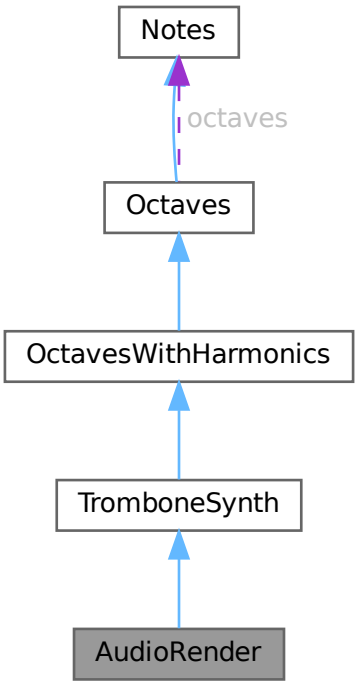
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/audio_output.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/drivers/audio_output.cpp

8.3 AudioRender Class Reference

Inheritance diagram for AudioRender:



Collaboration diagram for AudioRender:



Public Member Functions

- **AudioRender** (const AudioRender &)=delete
- AudioRender & **operator=** (const AudioRender &)=delete
- void **start** ()
- void **stop** ()
- bool **isRunning** () const
- void **noteOnFromMidi** (int midiNote)
- void **noteOffFromMidi** ()
- void **noteChangeFromMidi** (int midiNote)
- void **pitchBendFromMidi** (int bend)
- void **setDebugTone** (bool enabled)

Public Member Functions inherited from TromboneSynth

- TromboneSynth (int sampleRate=44100, float attack=10.0f, float decay=50.0f, float sustain=1.0f, float rest=10.0f)

  The constructor function for the `TromboneSynth` class.
- void StartTromboneNote (Notes::Notes_t note_in, int octave_in)

  A function which may be called to set the note and octave of a new sound by the user. If there is already a note playing, this will stop the previous not and play this one without phase shifting (initially).
- void ChangeTromboneNote (Notes::Notes_t note_in, int octave_in)

  A function that can change the current note without needed to change the stage that the envelope is in.

- float ReadTromboneAudio (void)

  A function which may be called to read the current amplitude of the audio signal being generated.
- void **StopTromboneNote** (void)

  A function which sends the synthesiser into its rest state.
- void NewTromboneNoteMIDI (int MIDINote)

  A function which can take in the MIDI equivalent note and map it to the Notes_t options to match the rest of the synth interface.
- void **HandleMIDINoteOn** ()

  A method to handle NoteOn messages as passed from the MIDI Coordinator.
- void **HandleMIDINoteOff** ()
- void **HandleMIDIPitchBend** (int midipitchbend)
- void **HandleMIDINoteChange** (int new_note)
- void setAttackMS (float attack_time)

  A function to set the amount of time in milliseconds that the attack stage takes.
- float getAttackMS (void)

  A function to get the current attack time in milliseconds.
- void setDecayMS (float decay_time)

  A function to set the amount of time in milliseconds that the decay stage takes.
- float getDecayMS (void)

  A function to get the current decay time in milliseconds.
- void setRestMS (float rest_time)

  A function to set the amount of time in milliseconds that the rest stage takes.
- float getRestMS (void)

  A function to get the current rest time in milliseconds.
- void setSustain (float sustain_in)

  A function to set the relative amplitude of the sustain stage.
- float getSustain (void)

  A function to get the current sustain value.
- void setNHarmonics (int n)

  A function to set the number of harmonics which are being generated for every note being played.
- int getNHarmonics (void)

  A function to get the number of harmonics which are being generated for every note being played.
- int getOctave (void)

  A function to get the current octave which the current note is in.
- Notes::Notes_t getNote (void)

  A function to get the current note which is being used.
- void setPitchBend (int bend)

  A function to set the pitch bend of the device.
- int getPitchBend (void)

  A function to get the pitch bend of the device.
- float getAdjustedFrequency (void)

  A function which determines the new frequency which should be being played depending upon the current note and octave as well as the pitch bend being input.
- Envelope **getEnvelope** (void)

Public Member Functions inherited from OctavesWithHarmonics

- **OctavesWithHarmonics** ()

  Creates an instance of the `OctavesWithHarmonics`.
- float PlayingNoteWithHarmonics (int n, int octave, Notes_t note, float phase)

  A function which may be called upon to stop playing a range of harmonic notes when a specific note is called upon to be played.
- float PlayingFrequencyWithHarmonics (int n, float phase)

  A function which may be called upon to stop playing a range of harmonic frequencies when a specific note is called upon to be played.
- float HarmonicDecay (int n)

  A function which calculates how much the amplitude of the harmonic should decay in comparison to the first harmonic.
- float getHarmonicDecayMax (int n)

  A function which returns the maximum amplitude possible for a set number of harmonics if they are all constructively interfering.
- void setDecayConstant (float decayC)

  A function that can change the value of the decay constant used to set the maxmimum amplitude of each successive harmonic.
- float getDecayConstant (void)

  A function which returns the decay constant being used to determine the relative ampltiude of each successive harmonic.

Public Member Functions inherited from Octaves

- float PlayingNote (int octave, Notes_t note, float phase)

  A function which may be called to determine the amplitude of a specified note at a specified phase.
- float PlayingFrequency (float phase)

  A function which may be called to determine the amplitude of a specified frequency at a specified time.

Public Member Functions inherited from Notes

- Notes (float C_in=16.35, float Db_in=17.32, float D_in=18.35, float Eb_in=19.45, float E_in=20.60, float F↩ _in=21.83, float Gb_in=23.12, float G_in=24.50, float Ab_in=25.96, float A_in=27.50, float Bb_in=29.14, float B_in=30.87)

  Constructor for the `Notes` class to create a new set of notes and assign specified frequencies to each. Each frequency is set to `Octave 0` by default.
- void setC (float frequency)

  A function that can change the value of the frequency that the C note plays.
- void setC_sharp (float frequency)

  A function that can change the value of the frequency that the C# note plays.
- void setDb (float frequency)

  A function that can change the value of the frequency that the Db note plays.
- void setD (float frequency)

  A function that can change the value of the frequency that the D note plays.
- void setD_sharp (float frequency)

  A function that can change the value of the frequency that the D# note plays.
- void setEb (float frequency)

  A function that can change the value of the frequency that the Eb note plays.
- void setE (float frequency)

  A function that can change the value of the frequency that the E note plays.
- void setF (float frequency)

A function that can change the value of the frequency that the F note plays.

- void setF_sharp (float frequency)

  A function that can change the value of the frequency that the F# note plays.
- void setGb (float frequency)

  A function that can change the value of the frequency that the Gb note plays.
- void setG (float frequency)

  A function that can change the value of the frequency that the G note plays.
- void setG_sharp (float frequency)

  A function that can change the value of the frequency that the G# note plays.
- void setAb (float frequency)

  A function that can change the value of the frequency that the Ab note plays.
- void setA (float frequency)

  A function that can change the value of the frequency that the A note plays.
- void setA_sharp (float frequency)

  A function that can change the value of the frequency that the A# note plays.
- void setBb (float frequency)

  A function that can change the value of the frequency that the Bb note plays.
- void setB (float frequency)

  A function that can change the value of the frequency that the B note plays.
- float setNote (Notes_t note, float freq)

  An alternative function which can allow for the user to alter the frequency of a specific note based upon the Notes_t enumerated values rather than specifying a function call.
- float getC (void)

  A function that returns the frequency of the C note for the specified object.
- float getC_sharp (void)

  A function that returns the frequency of the C# note for the specified object.
- float getDb (void)

  A function that returns the frequency of the Db note for the specified object.
- float getD (void)

  A function that returns the frequency of the D note for the specified object.
- float getD_sharp (void)

  A function that returns the frequency of the D# note for the specified object.
- float getEb (void)

  A function that returns the frequency of the Eb note for the specified object.
- float getE (void)

  A function that returns the frequency of the E note for the specified object.
- float getF (void)

  A function that returns the frequency of the F note for the specified object.
- float getF_sharp (void)

  A function that returns the frequency of the F# note for the specified object.
- float getGb (void)

  A function that returns the frequency of the Gb note for the specified object.
- float getG (void)

  A function that returns the frequency of the G note for the specified object.
- float getG_sharp (void)

  A function that returns the frequency of the G# note for the specified object.
- float getAb (void)

  A function that returns the frequency of the Ab note for the specified object.
- float getA (void)

  A function that returns the frequency of the A note for the specified object.
- float getA_sharp (void)

A function that returns the frequency of the A# note for the specified object.

- float getBb (void)

  A function that returns the frequency of the Bb note for the specified object.

- float getB (void)

  A function that returns the frequency of the B note for the specified object.

- float getNote (Notes_t note)

  An alternative function which can allow for the user to request a specific note based upon the Notes_t enumerated values rather than specifying a function call.

Static Public Attributes

- static constexpr int **frames** = 128
- static constexpr int **sample_rate** = 44100

Static Public Attributes inherited from Octaves

- static const char **nOctaves** = 9

  Stores the number of octaves being used.

Additional Inherited Members

Public Types inherited from Notes

- enum Notes_t {
  note_C , **note_Db** , **note_D** , **note_Eb** ,
  note_E , **note_F** , **note_Gb** , **note_G** ,
  note_Ab , **note_A** , **note_Bb** , **note_B** ,
  note_C_sharp = note_Db , **note_D_sharp** = note_Eb , **note_F_sharp** = note_Gb , **note_G_sharp** = note↩
  _Ab ,
  note_A_sharp = note_Bb }

  Creation of the enumerated list of possible notes.

Public Attributes inherited from Octaves

- Notes **octaves** [nOctaves]

  Initialising an array of the octaves and their notes.

Protected Member Functions inherited from OctavesWithHarmonics

- void **updateHarmonicCache** (int n)

Protected Attributes inherited from OctavesWithHarmonics

- float **decayConstant** = 0.1f
- std::array< float, maxHarmonics > **harmonicWeights** {}
- int **cachedNHarmonics** = 0
- bool **harmonicCacheDirty** = true

Static Protected Attributes inherited from OctavesWithHarmonics

- static constexpr int **maxHarmonics** = 64

The documentation for this class was generated from the following files:

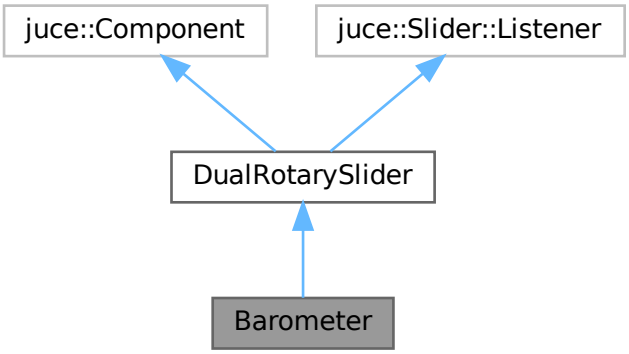
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/audioRender.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/audioRender.cpp

8.4 Barometer Class Reference

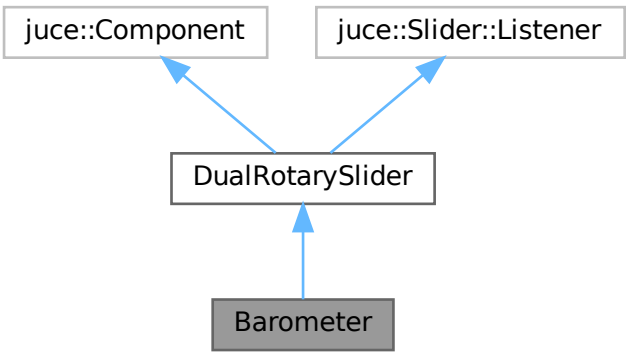
A class which can allow for the look and feel of a set of rotary sliders be modified to look like a barometer (pressure gauge).

`#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>`

Inheritance diagram for Barometer:



Collaboration diagram for Barometer:



Public Member Functions

- **Barometer** ()

  Constructor for the Barometer class which sets the look and feel of the sliders, attaches a listener to them both and sets the labels text to be the corresponding text.
- ~**Barometer** ()

  Decontructor for the Barometer class which removes the listeners from the sliders as well as resetting their look and feel.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A callback function which is automatically called upon when either of the sliders knobs are adjusted.
- void paint (juce::Graphics &g) override

  A function which is used to change the background of the component to look like it is a barometer/ pressure gauge.
- void setBackgroundColour (juce::Colour colour)

  A function to set the colour of the background of the slider.
- juce::Colour getBackgroundColour (void)

  A function to get the current colour of the background of the slider.
- void setTextColour (juce::Colour colour)

  A function to set the colour of the text around the slider.
- juce::Colour getTextColour (void)

  A function to get the current colour of the text around the slider.
- void setBoarderColour (juce::Colour colour)

  A function to set the colour of the boarder of the slider.
- juce::Colour getBoarderColour (void)

  A function to get the current colour of the boarder of the slider.
- void setEdgeColour (juce::Colour colour)

  A function to set the colour of the edge around the slider.
- juce::Colour getEdgeColour (void)

  A function to get the current colour of the edge around the slider.
- void setOuterRadius (float radius)

  A function to set the relative radius of the outer rotary slider.
- float getOuterRadius (void)

  A function to get the relative radius of the outer rotary slider.
- void setInnerRadius (float radius)

  A function to set the relative radius of the inner rotary slider.
- float getInnerRadius (void)

  A function to get the relative radius of the inner rotary slider.
- void setLabelRadius (float radius)

  A function to set the relative radius of the label.
- float getLabelRadius (void)

  A function to get the relative radius of the label.

Public Member Functions inherited from DualRotarySlider

- **DualRotarySlider** ()

  The constructor function for the `DualRotarySlider` class which sets the range of the sliders, minimum distance and adds listeners to them both so that when the sliders are moved, the corresponding variables may be changed.
- ~**DualRotarySlider** ()

  The destructor function for the `DualRotarySlider` class which removes the listeners from the sliders.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A function that is called upon when either of the fingers are moved.
- void **resized** () override

A callback function for when the window size has changed. This allows for the labels and sliders to be resized to be the best relative size. This can either be set to a predetermined size like 100px x 200px, but it is currrently set to be relative to the size of the screen.

- void setMinDifference (float difference)

  A function to set the minimum difference between the sliders 2 fingers.
- float getMinDifference (void)

  A function to get the current minimum difference between the 2 slider fingers.
- void setMinSliderRadius (float radius)

  A function to set the relative radius of the minimum finger slider. It should always be less than or equal to the maximum slider radius and be between 0 and 1.
- float getMinSliderRadius (void)

  A function to get the relative radius of the minimum finger slider for this object.
- void setMaxSliderRadius (float radius)

  A function to set the relative radius of the maximum finger slider. It should always be more than or equal to the minimum slider radius and be between 0 and 1.
- float getMaxSliderRadius (void)

  A function to get the relative radius of the maximum finger slider for this object.
- void setLabelHeight (float height)

  A function to set the maximum height of the label attached to the rotary slider.
- float getLabelHeight (void)

  A function to get the current height of the label which is attached to the rotary slider.
- void setMinLimit (float limit)

  A function to set the minimum limit of the rotary slider. This may be any floating point value that is less than the difference between the maximum value and the minimum difference between the slider fingers.
- float getMinLimit (void)

  A function to get the current minium limit of the rotary slider.
- void setMaxLimit (float limit)

  A function to set the maximum limit of the rotary slider. This may be any floating point value that is more than the sum of the minium value and the minimum difference between the slider fingers.
- float getMaxLimit (void)

  A function to get the current maximum limit of the rotary slider.
- void setInterval (float step)

  A function to set the minimum step the slider may be moved by the user. This value is in the same scale as the maxLimit and minLimit.
- float getInterval (void)

  A function to get the current minimum step which the slider may be adjusted by the user. This value is in the same scale as the maxLimit and minLimit.
- void setMinAngle (float angle)

  A function to allow for the minimum angle (furtherest anticlockwise) that the slider may be rotated to be set. This is measured in radians.
- float getMinAngle (void)

  A function to return the current minimum angle (furtherest anticlockwise) that the slider may be rotated. This is measured in radians.
- void setMaxAngle (float angle)

  A function to allow for the maximum angle (furtherest clockwise) that the slider may be rotated to be set. This is measured in radians.
- float getMaxAngle (void)

  A function to return the current maximum angle (furtherest clockwise) that the slider may be rotated. This is measured in radians.

Additional Inherited Members

Public Attributes inherited from DualRotarySlider

- juce::Slider maxSlider
- juce::Slider minSlider
- juce::Label rotaryLabel

8.4.1 Detailed Description

A class which can allow for the look and feel of a set of rotary sliders be modified to look like a barometer (pressure gauge).

Note

This is used in the Tromboneless project to calibrate the maximum and minimum gains on the mouthpiece.

8.4.2 Member Function Documentation

8.4.2.1 getBackgroundColour()

```
juce::Colour Barometer::getBackgroundColour (
            void  )
```

A function to get the current colour of the background of the slider.

Return values

Returns	the colour as a juce::Colour type.

8.4.2.2 getBoarderColour()

```
juce::Colour Barometer::getBoarderColour (
            void  )
```

A function to get the current colour of the boarder of the slider.

Return values

Returns	the colour as a juce::Colour type.

8.4.2.3 getEdgeColour()

```
juce::Colour Barometer::getEdgeColour (
            void  )
```

A function to get the current colour of the edge around the slider.

Return values

Returns	the colour as a juce::Colour type.

8.4.2.4 getInnerRadius()

```
float Barometer::getInnerRadius (
            void  )
```

A function to get the relative radius of the inner rotary slider.

Return values

The	floating point representation of the relative radius with respect to the widgets (barometer instance) bounds.

8.4.2.5 getLabelRadius()

```
float Barometer::getLabelRadius (
            void  )
```

A function to get the relative radius of the label.

Return values

The	floating point representation of the relative radius with respect to the widgets (barometer instance) bounds.

8.4.2.6 getOuterRadius()

```
float Barometer::getOuterRadius (
            void  )
```

A function to get the relative radius of the outer rotary slider.

Return values

The	floating point representation of the relative radius with respect to the widgets (barometer instance) bounds.

8.4.2.7 getTextColour()

```
juce::Colour Barometer::getTextColour (
            void  )
```

A function to get the current colour of the text around the slider.

Return values

Returns	the colour as a juce::Colour type.

8.4.2.8 paint()

```
void Barometer::paint (
            juce::Graphics & g )  [override]
```

A function which is used to change the background of the component to look like it is a barometer/ pressure gauge.

Parameters

g	A pointer to the juce::Graphics instance which is being drawn to. This is likely the parent component or the main window of the system.

Note

This function may be altered to change the look of the device but it currently adds a white circle as a background, golden outer circle for the edge of it, 2 arcs around it corresponding to bounds, with lines connecting them representing the possible settings.

8.4.2.9 setBackgroundColour()

```
void Barometer::setBackgroundColour (
            juce::Colour colour )
```

A function to set the colour of the background of the slider.

Parameters

colour	The colour to which the background should be set to.

8.4.2.10 setBoarderColour()

```
void Barometer::setBoarderColour (
            juce::Colour colour )
```

A function to set the colour of the boarder of the slider.

Parameters

colour	The colour to which the boarder should be set to.

8.4.2.11 setEdgeColour()

```
void Barometer::setEdgeColour (
            juce::Colour colour )
```

A function to set the colour of the edge around the slider.

Parameters

colour	The colour to which the edge should be set to.

8.4.2.12 setInnerRadius()

```
void Barometer::setInnerRadius (
            float radius )
```

A function to set the relative radius of the inner rotary slider.

Parameters

radius	The floating point representation of the relative radius with respect to the widgets (barometer instance) bounds.

8.4.2.13 setLabelRadius()

```
void Barometer::setLabelRadius (
            float radius )
```

A function to set the relative radius of the label.

Parameters

radius	The floating point representation of the relative radius with respect to the widgets (barometer instance) bounds.

8.4.2.14 setOuterRadius()

```
void Barometer::setOuterRadius (
            float radius )
```

A function to set the relative radius of the outer rotary slider.

Parameters

radius	The floating point representation of the relative radius with respect to the widgets (barometer instance) bounds.

8.4.2.15 setTextColour()

```
void Barometer::setTextColour (
            juce::Colour colour )
```

A function to set the colour of the text around the slider.

Parameters

colour	The colour to which the text should be set to.

8.4.2.16 sliderValueChanged()

```
void Barometer::sliderValueChanged (
            juce::Slider * sliderChanged )  [override]
```

A callback function which is automatically called upon when either of the sliders knobs are adjusted.

Parameters

sliderChanged	A pointer to the slider which has been changed by the user.

The documentation for this class was generated from the following files:

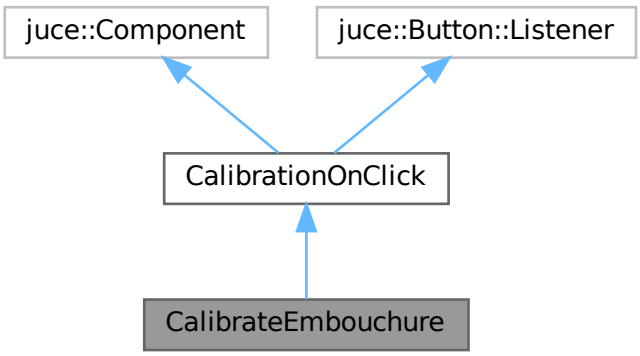
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.5 CalibrateEmbouchure Class Reference

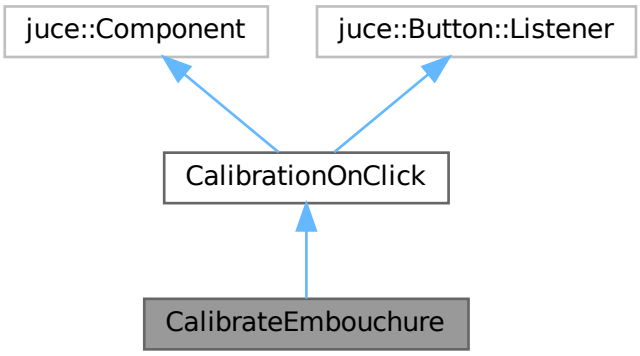
A class which is inherited from the CalibrationOnClick class that adds a dropdown menu next to the button for the user to select which option they wish to calibrate.

`#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>`

Inheritance diagram for CalibrateEmbouchure:



Collaboration diagram for CalibrateEmbouchure:



Public Member Functions

- **CalibrateEmbouchure** ()

  Constructor function for the CalibrateEmbouchure class which sets the specific parameters of the label, dropdown menu and button which are being used.
- ~**CalibrateEmbouchure** ()=default

  Destructor function for the CalibrateEmbouchure class which is currently set to default as there are mo specific methods or parameters which need to be dealt with before destructing the object.
- void **resized** () override

  A function which is automatically called upon when the user window is adjusted in size. This function will allow for the change in relative width and height of the components dependent upon screen size.
- void buttonClicked (juce::Button *thisButtonClicked) override

  Callback function which is automatically triggered when the button within this class is clicked by the user. As this class only has one instance of a button, there is no need to check which button has been pressed. However, a pointer to the button is still passed as default from the JUCE framework.

Public Member Functions inherited from CalibrationOnClick

- **CalibrationOnClick** ()

  A constructor for the CalibrationOnClick class which inherits characteristics from juce:: Component and Button::↩ Listener. It is used to receive a calibration message when clicked.
- ~**CalibrationOnClick** ()

  Destructor for the CalibrationOnClick class which removes the listener from the attached button.
- void resized () override

  A function which is called upon whenever the user interface window has changed in size. This can change the size of the component so that it can either be kept at a constant size of scaled to fit a specified area of the screen.

Additional Inherited Members

Public Attributes inherited from CalibrationOnClick

- juce::TextButton button

8.5.1 Detailed Description

A class which is inherited from the CalibrationOnClick class that adds a dropdown menu next to the button for the user to select which option they wish to calibrate.

Note

This class was specifically designed for the embachure selction as no other widget needed to have a calibration button, hence the name of the class and lack of an intermediate step from CalibrationOnClick.

8.5.2 Member Function Documentation

8.5.2.1 buttonClicked()

```
void CalibrateEmbouchure::buttonClicked (
            juce::Button * thisButtonClicked )  [override]
```

Callback function which is automatically triggered when the button within this class is clicked by the user. As this class only has one instance of a button, there is no need to check which button has been pressed. However, a pointer to the button is still passed as default from the JUCE framework.

Note

Currently this function updates the trombonelessParameters struct instance to change the calibrate embouchure flag to be on. This will allow the Core to read the ADC and set the corresponding value.

The documentation for this class was generated from the following files:

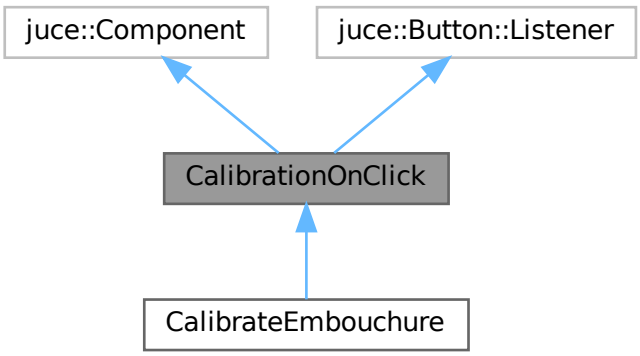
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.6 CalibrationOnClick Class Reference

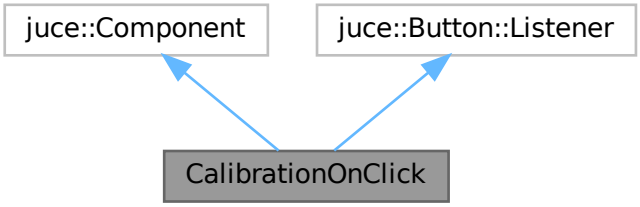
A class which allows for the additon of a button to the screen without inheriting the button directly. This allows for other widgets to be added to it later without compatibility issues.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for CalibrationOnClick:



Collaboration diagram for CalibrationOnClick:



Public Member Functions

- **CalibrationOnClick** ()

  A constructor for the CalibrationOnClick class which inherits characteristics from juce:: Component and Button::↩ Listener. It is used to receive a calibration message when clicked.
- ∼**CalibrationOnClick** ()

  Destructor for the CalibrationOnClick class which removes the listener from the attached button.
- void resized () override

  A function which is called upon whenever the user interface window has changed in size. This can change the size of the component so that it can either be kept at a constant size of scaled to fit a specified area of the screen.

Public Attributes

- juce::TextButton button

8.6.1 Detailed Description

A class which allows for the additon of a button to the screen without inheriting the button directly. This allows for other widgets to be added to it later without compatibility issues.

8.6.2 Member Function Documentation

8.6.2.1 resized()

```
void CalibrationOnClick::resized ( ) [override]
```

A function which is called upon whenever the user interface window has changed in size. This can change the size of the component so that it can either be kept at a constant size of scaled to fit a specified area of the screen.

Note

Currently, this adds a small margin around the local area and fills the rest of the area with the attached button.

8.6.3 Member Data Documentation

8.6.3.1 button

```
juce::TextButton CalibrationOnClick::button
```

Creating the TextButton instance for the CalibrationOnClick class to have a button within it.

The documentation for this class was generated from the following files:

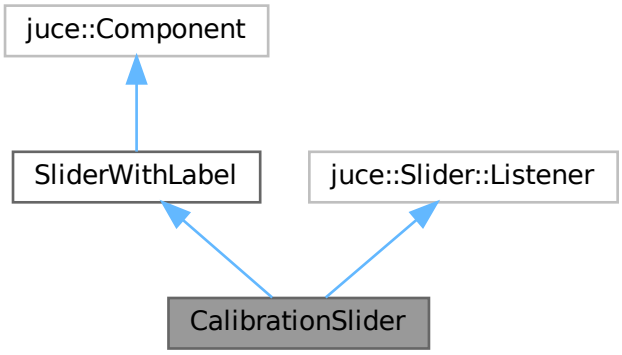
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.7 CalibrationSlider Class Reference

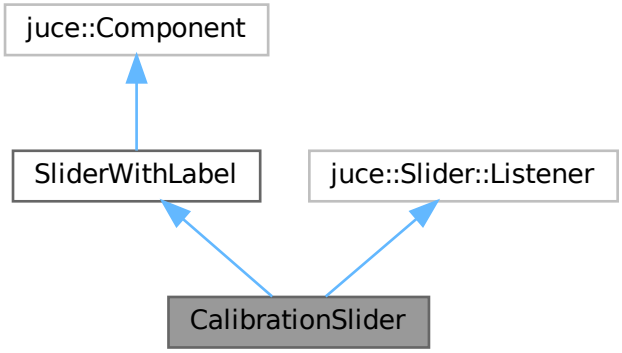
A class that adds more functionality to the slider class from Juce.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for CalibrationSlider:



Collaboration diagram for CalibrationSlider:



Public Member Functions

- **CalibrationSlider** ()

  The contructor function for the `CalibrationSlider` class which requires the type of slider being used to be initiated. This is set to `TwoValueHorizontal` by default.
- ~**CalibrationSlider** ()

  The destructor function for the `CalibrationSlider` class which removes the listener from the slider.
- void setMinDifference (float difference)

  A function which sets the minimum difference between the upper and lower fingers of the slider.
- float getMinDifference (void)

  A function which returns the current minimum distance that the slider fingers may be apart.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A function that is called upon when either of the fingers are moved.

Public Member Functions inherited from SliderWithLabel

- **SliderWithLabel** (juce::Slider::SliderStyle style)

  Constructor which sets the style of the slider being used and removes the text entry option.
- InitialiseLabelErrors_t CreateLabel (LabelPositions_t position, juce::String phrase, juce::NotificationType NotificationEnable=juce::dontSendNotification)

  A function which may be called to create or edit a specified label around the slider.
- void resized () override

  A function which is automatically called whenever the window size is modified. This automatically adjusts the bounds of each of the labels as well as the slider to ensure the size is correct with respect to the screen size for the component.
- void setTopLabelBounds (int bound)

  A function to set the bounds for which the top labels may be situated in.
- int getTopLabelBounds (void)

  A function to get the bounds for which the top labels are situated in.
- void setLeftLabelBounds (int bound)

  A function to set the bounds for which the labels on the left may be situated in.
- int getLeftLabelBounds (void)

A function to get the bounds for which the labels on the left are situated in.

- void setRightLabelBounds (int bound)

  A function to set the bounds for which the labels on the right may be situated in.
- int getRightLabelBounds (void)

  A function to get the bounds for which the labels on the right are situated in.
- void setBottomLabelBounds (int bound)

  A function to set the bounds for which the bottom labels may be situated in.
- int getBottomLabelBounds (void)

  A function to get the bounds for which the bottom labels are situated in.

Additional Inherited Members

Public Types inherited from SliderWithLabel

- enum LabelPositions_t {
  UpperLeft , **UpperCentre** , **UpperRight** , **MiddleLeft** ,
  MiddleRight , **LowerLeft** , **LowerCentre** , **LowerRight** }

  Enumerated list of possible positions for the label to be attached relative to the slider.
- enum InitialiseLabelErrors_t { **LabelOK** , **PositonNotAccepted** }

  Enumerated list of possible errors when initialising the labels within this component.

Public Attributes inherited from SliderWithLabel

- juce::Slider **slider**

  The slider which is having the labels added to it.

8.7.1 Detailed Description

A class that adds more functionality to the slider class from Juce.

8.7.2 Member Function Documentation

8.7.2.1 getMinDifference()

```
float CalibrationSlider::getMinDifference (
            void  )
```

A function which returns the current minimum distance that the slider fingers may be apart.

Return values

The	double precision representation of the minimum distance between the slider fingers.

8.7.2.2 setMinDifference()

```
void CalibrationSlider::setMinDifference (
```

```
float difference )
```

A function which sets the minimum difference between the upper and lower fingers of the slider.

Parameters

difference	The value of the minimum distance between the slider knobs.

8.7.2.3 sliderValueChanged()

```
void CalibrationSlider::sliderValueChanged (
            juce::Slider * sliderChanged )  [override]
```

A function that is called upon when either of the fingers are moved.

Note

This function automatically moves the fingers to ensure that they stay at least the minimum range apart from each other.

The documentation for this class was generated from the following files:

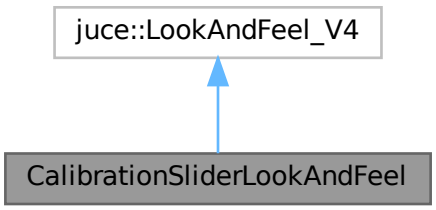
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.8 CalibrationSliderLookAndFeel Class Reference

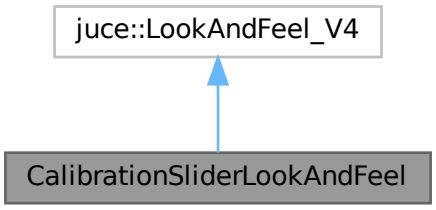
A class to set the look and feel of a two finger linear slider to look like a trombone where the left most finger is the main body and the right most is the end of the trombones slide.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.↩
hpp>
```

Inheritance diagram for CalibrationSliderLookAndFeel:



Collaboration diagram for CalibrationSliderLookAndFeel:



Public Member Functions

- virtual void drawLinearSlider (juce::Graphics &g, int x, int y, int width, int height, float sliderPos, float min↩ SliderPos, float maxSliderPos, const juce::Slider::SliderStyle style, juce::Slider &slider)

  When this look and feel is attached to an object which has a vertical slider in it, this function sets the look and feel of the slider being used. This makes it look like an average equalizer slider.
- void setEmptyTrackColour (juce::Colour colour)

  A function to allow for the empty track colour to be changed.
- juce::Colour getEmptyTrackColour (void)

  A function to get the current colour of the empty track of the slider.
- void setFullTrackColour (juce::Colour colour)

  A function to allow for the full track colour to be changed.
- juce::Colour getFullTrackColour (void)

  A function to get the current colour of the full track of the slider.
- void setMinThumbColour (juce::Colour colour)

  A function to allow for the colour of the first thumb to be changed.
- juce::Colour getMinThumbColour (void)

  A function to get the current colour of the minimum finger of the slider.
- void setMaxThumbColour (juce::Colour colour)

  A function to allow for the colour of the second thumb to be changed.
- juce::Colour getMaxThumbColour (void)

  A function to get the current colour of the maximum finger of the slider.
- void setTrackWidth (float width)

  A function to set the width of the track to indicate what step the slider is positioned in.
- float getTrackWidth (void)

  A function which may be used to find the current track width of the object.

8.8.1 Detailed Description

A class to set the look and feel of a two finger linear slider to look like a trombone where the left most finger is the main body and the right most is the end of the trombones slide.

Note

This is used in the Tromboneless project to calibrate the maximum and minimum distances of the slider by the users preference.

8.8.2 Member Function Documentation

8.8.2.1 drawLinearSlider()

```
void CalibrationSliderLookAndFeel::drawLinearSlider (
            juce::Graphics & g,
            int x,
            int y,
            int width,
            int height,
            float sliderPos,
            float minSliderPos,
            float maxSliderPos,
            const juce::Slider::SliderStyle style,
            juce::Slider & slider )  [virtual]
```

When this look and feel is attached to an object which has a vertical slider in it, this function sets the look and feel of the slider being used. This makes it look like an average equalizer slider.

Parameters

g	A pointer to the `juce::Graphics` instance which is being drawn to. This is likely the main window or else the parent component.
x	The integer representation of the slider's left corner in the global window.
y	The integer representation of the sliders top corner in the global window.
width	The integer representation of the width of the slider. i.e. the difference between the left most and right most part of the slider.
height	The integer representation of the width of the slider. i.e. the difference between the top most and bottom most parts of the slider's bounds.
sliderPos	The position of the sliders finger. The position represents x if the slider is horizontal and y if the slider is vertical.
minSliderPos	The minimum slider position.
maxSliderPos	The maximum slider position.
style	The style of the slider whose look and feel is getting modified. For example, LinearVertical or Rotary.
slider	A pointer to the slider which is being modified. This can allow for extraction of more data from the slider.

8.8.2.2 getEmptyTrackColour()

```
juce::Colour CalibrationSliderLookAndFeel::getEmptyTrackColour (
            void  )
```

A function to get the current colour of the empty track of the slider.

Return values

The	colour as a juce::Colour type.

8.8.2.3 getFullTrackColour()

```
juce::Colour CalibrationSliderLookAndFeel::getFullTrackColour (
            void  )
```

A function to get the current colour of the full track of the slider.

Return values

The	colour as a juce::Colour type.

Note

The full track refers to the part of the track between the two fingers/knobs

8.8.2.4 getMaxThumbColour()

```
juce::Colour CalibrationSliderLookAndFeel::getMaxThumbColour (
            void  )
```

A function to get the current colour of the maximum finger of the slider.

Return values

The	colour as a juce::Colour type.

8.8.2.5 getMinThumbColour()

```
juce::Colour CalibrationSliderLookAndFeel::getMinThumbColour (
            void  )
```

A function to get the current colour of the minimum finger of the slider.

Return values

The	colour as a juce::Colour type.

8.8.2.6 getTrackWidth()

```
float CalibrationSliderLookAndFeel::getTrackWidth (
            void  )
```

A function which may be used to find the current track width of the object.

Return values

Returns	the floating point representation of the track's width.

8.8.2.7 setEmptyTrackColour()

```
void CalibrationSliderLookAndFeel::setEmptyTrackColour (
            juce::Colour colour )
```

A function to allow for the empty track colour to be changed.

Parameters

colour	The colour which the emptry track should be set to.

8.8.2.8 setFullTrackColour()

```
void CalibrationSliderLookAndFeel::setFullTrackColour (
            juce::Colour colour )
```

A function to allow for the full track colour to be changed.

Parameters

colour	The colour which the full track should be set to.

Note

The full track refers to the part of the track between the two fingers/knobs

8.8.2.9 setMaxThumbColour()

```
void CalibrationSliderLookAndFeel::setMaxThumbColour (
            juce::Colour colour )
```

A function to allow for the colour of the second thumb to be changed.

Parameters

colour	The colour which the thumb should be set to.

8.8.2.10 setMinThumbColour()

```
void CalibrationSliderLookAndFeel::setMinThumbColour (
            juce::Colour colour )
```

A function to allow for the colour of the first thumb to be changed.

Parameters

colour	The colour which the thumb should be set to.

8.8.2.11 setTrackWidth()

```
void CalibrationSliderLookAndFeel::setTrackWidth (
            float width )
```

A function to set the width of the track to indicate what step the slider is positioned in.

Parameters

width	The width of the track in terms of the resolution.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/CustomStyles.cpp

8.9 CAP1188 Class Reference

Public Types

- using **CAP1188CallbackInterface** = std::function< void(uint8_t)>

Public Member Functions

- **CAP1188** (I2CBus &bus, uint8_t address=0x28, const std::string &gpioChipPath="/dev/gpiochip0", unsigned int gpioLine=27, unsigned int gpioReset=22, uint8_t enabledChannels=0x01, uint8_t interruptEnabled↩Channels=0x01)
- void **start** ()
- void **stop** ()
- bool **initialise** ()
- uint8_t **sensitivityGetter** ()
- void **sensitivitySetter** (uint8_t sensitivityVal)
- uint8_t **getAveraging** ()
- void **setAveraging** (uint8_t averageingVal)
- std::string **getSampleTime** ()
- void **setSampleTime** (std::string newSampleTime)
- std::string **getCycleTime** ()
- void **setCycleTime** (std::string newCycleTime)
- std::array< bool, 8 > **touched_pins** ()
- int **touched** ()
- int8_t **deltaCount** (int pin)
- void **recalibratePins** (uint8_t pins)
- std::array< uint8_t, 8 > **getThresholds** ()
- void **setThresholds** (std::array< uint8_t, 8 > newThresholds)
- void **worker** ()
- void **handleDataReady** ()
- void **disableStandby** ()
- void **clearInterrupt** ()
- void **registerCallback** (CAP1188CallbackInterface ci)
- void **tester** ()

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/cap_sensor.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/drivers/cap_sensor.cpp

8.10 CAP1188Channel Class Reference

Public Member Functions

- bool **isPinTouched** ()
- int8_t **pinDelta** ()
- int **getThreshold** ()
- void **setThreshold** (int threshold)
- void **pinRecalibrate** ()

Friends

- class **CAP1188**

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/cap_sensor.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/drivers/cap_sensor.cpp

8.11 CAP1188Printer Class Reference

Public Member Functions

- void **hasCAPSample** (int8_t delta)

The documentation for this class was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/Core/cap1188_test/cap1188_test.cpp

8.12 CoreWrapper Class Reference

Public Member Functions

- void **start** ()
- void **stop** ()

The documentation for this class was generated from the following files:

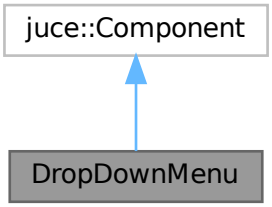
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/CoreWrapper.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/CoreWrapper.cpp

8.13 DropDownMenu Class Reference

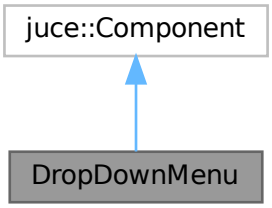
A class that creates a drop down menu or 'comboBox' with a label attached to it. This takes up 95% of the available local area. The label is set to have a maxmimum width of 200, but scales as the window size is reduced.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for DropDownMenu:



Collaboration diagram for DropDownMenu:



Public Member Functions

- DropDownMenu (juce::String labelPhrase="ComboBox")

  Constructor for the DropDownMenu class that creates an instance of the class. This makes a juce component that has a combobox or drop-down menu within it with a label attached to its left by default.
- ~**DropDownMenu** ()=default

  Destructor for the DropDownMenu which is set to default as there are no components within this class that need to be deinitialised before the deconstruction of the object.
- void **resized** () override

  Adding a function that will allow for the `DropDownMenu` to be resized when the window size is adjusted to fit best.
- void AddItem (juce::String itemName, int itemValue)

  A function which allows for the addition of more items within the drop down menu.
- template<typename T >
  void OnChange (T *updateVariable)

This function sets what will happen when the option wihtin the drop down menu has been changed by the user.

- void ChangeLabelText (juce::String text, juce::NotificationType notification=juce::dontSendNotification)

  This function allows for the editting of the main label displayed beside the drop down menu.

- juce::String getLabelText (void)

  A function which returns the current text which the label is displaying.

8.13.1 Detailed Description

A class that creates a drop down menu or 'comboBox' with a label attached to it. This takes up 95% of the available local area. The label is set to have a maxmimum width of 200, but scales as the window size is reduced.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 DropDownMenu()

```
DropDownMenu::DropDownMenu (
            juce::String labelPhrase = "ComboBox" )
```

Constructor for the DropDownMenu class that creates an instance of the class. This makes a juce component that has a combobox or drop-down menu within it with a label attached to its left by default.

Parameters

labelPhrase	This parameter should be of type juce::String and contain the phrase which should be displayed by the label. By default this is set to "ComboBox" so it may be viewed upon creation.

8.13.3 Member Function Documentation

8.13.3.1 AddItem()

```
void DropDownMenu::AddItem (
            juce::String itemName,
            int itemValue )
```

A function which allows for the addition of more items within the drop down menu.

Parameters

itemName	This parameter of type `juce::String` should contain the phrase which will be displayed to the end user on the display to select this option.
itemValue	This is the unique number assigned to each of the items. It must be more than 0 for it to be displayed. Each item must have a different value for them to be distinguishable from each other.

8.13.3.2 ChangeLabelText()

```
void DropDownMenu::ChangeLabelText (
```

```
            juce::String text,
            juce::NotificationType notification = juce::dontSendNotification )
```

This function allows for the editting of the main label displayed beside the drop down menu.

Parameters

text	This parameter should contain the text to which the label should be updated to display.
notification	This is the type of notification which should be attached to the drop down menu. By default this value is set to `juce::dontSendNotification`.

8.13.3.3 getLabelText()

```
juce::String DropDownMenu::getLabelText (
            void  )
```

A function which returns the current text which the label is displaying.

Return values

Returns	the text being distplayed as a type `juce::String`.

8.13.3.4 OnChange()

```
template<typename T >
void DropDownMenu::OnChange (
            T * updateVariable )  [inline]
```

This function sets what will happen when the option wihtin the drop down menu has been changed by the user.

Parameters

updateVariable	This should be a pointer to the variable which is being set to the item index which has been chosen.

Note

This function uses a lambda function to update the specified variable to ensure that the specified variable may be changed for different instances of the class.

The documentation for this class was generated from the following files:

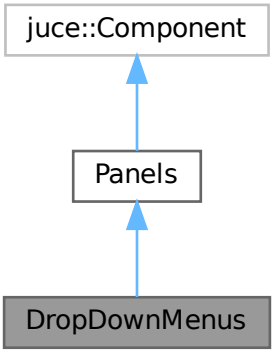
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.14 DropDownMenus Class Reference

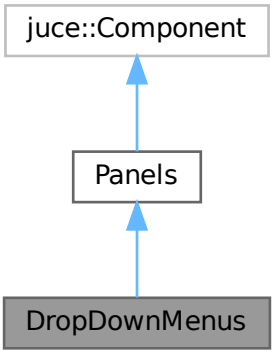
A class which is created to display the drop down menus required by the Tromboneless UI to choose the chosen option for calibration settings.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp>
```

Inheritance diagram for DropDownMenus:



Collaboration diagram for DropDownMenus:



Public Member Functions

- **DropDownMenus** ()

  The constructor function for the `DropDownMenus` class which creates an instance of the class and adds the chosen options to the shift keying drop down specifically. This also makes the components within it visible to the user.

- ~**DropDownMenus** ()=default

  The destructor function for the `DropDownMenus` class which is set to default as no child object within the instance need to be altered within the class.
- void **resized** () override

  Adding a function that will allow for the `DropDownMenus` to be resized when the window size is adjusted to fit best.

Public Member Functions inherited from Panels

- **Panels** ()=default

  The constructor function for the `Panels` class which currently provides no other function than creating an instance of the `Panels` class. I.e. no initialisation functions or settings.
- ~**Panels** ()=default

  The destructor function for the `Panels` class which currently does nothing extra as there is no initialisation within the class.
- void paint (juce::Graphics &g) override

  This function draws onto the parent function the default shape and colour of the panels designed.
- void setBackgroundColour (juce::Colour colour)

  A function to set the background colour of the panels for the main window.
- juce::Colour getBackgroundColour (void)

  A function to get the current background colour being used for the panels within the main window.

8.14.1 Detailed Description

A class which is created to display the drop down menus required by the Tromboneless UI to choose the chosen option for calibration settings.

The documentation for this class was generated from the following files:

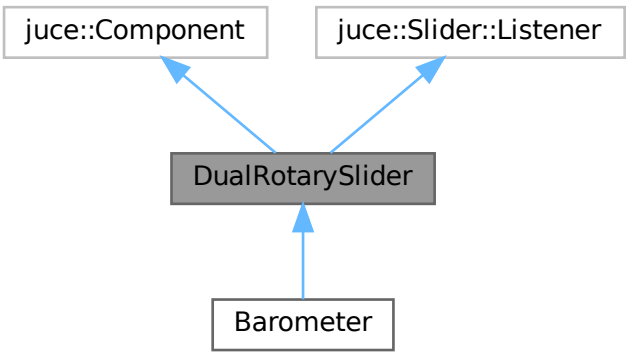
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Panels.cpp

8.15 DualRotarySlider Class Reference

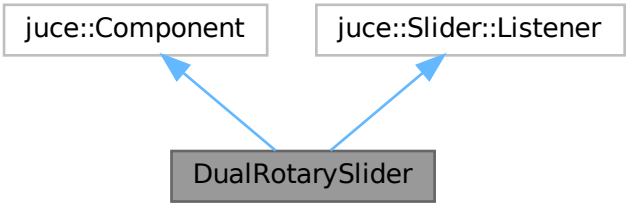
A class that adds more functionality to the slider class from Juce.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for DualRotarySlider:



Collaboration diagram for DualRotarySlider:



Public Member Functions

- **DualRotarySlider** ()

  The constructor function for the `DualRotarySlider` class which sets the range of the sliders, minimum distance and adds listeners to them both so that when the sliders are moved, the corresponding variables may be changed.
- ~**DualRotarySlider** ()

  The destructor function for the `DualRotarySlider` class which removes the listeners from the sliders.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A function that is called upon when either of the fingers are moved.
- void **resized** () override

  A callback function for when the window size has changed. This allows for the labels and sliders to be resized to be the best relative size. This can either be set to a predetermined size like 100px x 200px, but it is currrently set to be relative to the size of the screen.
- void setMinDifference (float difference)

  A function to set the minimum difference between the sliders 2 fingers.
- float getMinDifference (void)

  A function to get the current minimum difference between the 2 slider fingers.
- void setMinSliderRadius (float radius)

  A function to set the relative radius of the minimum finger slider. It should always be less than or equal to the maximum slider radius and be between 0 and 1.
- float getMinSliderRadius (void)

  A function to get the relative radius of the minimum finger slider for this object.
- void setMaxSliderRadius (float radius)

  A function to set the relative radius of the maximum finger slider. It should always be more than or equal to the minimum slider radius and be between 0 and 1.
- float getMaxSliderRadius (void)

  A function to get the relative radius of the maximum finger slider for this object.
- void setLabelHeight (float height)

  A function to set the maximum height of the label attached to the rotary slider.
- float getLabelHeight (void)

  A function to get the current height of the label which is attached to the rotary slider.
- void setMinLimit (float limit)

  A function to set the minimum limit of the rotary slider. This may be any floating point value that is less than the difference between the maximum value and the minimum difference between the slider fingers.
- float getMinLimit (void)

  A function to get the current minium limit of the rotary slider.

- void setMaxLimit (float limit)

  A function to set the maximum limit of the rotary slider. This may be any floating point value that is more than the sum of the minium value and the minimum difference between the slider fingers.
- float getMaxLimit (void)

  A function to get the current maximum limit of the rotary slider.
- void setInterval (float step)

  A function to set the minimum step the slider may be moved by the user. This value is in the same scale as the maxLimit and minLimit.
- float getInterval (void)

  A function to get the current minimum step which the slider may be adjusted by the user. This value is in the same scale as the maxLimit and minLimit.
- void setMinAngle (float angle)

  A function to allow for the minimum angle (furtherest anticlockwise) that the slider may be rotated to be set. This is measured in radians.
- float getMinAngle (void)

  A function to return the current minimum angle (furtherest anticlockwise) that the slider may be rotated. This is measured in radians.
- void setMaxAngle (float angle)

  A function to allow for the maximum angle (furtherest clockwise) that the slider may be rotated to be set. This is measured in radians.
- float getMaxAngle (void)

  A function to return the current maximum angle (furtherest clockwise) that the slider may be rotated. This is measured in radians.

Public Attributes

- juce::Slider maxSlider
- juce::Slider minSlider
- juce::Label rotaryLabel

8.15.1 Detailed Description

A class that adds more functionality to the slider class from Juce.

8.15.2 Member Function Documentation

8.15.2.1 getInterval()

```
float DualRotarySlider::getInterval (
            void  )
```

A function to get the current minimum step which the slider may be adjusted by the user. This value is in the same scale as the maxLimit and minLimit.

Return values

The	floating point representation of the current minimum step.

8.15.2.2 getLabelHeight()

```
float DualRotarySlider::getLabelHeight (
            void  )
```

A function to get the current height of the label which is attached to the rotary slider.

Return values

The	height of the label as a floating point value.

8.15.2.3 getMaxAngle()

```
float DualRotarySlider::getMaxAngle (
            void  )
```

A function to return the current maximum angle (furtherest clockwise) that the slider may be rotated. This is measured in radians.

Return values

The	angle in radians as a floating point value.

8.15.2.4 getMaxLimit()

```
float DualRotarySlider::getMaxLimit (
            void  )
```

A function to get the current maximum limit of the rotary slider.

Return values

The	floating point representation of the minimum limit.

8.15.2.5 getMaxSliderRadius()

```
float DualRotarySlider::getMaxSliderRadius (
            void  )
```

A function to get the relative radius of the maximum finger slider for this object.

Return values

Returns	the floating point representation of the relative radius.

8.15.2.6 getMinAngle()

```
float DualRotarySlider::getMinAngle (
            void  )
```

A function to return the current minimum angle (furtherest anticlockwise) that the slider may be rotated. This is measured in radians.

Return values

The	angle in radians as a floating point value.

8.15.2.7 getMinDifference()

```
float DualRotarySlider::getMinDifference (
            void  )
```

A function to get the current minimum difference between the 2 slider fingers.

Return values

Returns	the floating point representation of the minimum difference between the 2 sliders in terms of the chosen maximum and minimum values.

8.15.2.8 getMinLimit()

```
float DualRotarySlider::getMinLimit (
            void  )
```

A function to get the current minium limit of the rotary slider.

Return values

The	floating point representation of the minimum limit.

8.15.2.9 getMinSliderRadius()

```
float DualRotarySlider::getMinSliderRadius (
            void  )
```

A function to get the relative radius of the minimum finger slider for this object.

Return values

Returns	the floating point representation of the relative radius.

8.15.2.10 setInterval()

```
void DualRotarySlider::setInterval (
            float step )
```

A function to set the minimum step the slider may be moved by the user. This value is in the same scale as the maxLimit and minLimit.

Parameters

step	The minimum step as a floating point value.

8.15.2.11 setLabelHeight()

```
void DualRotarySlider::setLabelHeight (
            float height )
```

A function to set the maximum height of the label attached to the rotary slider.

Parameters

height	The height of the label in terms of the resolution as a floating point value.

8.15.2.12 setMaxAngle()

```
void DualRotarySlider::setMaxAngle (
            float angle )
```

A function to allow for the maximum angle (furtherest clockwise) that the slider may be rotated to be set. This is measured in radians.

Parameters

angle	The angle in radians as a floating point value.

8.15.2.13 setMaxLimit()

```
void DualRotarySlider::setMaxLimit (
            float limit )
```

A function to set the maximum limit of the rotary slider. This may be any floating point value that is more than the sum of the minium value and the minimum difference between the slider fingers.

Parameters

limit	The floating point representation of the limit.

8.15.2.14 setMaxSliderRadius()

```
void DualRotarySlider::setMaxSliderRadius (
            float radius )
```

A function to set the relative radius of the maximum finger slider. It should always be more than or equal to the minimum slider radius and be between 0 and 1.

Parameters

radius	The relative radius that the maximum slider should have.

8.15.2.15 setMinAngle()

```
void DualRotarySlider::setMinAngle (
            float angle )
```

A function to allow for the minimum angle (furtherest anticlockwise) that the slider may be rotated to be set. This is measured in radians.

Parameters

angle	The angle in radians as a floating point value.

8.15.2.16 setMinDifference()

```
void DualRotarySlider::setMinDifference (
            float difference )
```

A function to set the minimum difference between the sliders 2 fingers.

Parameters

difference	The difference in terms of the chosen maximum and minimum values as a floating point value.

8.15.2.17 setMinLimit()

```
void DualRotarySlider::setMinLimit (
            float limit )
```

A function to set the minimum limit of the rotary slider. This may be any floating point value that is less than the difference between the maximum value and the minimum difference between the slider fingers.

Parameters

limit	The floating point representation of the limit.

8.15.2.18 setMinSliderRadius()

```
void DualRotarySlider::setMinSliderRadius (
            float radius )
```

A function to set the relative radius of the minimum finger slider. It should always be less than or equal to the maximum slider radius and be between 0 and 1.

Parameters

radius	The relative radius that the minimum slider should have.

8.15.2.19 sliderValueChanged()

```
void DualRotarySlider::sliderValueChanged (
            juce::Slider * sliderChanged )  [override]
```

A function that is called upon when either of the fingers are moved.

Note

This function automatically moves the fingers to ensure that they stay at least the minimum range apart from each other.

8.15.3 Member Data Documentation

8.15.3.1 maxSlider

```
juce::Slider DualRotarySlider::maxSlider
```

The slider which will represent the high end of the calibration.

8.15.3.2 minSlider

```
juce::Slider DualRotarySlider::minSlider
```

The slider which will represent the low end of the calibration.

8.15.3.3 rotaryLabel

```
juce::Label DualRotarySlider::rotaryLabel
```

The label of the component.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.16 Envelope Class Reference

A class which can allow for the create of an envelope function which outputs a value between 0 and 1 depending on what stage the envelope is in. This uses the ADSR standard in music synthesis where A is the Attack, D is the decay, S is the saturation and R is the rest stage.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/envelope.↩
hpp>
```

Public Types

- enum envelope_stages_t {
  no_stage , **attack_stage** , **decay_stage** , **sustain_stage** ,
  rest_stage }

  Enumerated type to describe which part of the ADSR the object is in.

Public Member Functions

- Envelope (int attack_in=10, int decay_in=20, float sustain_in=0.98, int rest_in=10)

  The constructor function for the `Envelope` class which sets the values of the ADSR of the envelope and initialises the beginning of it.
- ~**Envelope** ()=default

  The destructor function for the `Envelope` class which is currently just set to default.
- float getAmplitude (void)

  A function that will return the current maximum amplitude of the envelope function. Depending one how many times this has been called sets at which stage the envelope is in.
- void **startEnvelope** (void)

  A function that enters the device into the attack stage.
- void **endEnvelope** (void)

  A function which sets the envelope to enter the rest stage and begin to approach 0.
- void setAttack (int attack_in)

  A function which can allow for the attack value to be changed externally from the class.
- int getAttack (void)

  A function which returns the number of times the `getAmplitude()` function needs to be called to go from 0 to 1 for the envelope gain value.
- void setDecay (int decay_in)

  A function which can allow for the decay value to be changed externally from the class.
- int getDecay (void)

  A function which returns the number of times the `getAmplitude()` function needs to be called to go from 1 to the sustain level for the envelope gain value.
- void setSustain (float sustain_in)

  A function which can allow for the sustain value to be changed externally from the class.
- float getSustain (void)

  A function which returns the value of the sustain level currently set for the object.
- void setRest (int rest_in)

  A function which can allow for the rest value to be changed externally from the class.
- int getRest (void)

  A function which can allow for the rest value to be used outside of the object without affecting the objects instance of the value from changing.
- float attackFunction (void)

  This function determines the shape of the attack curve. It needs to start from 0 and end at 1.

- float decayFunction (void)

  This function determines the shape of the decay curve. It needs to start from 1 and finish at the sustain value which has been set by the object.
- float restFunction (void)

  This function determines the shape of the rest curve. It needs to start from the sustain value and end at 0.
- envelope_stages_t getStage (void)

  A function which gets the current stage which the envelope is in.
- void setStage (envelope_stages_t stage_in)

  A function which sets the current stage which the envelope is in.

8.16.1 Detailed Description

A class which can allow for the create of an envelope function which outputs a value between 0 and 1 depending on what stage the envelope is in. This uses the ADSR standard in music synthesis where A is the Attack, D is the decay, S is the saturation and R is the rest stage.

Note

The public functions to set up the envelope take the time in milliseconds for each parameter as well as the sample rate to determine the number of samples each stage should last for.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 Envelope()

```
Envelope::Envelope (
            int attack_in = 10,
            int decay_in = 20,
            float sustain_in = 0.98,
            int rest_in = 10 )
```

The constructor function for the `Envelope` class which sets the values of the ADSR of the envelope and initialises the beginning of it.

Parameters

attack_in	The number of times the `getAmplitude()` function should be called before the envelope reaches its peak of 1.
decay_in	The number of times the `getAmplitude()` function should be called after the attack stage to reach the sustain level.
sustain↩_in	The relative amplitude of the signal after the decay stage.
rest_in	The number of times the `getAmplitde()` function should be called once the endNote flag has been triggered before the noise reaches an amplitude of 0.

Note

This class does not depend upon the real time or clock of the device, but instead the number of times the `getAmplitude()` function is called upon.

8.16.3 Member Function Documentation

8.16.3.1 attackFunction()

```
float Envelope::attackFunction (
            void  )
```

This function determines the shape of the attack curve. It needs to start from 0 and end at 1.

Return values

Returns	the amplitude at the time of calling.

Note

This is currently a linear ramp function between 0 and 1.

8.16.3.2 decayFunction()

```
float Envelope::decayFunction (
            void  )
```

This function determines the shape of the decay curve. It needs to start from 1 and finish at the sustain value which has been set by the object.

Return values

Returns	the floating point representation of the ampltiude at the time of request.

Note

This is currently a linear ramp function between 1 and the sustain value which has been set.

8.16.3.3 getAmplitude()

```
float Envelope::getAmplitude (
            void  )
```

A function that will return the current maximum amplitude of the envelope function. Depending one how many times this has been called sets at which stage the envelope is in.

Return values

A	floating point value between 0 and 1 representing the gain which would be applied to a signal it is attached to.

8.16.3.4 getAttack()

```
int Envelope::getAttack (
            void  )
```

A function which returns the number of times the `getAmplitude()` function needs to be called to go from 0 to 1 for the envelope gain value.

Return values

The	integer representation of how many times the the `getAmplitude()` function needs to be called to exit the attack stage.

8.16.3.5 getDecay()

```
int Envelope::getDecay (
            void  )
```

A function which returns the number of times the `getAmplitude()` function needs to be called to go from 1 to the sustain level for the envelope gain value.

Return values

The	integer representation of how many times the the `getAmplitude()` function needs to be called to exit the decay stage and enter sustain.

8.16.3.6 getRest()

```
int Envelope::getRest (
            void  )
```

A function which can allow for the rest value to be used outside of the object without affecting the objects instance of the value from changing.

Return values

Returns	the number of times the `getAmplitude()` function needs to be called before the ampltude reaches 0 from the sustain value.

8.16.3.7 getStage()

```
Envelope::envelope_stages_t Envelope::getStage (
            void  )
```

A function which gets the current stage which the envelope is in.

Return values

The	stage at which the envelope is in as type `envelope_stages_t`.

8.16.3.8 getSustain()

```
float Envelope::getSustain (
            void  )
```

A function which returns the value of the sustain level currently set for the object.

Return values

A	floating point representation of the sustian amplitude.

8.16.3.9 restFunction()

```
float Envelope::restFunction (
            void  )
```

This function determines the shape of the rest curve. It needs to start from the sustain value and end at 0.

Return values

Returns	the floating point representation of the amplitude at the time of request.

Note

This is currently a linear ramp function between the sustain value set and 0. This may be modified to match the shape of the instrument better.

8.16.3.10 setAttack()

```
void Envelope::setAttack (
            int attack_in )
```

A function which can allow for the attack value to be changed externally from the class.

Parameters

attack↩ _in	The number of times the `getAmplitude()` function should be called before the envelope reaches its peak of 1.

8.16.3.11 setDecay()

```
void Envelope::setDecay (
            int decay_in )
```

A function which can allow for the decay value to be changed externally from the class.

Parameters

decay_in	The number of times the `getAmplitude()` function should be called before the output goes from 1 to the chosen sustain value.

8.16.3.12 setRest()

```
void Envelope::setRest (
            int rest_in )
```

A function which can allow for the rest value to be changed externally from the class.

Parameters

rest_in	The number of times the `getAmplitude()` function should be called before the envelope reaches its peak of 0 from the sustain value.

8.16.3.13 setStage()

```
void Envelope::setStage (
            envelope_stages_t stage_in )
```

A function which sets the current stage which the envelope is in.

Parameters

stage_in	The stage at which the envelope should enter or already be in.

8.16.3.14 setSustain()

```
void Envelope::setSustain (
            float sustain_in )
```

A function which can allow for the sustain value to be changed externally from the class.

Parameters

sustain_in	The relative amplitude that the envelope should hold after the decay stage. This should be a floating point value between 0 and 1.

The documentation for this class was generated from the following files:

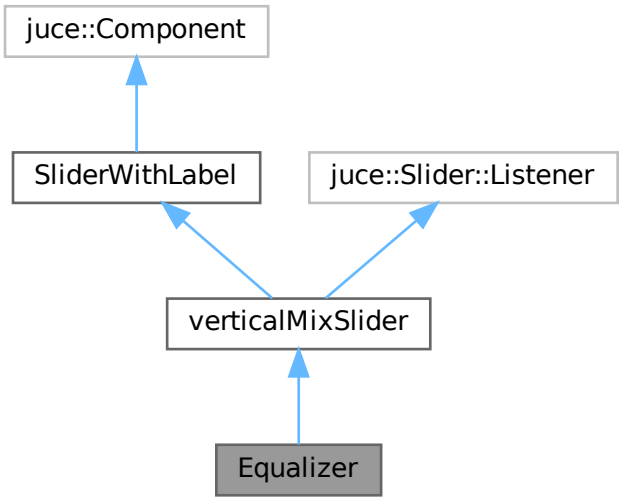
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/envelope.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/envelope.cpp

8.17 Equalizer Class Reference

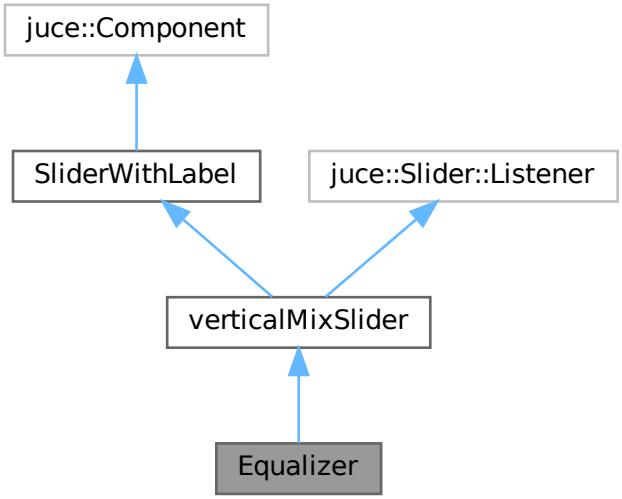
A class which inherits its sliders from `verticalMixSlider` made to act as an equalizer.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for Equalizer:



Collaboration diagram for Equalizer:



Public Member Functions

- Equalizer (char numberOfSliders=10)

  Contructor for the Equalizer class which creates a set of n vertical sliders in the style of an equalizer.
- ~**Equalizer** ()

  Destructor for the Equalizer which removes the slider listeners from each of the sliders as well as resetting the look and feel of each of them.
- void **resized** () override

  A function to set the size of the sliders with respect to the equalizer size. Whenever the parent component changes size, this function is called upon.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A callback function for when one of the sliders within the equalizer has been detected to have changed in value. This is where the corresponding variables the sliders affect may be changed.

Public Member Functions inherited from verticalMixSlider

- **verticalMixSlider** ()

  Constructor function for the vericalMixSlider which creates an instance of the class and sets the parameters to match it best such as setting the range to be between -9 dB and 9 dB with a step size of 3 dB as well as label size.
- ~**verticalMixSlider** ()

  Destructor function for the verticalMixSlider class which removes the listener from the slider before killing the object.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A callback function for when the slider used has been adjusted and its value may have been changed.

Public Member Functions inherited from SliderWithLabel

- **SliderWithLabel** (juce::Slider::SliderStyle style)

  Constructor which sets the style of the slider being used and removes the text entry option.
- InitialiseLabelErrors_t CreateLabel (LabelPositions_t position, juce::String phrase, juce::NotificationType NotificationEnable=juce::dontSendNotification)

  A function which may be called to create or edit a specified label around the slider.
- void resized () override

  A function which is automatically called whenever the window size is modified. This automatically adjusts the bounds of each of the labels as well as the slider to ensure the size is correct with respect to the screen size for the component.
- void setTopLabelBounds (int bound)

  A function to set the bounds for which the top labels may be situated in.
- int getTopLabelBounds (void)

  A function to get the bounds for which the top labels are situated in.
- void setLeftLabelBounds (int bound)

  A function to set the bounds for which the labels on the left may be situated in.
- int getLeftLabelBounds (void)

  A function to get the bounds for which the labels on the left are situated in.
- void setRightLabelBounds (int bound)

  A function to set the bounds for which the labels on the right may be situated in.
- int getRightLabelBounds (void)

  A function to get the bounds for which the labels on the right are situated in.
- void setBottomLabelBounds (int bound)

  A function to set the bounds for which the bottom labels may be situated in.
- int getBottomLabelBounds (void)

  A function to get the bounds for which the bottom labels are situated in.

Public Attributes

- juce::OwnedArray< verticalMixSlider > **eqSliders**

  Creating an array of sliders for the equalizer.

Public Attributes inherited from SliderWithLabel

- juce::Slider **slider**

  The slider which is having the labels added to it.

Additional Inherited Members

Public Types inherited from SliderWithLabel

- enum LabelPositions_t {
  UpperLeft , **UpperCentre** , **UpperRight** , **MiddleLeft** ,
  MiddleRight , **LowerLeft** , **LowerCentre** , **LowerRight** }

  Enumerated list of possible positions for the label to be attached relative to the slider.
- enum InitialiseLabelErrors_t { **LabelOK** , **PositonNotAccepted** }

  Enumerated list of possible errors when initialising the labels within this component.

8.17.1 Detailed Description

A class which inherits its sliders from `verticalMixSlider` made to act as an equalizer.

Parameters

nSliders	The number of sliders which should be added to the object. 10 by default.

8.17.2 Constructor & Destructor Documentation

8.17.2.1 Equalizer()

```
Equalizer::Equalizer (
            char numberOfSliders = 10 )
```

Contructor for the Equalizer class which creates a set of n vertical sliders in the style of an equalizer.

Parameters

numberOfSliders	The number of sliders which are present within the equalizer. This is set to 10 by default by may be altered.

Note

Caution: If the nuimberOfSliders is set to be higher than the number of elements in the `gains` variable in `SynthesiserParameters_t`, then the extra sliders will not affect any variable.

8.17.3 Member Function Documentation

8.17.3.1 sliderValueChanged()

```
void Equalizer::sliderValueChanged (
            juce::Slider * sliderChanged )  [override]
```

A callback function for when one of the sliders within the equalizer has been detected to have changed in value. This is where the corresponding variables the sliders affect may be changed.

Parameters

sliderChanged	A pointer to the slider which has been changed.

Note

As the equalizer has multiple sliders within it, it is important to check which of the sliders has been changed before altering any values. This function uses predefined variables which are affected by each of the sliders. If you wish to change this, you must change it within the function defintion as there are no functions to call to change the variable.

The documentation for this class was generated from the following files:

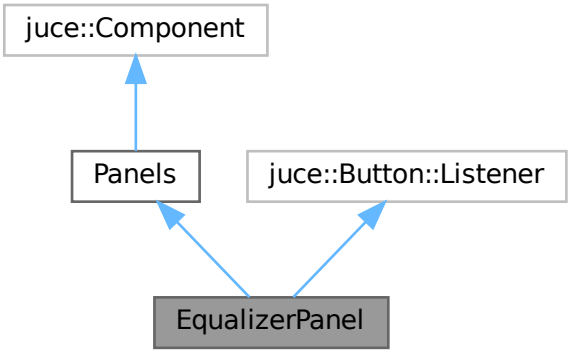
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.18 EqualizerPanel Class Reference

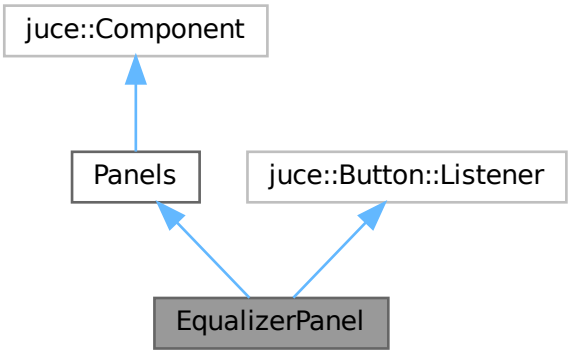
A class which may be called to add a panel to the window contains a series of sliders for the equalizer allowing for live editting of the synthesisers sound to ensure it sounds how the Trombonist wishes it to.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp>
```

Inheritance diagram for EqualizerPanel:



Collaboration diagram for EqualizerPanel:



Public Member Functions

- **EqualizerPanel** ()

  The constructor function for the `EqualizerPanel` class which adds the child components to the parent window and attaches a listener to the equalizer to detect when the slider values have been adjusted by the user.

- ~**EqualizerPanel** ()

  The destructor function which for the `EqualizerPanel` class which removes the listener from the equalizer before deinitialising.
- void resized () override

  A function which changes the size of the child components when the parent window has changed in size.
- void buttonClicked (juce::Button *buttonClicked) override

  A callback function which is called upon automatically when the user has pressed a button within this class.

Public Member Functions inherited from Panels

- **Panels** ()=default

  The constructor function for the `Panels` class which currently provides no other function than creating an instance of the `Panels` class. I.e. no initialisation functions or settings.
- ~**Panels** ()=default

  The destructor function for the `Panels` class which currently does nothing extra as there is no initialisation within the class.
- void paint (juce::Graphics &g) override

  This function draws onto the parent function the default shape and colour of the panels designed.
- void setBackgroundColour (juce::Colour colour)

  A function to set the background colour of the panels for the main window.
- juce::Colour getBackgroundColour (void)

  A function to get the current background colour being used for the panels within the main window.

8.18.1 Detailed Description

A class which may be called to add a panel to the window contains a series of sliders for the equalizer allowing for live editting of the synthesisers sound to ensure it sounds how the Trombonist wishes it to.

8.18.2 Member Function Documentation

8.18.2.1 buttonClicked()

```
void EqualizerPanel::buttonClicked (
            juce::Button * buttonClicked )  [override]
```

A callback function which is called upon automatically when the user has pressed a button within this class.

Parameters

buttonClicked	A pointer to the button which has been clicked.

Note

As this class only has one button, there is no need to check which button has been clicked but it is predefined to pass a parameter in the juce library. This button is used to show or hide the equalizer from the user in the parent window.

8.18.2.2 resized()

```
void EqualizerPanel::resized ( ) [override]
```

A function which changes the size of the child components when the parent window has changed in size.

Note

This function currently adds a slight margin around the edges of the bounds to ensure that the child components do not overlap with any external components.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Panels.cpp

8.19 EventHandler Class Reference

Public Types

- using **EventQueue** = std::queue< RawInputEvent >

Public Member Functions

- **EventHandler** (EventQueue &queue, std::mutex &queueMutex, std::condition_variable &queueCv)

  Constructor sets up event queuing, conditional blocking, initialises drivers for inputs.
- bool **initialise** ()
- void **start** ()

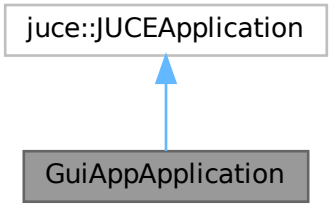
The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/eventHandler.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/eventHandler.cpp

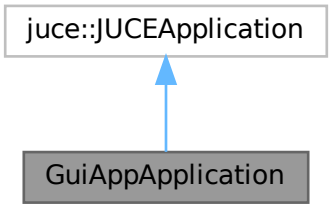
8.20 GuiAppApplication Class Reference

Creation of main application class.

Inheritance diagram for GuiAppApplication:



Collaboration diagram for GuiAppApplication:



Classes

- class MainWindow

  This class implements the desktop window that contains an instance of the MainComponent class.

Public Member Functions

- **GuiAppApplication** ()

  Defining the GuiAppApplication function.
- const juce::String getApplicationName () override

  A function which returns the name of the applicaiton.
- const juce::String getApplicationVersion () override

  A function which gets the current juce version and returns the name of the one being used.
- bool **moreThanOneInstanceAllowed** () override

A function which determines whether multiple instances of the window may be opened simultaneously. This is currently set to be false so that there may on be one window in use at a time.

- void initialise (const juce::String &commandLine) override

  The function which initialises the main window when the code is executed.
- void shutdown () override

  Function used to control what happens when the user wants to close the application.
- void **systemRequestedQuit** () override

  A function used to close the window. This is requested when the user presses the 'X' button.
- void anotherInstanceStarted (const juce::String &commandLine) override

  Handler for when the user tries to open mutliple instances of the application.

8.20.1 Detailed Description

Creation of main application class.

Note

This is called upon when the code is run for the first time.

8.20.2 Member Function Documentation

8.20.2.1 anotherInstanceStarted()

```
void GuiAppApplication::anotherInstanceStarted (
            const juce::String & commandLine )  [inline], [override]
```

Handler for when the user tries to open mutliple instances of the application.

Note

This function will return the command line parameters passed for the new instance were and may be used within the current window.

8.20.2.2 getApplicationName()

```
const juce::String GuiAppApplication::getApplicationName ( )  [inline], [override]
```

A function which returns the name of the applicaiton.

Note

We inject the name as compile definitions from the CMakeLists.txt

8.20.2.3 getApplicationVersion()

```
const juce::String GuiAppApplication::getApplicationVersion ( )  [inline], [override]
```

A function which gets the current juce version and returns the name of the one being used.

Note

We inject the name as compile definitions from the CMakeLists.txt

8.20.2.4 initialise()

```
void GuiAppApplication::initialise (
            const juce::String & commandLine )  [inline], [override]
```

The function which initialises the main window when the code is executed.

Parameters

commandLine	The arguments which the user has input on the terminal when executing the code. This is currently not used.

8.20.2.5 shutdown()

```
void GuiAppApplication::shutdown ( ) [inline], [override]
```

Function used to control what happens when the user wants to close the application.

Note

Currently nothing extra is happening here other than closing the window

The documentation for this class was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/App/src/main.cpp

8.21 I2CBus Class Reference

Public Member Functions

- **I2CBus** (const char ∗devicePath="/dev/i2c-1")
- bool **openBus** ()
- void **closeBus** ()
- int **writeBlock16** (uint8_t address, uint16_t reg, const uint8_t ∗data, uint8_t length)
- int **readBlock16** (uint8_t address, uint16_t reg, uint8_t ∗data, uint8_t length)
- int **writeBlock8** (uint8_t address, uint8_t reg, const uint8_t ∗data, uint8_t length)
- int **readBlock8** (uint8_t address, uint8_t reg, uint8_t ∗data, uint8_t length)

The documentation for this class was generated from the following files:

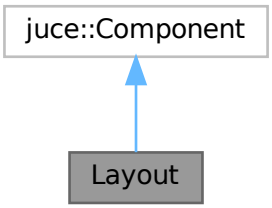
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/i2c_bus.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/drivers/i2c_bus.cpp

8.22 Layout Class Reference

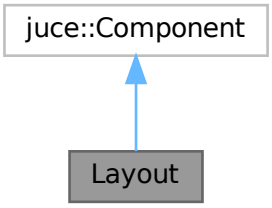
Main class that creates the window.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Layout.hpp>
```

Inheritance diagram for Layout:



Collaboration diagram for Layout:



Public Member Functions

- void paint (juce::Graphics &g) override

  Function that sets the background colour of the window.
- void resized () override

  Function that changes the size of the profiler.

8.22.1 Detailed Description

Main class that creates the window.

8.22.2 Member Function Documentation

8.22.2.1 paint()

```
void Layout::paint (
            juce::Graphics & g )  [override]
```

Function that sets the background colour of the window.

Parameters

g	The panel which should be modified.

Note

Always sets the background to the user preference for the background. This function also adds a title to be within the window.

8.22.2.2 resized()

```
void Layout::resized ( ) [override]
```

Function that changes the size of the profiler.

Note

Resizes the contents of the window including dropdown menus, labels and buttons

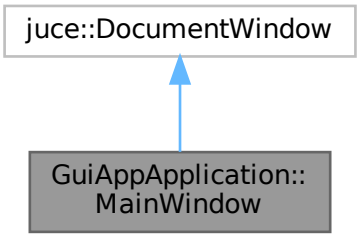
The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Layout.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Layout.cpp

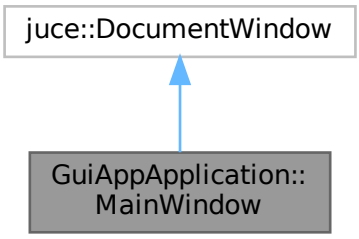
8.23 GuiAppApplication::MainWindow Class Reference

This class implements the desktop window that contains an instance of the MainComponent class.

Inheritance diagram for GuiAppApplication::MainWindow:



Collaboration diagram for GuiAppApplication::MainWindow:



Public Member Functions

- MainWindow (juce::String name)
- void closeButtonPressed () override

  function handler for when the user tries to close the program

8.23.1 Detailed Description

This class implements the desktop window that contains an instance of the MainComponent class.

Note

Inherits the juce::DocumentWindow class and functions.

8.23.2 Constructor & Destructor Documentation

8.23.2.1 MainWindow()

```
GuiAppApplication::MainWindow::MainWindow (
            juce::String name )  [inline], [explicit]
```

Generation of what the main window will look like.

8.23.3 Member Function Documentation

8.23.3.1 closeButtonPressed()

```
void GuiAppApplication::MainWindow::closeButtonPressed ( )  [inline], [override]
```

function handler for when the user tries to close the program

Note

Currently does not do anything other than request the system to close the application.

The documentation for this class was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/App/src/main.cpp

8.24 MidiCoordinator Class Reference

Class to coordinate Midi message construction based on output from sensor mappers. This class is stateful and responsible for translating high-level sensor data into Midi messages. Utilises MidiMessageBuilder for external synths and directly controls internal synth.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiCoordinator.↩
hpp>
```

Public Member Functions

- MidiCoordinator (AudioRender &synth)
- void **RegisterCallback** (CallbackInterface cb)
- void PressureEdge (bool on)
- void ChangeNote (int note)

  Method to update note from mouthpiece.
- void setBend (int bend)
- void setExpr (int expr)
- void setState (State newstate)
- void **setDevice** (bool external_device_present)
- AudioRender & **getSynth** (void)

8.24.1 Detailed Description

Class to coordinate Midi message construction based on output from sensor mappers. This class is stateful and responsible for translating high-level sensor data into Midi messages. Utilises MidiMessageBuilder for external synths and directly controls internal synth.

8.24.2 Constructor & Destructor Documentation

8.24.2.1 MidiCoordinator()

```
MidiCoordinator::MidiCoordinator (
            AudioRender & synth )
```

Default constructor.

8.24.3 Member Function Documentation

8.24.3.1 ChangeNote()

```
void MidiCoordinator::ChangeNote (
            int note )
```

Method to update note from mouthpiece.

Parameters

note	MIDI standard 0-127 notes.

8.24.3.2 PressureEdge()

```
void MidiCoordinator::PressureEdge (
            bool on )
```

Method to update gate on/off.

Parameters

on	Boolean: Decides whether

8.24.3.3 setBend()

```
void MidiCoordinator::setBend (
            int bend )
```

Method to update pitch bend from slide.

Parameters

bend	Values from -8192 to 0 (so only downward bends possible)

8.24.3.4 setExpr()

```
void MidiCoordinator::setExpr (
            int expr )
```

Method to update MIDI expression.

Parameters

expr	Expression value, 0-127.

8.24.3.5 setState()

```
void MidiCoordinator::setState (
            State newstate )
```

Method to set state of instrument.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiCoordinator.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/MidiCoordinator.cpp

8.25 MidiMessage Class Reference

Represents a raw MIDI message as a sequence of bytes.

`#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiMessage.↩ hpp>`

Public Types

- using **Byte** = unsigned char

Public Member Functions

- MidiMessage ()=default

  Default constructor.
- MidiMessage (std::vector< Byte > bytes)

  Construct a MIDI message from an existing byte vector.
- MidiMessage (std::initializer_list< Byte > bytes)

  Construct a MIDI message from a list of bytes.
- const std::vector< Byte > & bytes () const noexcept

  Access the raw byte buffer.
- std::size_t size () const noexcept

  Get number of bytes in the message.
- bool empty () const noexcept

  Check whether the message contains any bytes.

8.25.1 Detailed Description

Represents a raw MIDI message as a sequence of bytes.

This class is a lightweight value type used to store MIDI messages before transmission. It does not perform validation beyond basic structural construction.

A MIDI message consists of:

- A status byte
- Zero, one, or two data bytes (for standard channel messages)
- Or an arbitrary number of bytes (e.g. SysEx)

The underlying storage uses std::vector<unsigned char> to allow variable-length messages for future extensibility.

8.25.2 Constructor & Destructor Documentation

8.25.2.1 MidiMessage() [1/3]

```
MidiMessage::MidiMessage ( ) [default]
```

Default constructor.

Creates an empty MIDI message.

8.25.2.2 MidiMessage() [2/3]

```
MidiMessage::MidiMessage (
            std::vector< Byte > bytes ) [explicit]
```

Construct a MIDI message from an existing byte vector.

Parameters

bytes	Vector containing raw MIDI bytes.

8.25.2.3 MidiMessage() [3/3]

```
MidiMessage::MidiMessage (
            std::initializer_list< Byte > bytes )
```

Construct a MIDI message from a list of bytes.

Example:
```
MidiMessage msg{0x90, 0x3C, 0x64};
```

Parameters

bytes	Initialiser list of raw MIDI bytes.

8.25.3 Member Function Documentation

8.25.3.1 bytes()

```
const std::vector< MidiMessage::Byte > & MidiMessage::bytes ( ) const  [noexcept]
```

Access the raw byte buffer.

Returns

Const reference to internal byte vector.

8.25.3.2 empty()

```
bool MidiMessage::empty ( ) const  [noexcept]
```

Check whether the message contains any bytes.

Returns

True if empty, false otherwise.

8.25.3.3 size()

```
std::size_t MidiMessage::size ( ) const  [noexcept]
```

Get number of bytes in the message.

Returns

Size of the message in bytes.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiMessage.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/MidiMessage.cpp

8.26 MidiMessageBuilder Class Reference

Factory class for constructing common MIDI channel messages.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiMessage↩
Builder.hpp>
```

Public Types

- using **Byte** = MidiMessage::Byte

Public Member Functions

- MidiMessage noteOn (int channel, Byte note, Byte velocity) const

  Create a MIDI Note On message.
- MidiMessage noteOff (int channel, Byte note, Byte velocity=0) const

  Create a MIDI Note Off message.
- MidiMessage pitchBend (int channel, int value) const

  Pitch bend message constructor.
- MidiMessage **expr** (int channel, int value) const

8.26.1 Detailed Description

Factory class for constructing common MIDI channel messages.

This class encapsulates the encoding logic required to construct correctly formatted MIDI channel voice messages.

It separates message construction from message transmission, supporting a clean single-responsibility design.

Channels are specified using human-friendly numbering (1–16) and internally converted to zero-based encoding (0–15).

8.26.2 Member Function Documentation

8.26.2.1 noteOff()

```
MidiMessage MidiMessageBuilder::noteOff (
            int channel,
            Byte note,
            Byte velocity = 0 ) const
```

Create a MIDI Note Off message.

This implementation uses the dedicated Note Off status byte (0x8n) rather than Note On with velocity 0.

Parameters

channel	MIDI channel number (1–16).
note	MIDI note number (0–127).
velocity	Release velocity (default 0).

Returns

Constructed MidiMessage.

8.26.2.2 noteOn()

```
MidiMessage MidiMessageBuilder::noteOn (
            int channel,
            Byte note,
            Byte velocity ) const
```

Create a MIDI Note On message.

Parameters

channel	MIDI channel number (1–16).
note	MIDI note number (0–127).
velocity	Note velocity (0–127).

Returns

Constructed MidiMessage.

8.26.2.3 pitchBend()

```
MidiMessage MidiMessageBuilder::pitchBend (
            int channel,
            int value ) const
```

Pitch bend message constructor.

Parameters

channel	MIDI channel number (1–16).
value	Pitch bend value (0–8192).

Returns

Constructed MidiMessage.

The documentation for this class was generated from the following files:

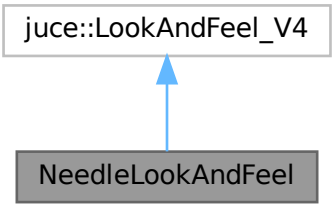
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/MidiMessageBuilder.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/MidiMessageBuilder.cpp

8.27 NeedleLookAndFeel Class Reference

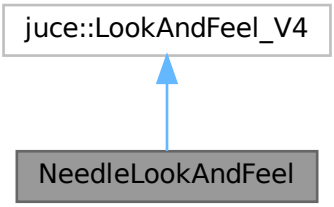
A class which may be attached to a rotary slider from the juce framework to change its look and feel. This changes the look of the slider to have an arrow rotate about the centre of the slider pointing to the different values rather than a knob moving in an arc as the basic rotary slider.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.↩
hpp>
```

Inheritance diagram for NeedleLookAndFeel:



Collaboration diagram for NeedleLookAndFeel:



Public Member Functions

- **NeedleLookAndFeel** ()=default

  The constructor function for the `NeedleLookAndFeel` class which is set to default.
- virtual ~**NeedleLookAndFeel** ()=default

  The destructor function for the `NeedleLookAndFeel` class which is set to default.
- virtual void drawRotarySlider (juce::Graphics &g, int x, int y, int width, int height, float sliderPos, const float rotaryStartAngle, const float rotaryEndAngle, juce::Slider &slider) override

  A function which is used to redraw the rotary slider whose look and feel is being changed. This will full describe how it should appear to the user.
- void setNeedleColour (juce::Colour colour)

  A function to set the colour of the needle which represents the position of the slider.
- juce::Colour getNeedleColour (void)

  A function to get the current colour of the needle which represents the position of the slider.

8.27.1 Detailed Description

A class which may be attached to a rotary slider from the juce framework to change its look and feel. This changes the look of the slider to have an arrow rotate about the centre of the slider pointing to the different values rather than a knob moving in an arc as the basic rotary slider.

8.27.2 Member Function Documentation

8.27.2.1 drawRotarySlider()

```
void NeedleLookAndFeel::drawRotarySlider (
            juce::Graphics & g,
            int x,
            int y,
            int width,
            int height,
            float sliderPos,
            const float rotaryStartAngle,
            const float rotaryEndAngle,
            juce::Slider & slider )  [override], [virtual]
```

A function which is used to redraw the rotary slider whose look and feel is being changed. This will full describe how it should appear to the user.

Parameters

g	A pointer to the `juce::Graphics` instance which is being updated. This is the parent component or the main window.
x	The x position of the slider.
y	The y position of the slider.
width	The width of the slider.
height	The maximum height of the slider.
sliderPos	The position of the slider finger.
rotaryStartAngle	The minimum angle which the slider starts at. This represents the minimum value, typically bottom left.
rotaryEndAngle	The maximum angle which the slider ends at. This represents the maximum value, typically bottom right.
slider	This is a pointer to the slider which is being drawn. Extra data may be extracted from it if necessary.

Note

This function is a call back which is automatically called to redraw the slider to look how described. It should not need to be called by any high level function.

8.27.2.2 getNeedleColour()

```
juce::Colour NeedleLookAndFeel::getNeedleColour (
            void  )
```

A function to get the current colour of the needle which represents the position of the slider.

Return values

The	colour of the needle as a juce::Colour type.

8.27.2.3 setNeedleColour()

```
void NneedleLookAndFeel::setNeedleColour (
            juce::Colour colour )
```

A function to set the colour of the needle which represents the position of the slider.

Parameters

colour	The colour which the needle should be set to.

The documentation for this class was generated from the following files:

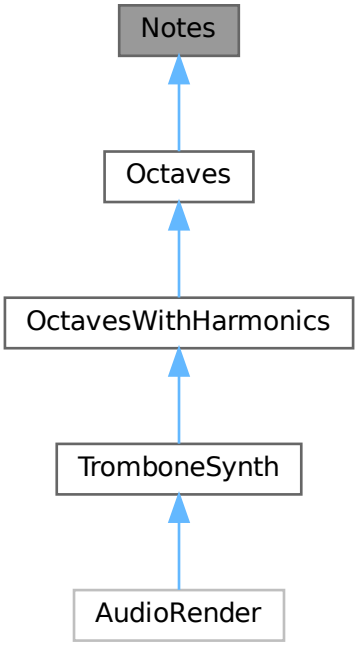
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/CustomStyles.cpp

8.28 Notes Class Reference

A class to store the frequency of the notes which may be referenced. The notes are stored in an enum `Notes_t` which allow for the higher level functions to call upon the specific note as opposed to frequency.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp>
```

Inheritance diagram for Notes:



Public Types

- enum Notes_t {
  note_C , **note_Db** , **note_D** , **note_Eb** ,
  note_E , **note_F** , **note_Gb** , **note_G** ,
  note_Ab , **note_A** , **note_Bb** , **note_B** ,
  note_C_sharp = note_Db , **note_D_sharp** = note_Eb , **note_F_sharp** = note_Gb , **note_G_sharp** = note↩_Ab ,
  note_A_sharp = note_Bb }

  Creation of the enumerated list of possible notes.

Public Member Functions

- Notes (float C_in=16.35, float Db_in=17.32, float D_in=18.35, float Eb_in=19.45, float E_in=20.60, float F↩_in=21.83, float Gb_in=23.12, float G_in=24.50, float Ab_in=25.96, float A_in=27.50, float Bb_in=29.14, float B_in=30.87)

  Constructor for the `Notes` class to create a new set of notes and assign specified frequencies to each. Each frequency is set to `Octave 0` by default.
- void setC (float frequency)

  A function that can change the value of the frequency that the C note plays.
- void setC_sharp (float frequency)

  A function that can change the value of the frequency that the C# note plays.
- void setDb (float frequency)

  A function that can change the value of the frequency that the Db note plays.

- void setD (float frequency)

  A function that can change the value of the frequency that the D note plays.
- void setD_sharp (float frequency)

  A function that can change the value of the frequency that the D# note plays.
- void setEb (float frequency)

  A function that can change the value of the frequency that the Eb note plays.
- void setE (float frequency)

  A function that can change the value of the frequency that the E note plays.
- void setF (float frequency)

  A function that can change the value of the frequency that the F note plays.
- void setF_sharp (float frequency)

  A function that can change the value of the frequency that the F# note plays.
- void setGb (float frequency)

  A function that can change the value of the frequency that the Gb note plays.
- void setG (float frequency)

  A function that can change the value of the frequency that the G note plays.
- void setG_sharp (float frequency)

  A function that can change the value of the frequency that the G# note plays.
- void setAb (float frequency)

  A function that can change the value of the frequency that the Ab note plays.
- void setA (float frequency)

  A function that can change the value of the frequency that the A note plays.
- void setA_sharp (float frequency)

  A function that can change the value of the frequency that the A# note plays.
- void setBb (float frequency)

  A function that can change the value of the frequency that the Bb note plays.
- void setB (float frequency)

  A function that can change the value of the frequency that the B note plays.
- float setNote (Notes_t note, float freq)

  An alternative function which can allow for the user to alter the frequency of a specific note based upon the Notes_t enumerated values rather than specifying a function call.
- float getC (void)

  A function that returns the frequency of the C note for the specified object.
- float getC_sharp (void)

  A function that returns the frequency of the C# note for the specified object.
- float getDb (void)

  A function that returns the frequency of the Db note for the specified object.
- float getD (void)

  A function that returns the frequency of the D note for the specified object.
- float getD_sharp (void)

  A function that returns the frequency of the D# note for the specified object.
- float getEb (void)

  A function that returns the frequency of the Eb note for the specified object.
- float getE (void)

  A function that returns the frequency of the E note for the specified object.
- float getF (void)

  A function that returns the frequency of the F note for the specified object.
- float getF_sharp (void)

  A function that returns the frequency of the F# note for the specified object.
- float getGb (void)

  A function that returns the frequency of the Gb note for the specified object.

- float getG (void)

  A function that returns the frequency of the G note for the specified object.
- float getG_sharp (void)

  A function that returns the frequency of the G# note for the specified object.
- float getAb (void)

  A function that returns the frequency of the Ab note for the specified object.
- float getA (void)

  A function that returns the frequency of the A note for the specified object.
- float getA_sharp (void)

  A function that returns the frequency of the A# note for the specified object.
- float getBb (void)

  A function that returns the frequency of the Bb note for the specified object.
- float getB (void)

  A function that returns the frequency of the B note for the specified object.
- float getNote (Notes_t note)

  An alternative function which can allow for the user to request a specific note based upon the Notes_t enumerated values rather than specifying a function call.

8.28.1 Detailed Description

A class to store the frequency of the notes which may be referenced. The notes are stored in an enum `Notes_t` which allow for the higher level functions to call upon the specific note as opposed to frequency.

8.28.2 Constructor & Destructor Documentation

8.28.2.1 Notes()

```
Notes::Notes (
            float C_in = 16.35,
            float Db_in = 17.32,
            float D_in = 18.35,
            float Eb_in = 19.45,
            float E_in = 20.60,
            float F_in = 21.83,
            float Gb_in = 23.12,
            float G_in = 24.50,
            float Ab_in = 25.96,
            float A_in = 27.50,
            float Bb_in = 29.14,
            float B_in = 30.87 )
```

Constructor for the `Notes` class to create a new set of notes and assign specified frequencies to each. Each frequency is set to `Octave 0` by default.

Parameters

C_in	The frequency of the C key.
Db↩ _in	The frequency of the Db key.
D_in	The frequency of the D key.

Parameters

Eb↩ _in	The frequency of the Eb key.
E_in	The frequency of the E key.
F_in	The frequency of the F key.
Gb↩ _in	The frequency of the Gb key.
G_in	The frequency of the G key.
Ab↩ _in	The frequency of the Ab key.
A_in	The frequency of the A key.
Bb↩ _in	The frequency of the Bb key.
B_in	The frequency of the B key.

Note

This function simply sets the local frequenies to be equal to the passed in frequencies.

8.28.3 Member Function Documentation

8.28.3.1 getA()

```
float Notes::getA (
            void  )
```

A function that returns the frequency of the A note for the specified object.

Return values

The	frequency of the A note as a floating point value.

8.28.3.2 getA_sharp()

```
float Notes::getA_sharp (
            void  )
```

A function that returns the frequency of the A# note for the specified object.

Return values

The	frequency of the A# note as a floating point value.

8.28.3.3 getAb()

```
float Notes::getAb (
```

```
            void  )
```

A function that returns the frequency of the Ab note for the specified object.

Return values

The	frequency of the Ab note as a floating point value.

8.28.3.4 getB()

```
float Notes::getB (
            void  )
```

A function that returns the frequency of the B note for the specified object.

Return values

The	frequency of the C note as a floating point value.

8.28.3.5 getBb()

```
float Notes::getBb (
            void  )
```

A function that returns the frequency of the Bb note for the specified object.

Return values

The	frequency of the Bb note as a floating point value.

8.28.3.6 getC()

```
float Notes::getC (
            void  )
```

A function that returns the frequency of the C note for the specified object.

Return values

The	frequency of the C note as a floating point value.

8.28.3.7 getC_sharp()

```
float Notes::getC_sharp (
            void  )
```

A function that returns the frequency of the C# note for the specified object.

Return values

The	frequency of the C# note as a floating point value.

8.28.3.8 getD()

```
float Notes::getD (
            void  )
```

A function that returns the frequency of the D note for the specified object.

Return values

The	frequency of the D note as a floating point value.

8.28.3.9 getD_sharp()

```
float Notes::getD_sharp (
            void  )
```

A function that returns the frequency of the D# note for the specified object.

Return values

The	frequency of the D# note as a floating point value.

8.28.3.10 getDb()

```
float Notes::getDb (
            void  )
```

A function that returns the frequency of the Db note for the specified object.

Return values

The	frequency of the Db note as a floating point value.

8.28.3.11 getE()

```
float Notes::getE (
            void  )
```

A function that returns the frequency of the E note for the specified object.

Return values

The	frequency of the E note as a floating point value.

8.28.3.12 getEb()

```
float Notes::getEb (
            void  )
```

A function that returns the frequency of the Eb note for the specified object.

Return values

The	frequency of the Eb note as a floating point value.

8.28.3.13 getF()

```
float Notes::getF (
            void  )
```

A function that returns the frequency of the F note for the specified object.

Return values

The	frequency of the F note as a floating point value.

8.28.3.14 getF_sharp()

```
float Notes::getF_sharp (
            void  )
```

A function that returns the frequency of the F# note for the specified object.

Return values

The	frequency of the F# note as a floating point value.

8.28.3.15 getG()

```
float Notes::getG (
            void  )
```

A function that returns the frequency of the G note for the specified object.

Return values

The	frequency of the G note as a floating point value.

8.28.3.16 getG_sharp()

```
float Notes::getG_sharp (
            void  )
```

A function that returns the frequency of the G# note for the specified object.

Return values

The	frequency of the G# note as a floating point value.

8.28.3.17 getGb()

```
float Notes::getGb (
            void  )
```

A function that returns the frequency of the Gb note for the specified object.

Return values

The	frequency of the Gb note as a floating point value.

8.28.3.18 getNote()

```
float Notes::getNote (
            Notes_t note )
```

An alternative function which can allow for the user to request a specific note based upon the Notes_t enumerated values rather than specifying a function call.

Parameters

note	The note of which the user wants to receive the frequency of.

Return values

Returns	a floating point value representing the frequency in Hz of the note specified.

8.28.3.19 setA()

```
void Notes::setA (
```

```
            float frequency )
```

A function that can change the value of the frequency that the A note plays.

Parameters

frequency	The frequency which the A not should play.

Note

There is no return value from this function.

8.28.3.20 setA_sharp()

```
void Notes::setA_sharp (
            float frequency )
```

A function that can change the value of the frequency that the A# note plays.

Parameters

frequency	The frequency which the A# not should play.

Note

There is no return value from this function.

8.28.3.21 setAb()

```
void Notes::setAb (
            float frequency )
```

A function that can change the value of the frequency that the Ab note plays.

Parameters

frequency	The frequency which the Ab not should play.

Note

There is no return value from this function.

8.28.3.22 setB()

```
void Notes::setB (
            float frequency )
```

A function that can change the value of the frequency that the B note plays.

Parameters

frequency	The frequency which the B not should play.

Note

There is no return value from this function.

8.28.3.23 setBb()

```
void Notes::setBb (
            float frequency )
```

A function that can change the value of the frequency that the Bb note plays.

Parameters

frequency	The frequency which the Bb not should play.

Note

There is no return value from this function.

8.28.3.24 setC()

```
void Notes::setC (
            float frequency )
```

A function that can change the value of the frequency that the C note plays.

Parameters

frequency	The frequency which the C not should play.

Note

There is no return value from this function.

8.28.3.25 setC_sharp()

```
void Notes::setC_sharp (
            float frequency )
```

A function that can change the value of the frequency that the C# note plays.

Parameters

frequency	The frequency which the C# not should play.

Note

There is no return value from this function.

8.28.3.26 setD()

```
void Notes::setD (
            float frequency )
```

A function that can change the value of the frequency that the D note plays.

Parameters

frequency	The frequency which the D not should play.

Note

There is no return value from this function.

8.28.3.27 setD_sharp()

```
void Notes::setD_sharp (
            float frequency )
```

A function that can change the value of the frequency that the D# note plays.

Parameters

frequency	The frequency which the D# not should play.

Note

There is no return value from this function.

8.28.3.28 setDb()

```
void Notes::setDb (
            float frequency )
```

A function that can change the value of the frequency that the Db note plays.

Parameters

frequency	The frequency which the Db not should play.

Note

There is no return value from this function.

8.28.3.29 setE()

```
void Notes::setE (
            float frequency )
```

A function that can change the value of the frequency that the E note plays.

Parameters

frequency	The frequency which the E not should play.

Note

There is no return value from this function.

8.28.3.30 setEb()

```
void Notes::setEb (
            float frequency )
```

A function that can change the value of the frequency that the Eb note plays.

Parameters

frequency	The frequency which the Eb not should play.

Note

There is no return value from this function.

8.28.3.31 setF()

```
void Notes::setF (
            float frequency )
```

A function that can change the value of the frequency that the F note plays.

Parameters

frequency	The frequency which the F not should play.

Note

There is no return value from this function.

8.28.3.32 setF_sharp()

```
void Notes::setF_sharp (
            float frequency )
```

A function that can change the value of the frequency that the F# note plays.

Parameters

frequency	The frequency which the F# not should play.

Note

There is no return value from this function.

8.28.3.33 setG()

```
void Notes::setG (
            float frequency )
```

A function that can change the value of the frequency that the G note plays.

Parameters

frequency	The frequency which the G not should play.

Note

There is no return value from this function.

8.28.3.34 setG_sharp()

```
void Notes::setG_sharp (
            float frequency )
```

A function that can change the value of the frequency that the G# note plays.

Parameters

frequency	The frequency which the G# not should play.

Note

There is no return value from this function.

8.28.3.35 setGb()

```
void Notes::setGb (
            float frequency )
```

A function that can change the value of the frequency that the Gb note plays.

Parameters

frequency	The frequency which the Gb not should play.

Note

There is no return value from this function.

8.28.3.36 setNote()

```
float Notes::setNote (
            Notes_t note,
            float freq )
```

An alternative function which can allow for the user to alter the frequency of a specific note based upon the Notes_t enumerated values rather than specifying a function call.

Parameters

note	The note of which the user wants to change the frequency of.
freq	The frequency which the specified note should be changed to.

The documentation for this class was generated from the following files:

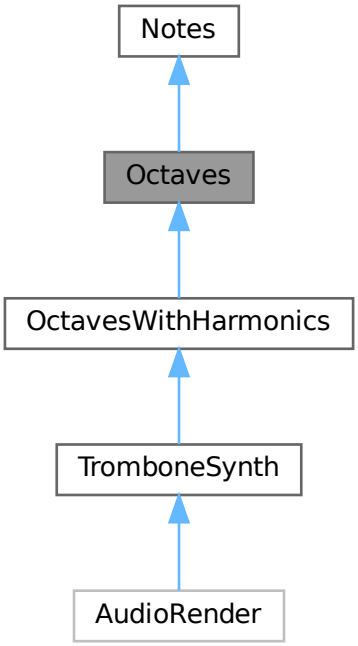
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/synth.cpp

8.29 Octaves Class Reference

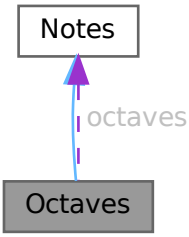
Creation of a class to store the frequencies of notes at different octaves. Inherits from `Notes` class.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp>
```

Inheritance diagram for Octaves:



Collaboration diagram for Octaves:



Public Member Functions

- float PlayingNote (int octave, Notes_t note, float phase)

  A function which may be called to determine the amplitude of a specified note at a specified phase.
- float PlayingFrequency (float phase)

  A function which may be called to determine the amplitude of a specified frequency at a specified time.

Public Member Functions inherited from Notes

- Notes (float C_in=16.35, float Db_in=17.32, float D_in=18.35, float Eb_in=19.45, float E_in=20.60, float F↩ _in=21.83, float Gb_in=23.12, float G_in=24.50, float Ab_in=25.96, float A_in=27.50, float Bb_in=29.14, float B_in=30.87)

  Constructor for the `Notes` class to create a new set of notes and assign specified frequencies to each. Each frequency is set to `Octave 0` by default.
- void setC (float frequency)

  A function that can change the value of the frequency that the C note plays.
- void setC_sharp (float frequency)

  A function that can change the value of the frequency that the C# note plays.
- void setDb (float frequency)

  A function that can change the value of the frequency that the Db note plays.
- void setD (float frequency)

  A function that can change the value of the frequency that the D note plays.
- void setD_sharp (float frequency)

  A function that can change the value of the frequency that the D# note plays.
- void setEb (float frequency)

  A function that can change the value of the frequency that the Eb note plays.
- void setE (float frequency)

  A function that can change the value of the frequency that the E note plays.
- void setF (float frequency)

  A function that can change the value of the frequency that the F note plays.
- void setF_sharp (float frequency)

  A function that can change the value of the frequency that the F# note plays.
- void setGb (float frequency)

  A function that can change the value of the frequency that the Gb note plays.
- void setG (float frequency)

  A function that can change the value of the frequency that the G note plays.
- void setG_sharp (float frequency)

  A function that can change the value of the frequency that the G# note plays.
- void setAb (float frequency)

  A function that can change the value of the frequency that the Ab note plays.
- void setA (float frequency)

  A function that can change the value of the frequency that the A note plays.
- void setA_sharp (float frequency)

  A function that can change the value of the frequency that the A# note plays.
- void setBb (float frequency)

  A function that can change the value of the frequency that the Bb note plays.
- void setB (float frequency)

  A function that can change the value of the frequency that the B note plays.
- float setNote (Notes_t note, float freq)

  An alternative function which can allow for the user to alter the frequency of a specific note based upon the Notes_t enumerated values rather than specifying a function call.
- float getC (void)

  A function that returns the frequency of the C note for the specified object.
- float getC_sharp (void)

  A function that returns the frequency of the C# note for the specified object.
- float getDb (void)

  A function that returns the frequency of the Db note for the specified object.
- float getD (void)

  A function that returns the frequency of the D note for the specified object.

- float getD_sharp (void)

  A function that returns the frequency of the D# note for the specified object.
- float getEb (void)

  A function that returns the frequency of the Eb note for the specified object.
- float getE (void)

  A function that returns the frequency of the E note for the specified object.
- float getF (void)

  A function that returns the frequency of the F note for the specified object.
- float getF_sharp (void)

  A function that returns the frequency of the F# note for the specified object.
- float getGb (void)

  A function that returns the frequency of the Gb note for the specified object.
- float getG (void)

  A function that returns the frequency of the G note for the specified object.
- float getG_sharp (void)

  A function that returns the frequency of the G# note for the specified object.
- float getAb (void)

  A function that returns the frequency of the Ab note for the specified object.
- float getA (void)

  A function that returns the frequency of the A note for the specified object.
- float getA_sharp (void)

  A function that returns the frequency of the A# note for the specified object.
- float getBb (void)

  A function that returns the frequency of the Bb note for the specified object.
- float getB (void)

  A function that returns the frequency of the B note for the specified object.
- float getNote (Notes_t note)

  An alternative function which can allow for the user to request a specific note based upon the Notes_t enumerated values rather than specifying a function call.

Public Attributes

- Notes **octaves** [nOctaves]

  Initialising an array of the octaves and their notes.

Static Public Attributes

- static const char **nOctaves** = 9

  Stores the number of octaves being used.

Additional Inherited Members

Public Types inherited from Notes

- enum Notes_t {
  note_C , **note_Db** , **note_D** , **note_Eb** ,
  note_E , **note_F** , **note_Gb** , **note_G** ,
  note_Ab , **note_A** , **note_Bb** , **note_B** ,
  note_C_sharp = note_Db , **note_D_sharp** = note_Eb , **note_F_sharp** = note_Gb , **note_G_sharp** = note↩
  _Ab ,
  note_A_sharp = note_Bb }

  Creation of the enumerated list of possible notes.

8.29.1 Detailed Description

Creation of a class to store the frequencies of notes at different octaves. Inherits from `Notes` class.

8.29.2 Member Function Documentation

8.29.2.1 PlayingFrequency()

```
float Octaves::PlayingFrequency (
            float phase )
```

A function which may be called to determine the amplitude of a specified frequency at a specified time.

Parameters

phase	The phase of the frequency at the time of sampling.

Return values

Returns	a floating point value representing the ampliude of the note at the time it was sampled.

Note

This function only uses unit amplitudes and does not change depending on the volume. The maxmimum and minimum will always be 1 and -1 respectively.

8.29.2.2 PlayingNote()

```
float Octaves::PlayingNote (
            int octave,
            Notes_t note,
            float phase )
```

A function which may be called to determine the amplitude of a specified note at a specified phase.

Parameters

octave	The octave in which the note being played is in.
note	The note which is currently being played.
phase	The phase of the wave at the time of sampling.

Return values

Returns	a floating point value representing the ampliude of the note at the time it was sampled.

Note

This function only uses unit amplitudes and does not change depending on the volume. The maxmimum and minimum will always be 1 and -1 respectively.

The documentation for this class was generated from the following files:

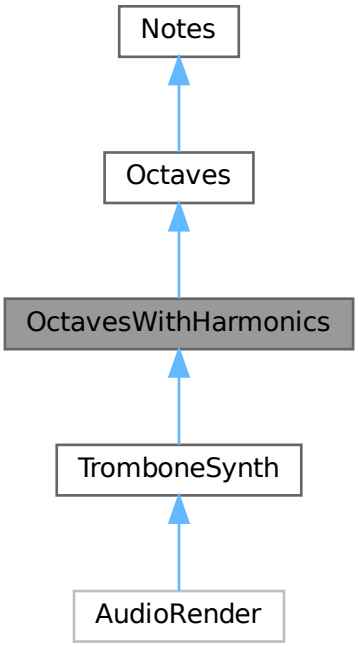
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/synth.cpp

8.30 OctavesWithHarmonics Class Reference

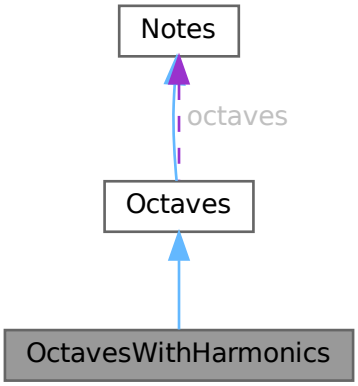
A class which inherits functions and variables from the `Octaves` class allowing for the notes being output to have a set of harmonic frequencies attached to them too.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp>
```

Inheritance diagram for OctavesWithHarmonics:



Collaboration diagram for OctavesWithHarmonics:



Public Member Functions

- **OctavesWithHarmonics** ()

  Creates an instance of the `OctavesWithHarmonics`.
- float PlayingNoteWithHarmonics (int n, int octave, Notes_t note, float phase)

  A function which may be called upon to stop playing a range of harmonic notes when a specific note is called upon to be played.
- float PlayingFrequencyWithHarmonics (int n, float phase)

  A function which may be called upon to stop playing a range of harmonic frequencies when a specific note is called upon to be played.
- float HarmonicDecay (int n)

  A function which calculates how much the amplitude of the harmonic should decay in comparison to the first harmonic.
- float getHarmonicDecayMax (int n)

  A function which returns the maximum amplitude possible for a set number of harmonics if they are all constructively interfering.
- void setDecayConstant (float decayC)

  A function that can change the value of the decay constant used to set the maxmimum amplitude of each successive harmonic.
- float getDecayConstant (void)

  A function which returns the decay constant being used to determine the relative ampltiude of each successive harmonic.

Public Member Functions inherited from Octaves

- float PlayingNote (int octave, Notes_t note, float phase)

  A function which may be called to determine the amplitude of a specified note at a specified phase.
- float PlayingFrequency (float phase)

  A function which may be called to determine the amplitude of a specified frequency at a specified time.

Public Member Functions inherited from Notes

- Notes (float C_in=16.35, float Db_in=17.32, float D_in=18.35, float Eb_in=19.45, float E_in=20.60, float F↩ _in=21.83, float Gb_in=23.12, float G_in=24.50, float Ab_in=25.96, float A_in=27.50, float Bb_in=29.14, float B_in=30.87)

  Constructor for the `Notes` class to create a new set of notes and assign specified frequencies to each. Each frequency is set to `Octave 0` by default.
- void setC (float frequency)

  A function that can change the value of the frequency that the C note plays.
- void setC_sharp (float frequency)

  A function that can change the value of the frequency that the C# note plays.
- void setDb (float frequency)

  A function that can change the value of the frequency that the Db note plays.
- void setD (float frequency)

  A function that can change the value of the frequency that the D note plays.
- void setD_sharp (float frequency)

  A function that can change the value of the frequency that the D# note plays.
- void setEb (float frequency)

  A function that can change the value of the frequency that the Eb note plays.
- void setE (float frequency)

  A function that can change the value of the frequency that the E note plays.
- void setF (float frequency)

  A function that can change the value of the frequency that the F note plays.
- void setF_sharp (float frequency)

  A function that can change the value of the frequency that the F# note plays.
- void setGb (float frequency)

  A function that can change the value of the frequency that the Gb note plays.
- void setG (float frequency)

  A function that can change the value of the frequency that the G note plays.
- void setG_sharp (float frequency)

  A function that can change the value of the frequency that the G# note plays.
- void setAb (float frequency)

  A function that can change the value of the frequency that the Ab note plays.
- void setA (float frequency)

  A function that can change the value of the frequency that the A note plays.
- void setA_sharp (float frequency)

  A function that can change the value of the frequency that the A# note plays.
- void setBb (float frequency)

  A function that can change the value of the frequency that the Bb note plays.
- void setB (float frequency)

  A function that can change the value of the frequency that the B note plays.
- float setNote (Notes_t note, float freq)

  An alternative function which can allow for the user to alter the frequency of a specific note based upon the Notes_t enumerated values rather than specifying a function call.
- float getC (void)

  A function that returns the frequency of the C note for the specified object.
- float getC_sharp (void)

  A function that returns the frequency of the C# note for the specified object.
- float getDb (void)

  A function that returns the frequency of the Db note for the specified object.
- float getD (void)

  A function that returns the frequency of the D note for the specified object.

- float getD_sharp (void)

  A function that returns the frequency of the D# note for the specified object.
- float getEb (void)

  A function that returns the frequency of the Eb note for the specified object.
- float getE (void)

  A function that returns the frequency of the E note for the specified object.
- float getF (void)

  A function that returns the frequency of the F note for the specified object.
- float getF_sharp (void)

  A function that returns the frequency of the F# note for the specified object.
- float getGb (void)

  A function that returns the frequency of the Gb note for the specified object.
- float getG (void)

  A function that returns the frequency of the G note for the specified object.
- float getG_sharp (void)

  A function that returns the frequency of the G# note for the specified object.
- float getAb (void)

  A function that returns the frequency of the Ab note for the specified object.
- float getA (void)

  A function that returns the frequency of the A note for the specified object.
- float getA_sharp (void)

  A function that returns the frequency of the A# note for the specified object.
- float getBb (void)

  A function that returns the frequency of the Bb note for the specified object.
- float getB (void)

  A function that returns the frequency of the B note for the specified object.
- float getNote (Notes_t note)

  An alternative function which can allow for the user to request a specific note based upon the Notes_t enumerated values rather than specifying a function call.

Protected Member Functions

- void **updateHarmonicCache** (int n)

Protected Attributes

- float **decayConstant** = 0.1f
- std::array< float, maxHarmonics > **harmonicWeights** {}
- int **cachedNHarmonics** = 0
- bool **harmonicCacheDirty** = true

Static Protected Attributes

- static constexpr int **maxHarmonics** = 64

Additional Inherited Members

Public Types inherited from Notes

- enum Notes_t {
  note_C , **note_Db** , **note_D** , **note_Eb** ,
  note_E , **note_F** , **note_Gb** , **note_G** ,
  note_Ab , **note_A** , **note_Bb** , **note_B** ,
  note_C_sharp = note_Db , **note_D_sharp** = note_Eb , **note_F_sharp** = note_Gb , **note_G_sharp** = note↩_Ab ,
  note_A_sharp = note_Bb }

  Creation of the enumerated list of possible notes.

Public Attributes inherited from Octaves

- Notes **octaves** [nOctaves]

  Initialising an array of the octaves and their notes.

Static Public Attributes inherited from Octaves

- static const char **nOctaves** = 9

  Stores the number of octaves being used.

8.30.1 Detailed Description

A class which inherits functions and variables from the `Octaves` class allowing for the notes being output to have a set of harmonic frequencies attached to them too.

8.30.2 Member Function Documentation

8.30.2.1 getDecayConstant()

```
float OctavesWithHarmonics::getDecayConstant (
            void  )
```

A function which returns the decay constant being used to determine the relative ampltiude of each successive harmonic.

Return values

The	floating point representation of the decay constant.

8.30.2.2 getHarmonicDecayMax()

```
float OctavesWithHarmonics::getHarmonicDecayMax (
            int n )
```

A function which returns the maximum amplitude possible for a set number of harmonics if they are all constructively interfering.

Parameters

n	The number of harmonics

Return values

The	maximum amplitude possible as a floating point value.

8.30.2.3 HarmonicDecay()

```
float OctavesWithHarmonics::HarmonicDecay (
            int n )
```

A function which calculates how much the amplitude of the harmonic should decay in comparison to the first harmonic.

Parameters

n	The harmonic which is being used.

Return values

Returns	a floating point value which represents the ratio of amplitudes.

Note

This function needs to be modified to better match the relationship between the frequency and the decay rate over the various harmonic frequencies.

8.30.2.4 PlayingFrequencyWithHarmonics()

```
float OctavesWithHarmonics::PlayingFrequencyWithHarmonics (
            int n,
            float phase )
```

A function which may be called upon to stop playing a range of harmonic frequencies when a specific note is called upon to be played.

Parameters

freq	The frequency in Hz that should be played.
time	The amount of time the signal has been playing. This corresponds to the phase that the waveform is at.

Return values

This	function will return a floating point value representing the normalised changed amplitude of the sigal at that point in time.

8.30.2.5 PlayingNoteWithHarmonics()

```
float OctavesWithHarmonics::PlayingNoteWithHarmonics (
            int n,
            int octave,
            Notes_t note,
            float phase )
```

A function which may be called upon to stop playing a range of harmonic notes when a specific note is called upon to be played.

Parameters

n	The number of harmonics which should be applied to the signal.
octave	The octave in which the first harmonic is in.
note	The note which is being played.
phase	The phase of the waveform at the time of sampling.

Return values

This	function will return a floating point value representing the normalised changed amplitude of the signal at that point in time.

Note

This function could be made to produce a much cleaner response by increasing the number of harmonics present and using non integer harmonics too.

8.30.2.6 setDecayConstant()

```
void OctavesWithHarmonics::setDecayConstant (
            float decayC )
```

A function that can change the value of the decay constant used to set the maxmimum amplitude of each successive harmonic.

Parameters

decayC	The floating point value which the decay constant should be set to.

The documentation for this class was generated from the following files:

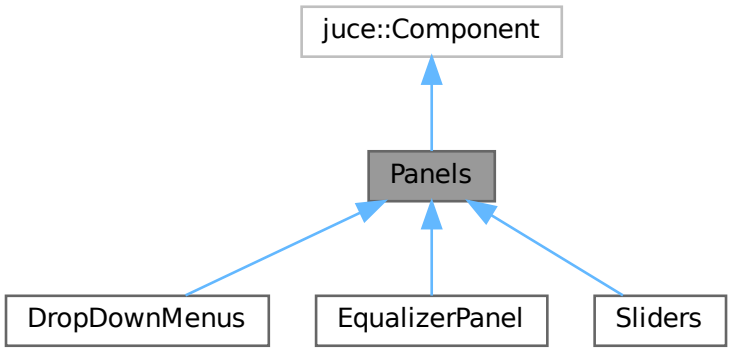
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/synth.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/synth.cpp

8.31 Panels Class Reference

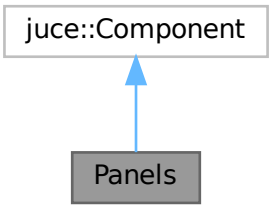
A class which is inherited from Juce:Component reponsible for setting the general style of a panel within the main window.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp>
```

Inheritance diagram for Panels:



Collaboration diagram for Panels:



Public Member Functions

- **Panels** ()=default

  The constructor function for the `Panels` class which currently provides no other function than creating an instance of the `Panels` class. I.e. no initialisation functions or settings.
- ~**Panels** ()=default

  The destructor function for the `Panels` class which currently does nothing extra as there is no initialisation within the class.
- void paint (juce::Graphics &g) override

  This function draws onto the parent function the default shape and colour of the panels designed.

- void setBackgroundColour (juce::Colour colour)

  A function to set the background colour of the panels for the main window.
- juce::Colour getBackgroundColour (void)

  A function to get the current background colour being used for the panels within the main window.

8.31.1 Detailed Description

A class which is inherited from Juce:Component reponsible for setting the general style of a panel within the main window.

Note

A panel should represent at least one component such as an image, label, slider or another widget type. Similar types should be grouped together such as all calibration settings.

8.31.2 Member Function Documentation

8.31.2.1 getBackgroundColour()

```
juce::Colour Panels::getBackgroundColour (
            void  )
```

A function to get the current background colour being used for the panels within the main window.

Return values

The	colour being used as a juce::Colour type.

8.31.2.2 paint()

```
void Panels::paint (
            juce::Graphics & g )  [override]
```

This function draws onto the parent function the default shape and colour of the panels designed.

Parameters

g	The point to the `juce::Graphics` instance which is being modified.

Note

The current design uses a rounded rectangle with a dark colour and a slight transparency to have the panel have a slight contrast to the background.

8.31.2.3 setBackgroundColour()

```
void Panels::setBackgroundColour (
            juce::Colour colour )
```

A function to set the background colour of the panels for the main window.

Parameters

colour	The colour which the background should be set to. Must be of type juce::Colour.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Panels.cpp

8.32 PitchMapper::PartialDef Struct Reference

Public Attributes

- int **minDelta**
- int **hysteresisDelta**
- int **partial**
- int **midiNote**

The documentation for this struct was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/PitchMapper.hpp

8.33 PitchMapper Class Reference

Takes raw input values from the LIDAR and mouthpiece/embouchure sensors and maps them to appropriate values for the musical coordinator to work with.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/Pitch↩
Mapper.hpp>
```

Classes

- struct PartialDef

Public Member Functions

- **PitchMapper** ()

  Constructor.
- int tof_to_MIDI_bend (uint16_t tof_distance)

  Maps raw LiDAR time-of-flight values to MIDI pitch bend scaled and clamped to range of 0-8192.
- int mouthpiece_to_MIDI_note (int8_t delta)

  Maps capacitative mouthpiece readings to discrete MIDI notes.
- void **SetSlideMaxLimit** (int new_slide_max_limit_mm)

  Set maximum trombone slide limit in mm.
- int **GetSlideMaxLimit** ()

  Return current max trombone slide limit in mm.
- void **SetSlideMinLimit** (int new_slide_min_limit_mm)

  Set minimum trombone slide limit in mm.
- int **GetSlideMinLimit** ()

  Return current min trombone slide limit in mm.
- void **SetMouthMIDI** (int new_MIDI_note)

  Set current mouthpiece MIDI note, bypassing mapping function.
- int **GetMouthMIDI** ()

  Return current mouthpiece MIDI note.
- void **SetTromboneType** (ShiftKeyingOptions_t trombone_choice)

8.33.1 Detailed Description

Takes raw input values from the LIDAR and mouthpiece/embouchure sensors and maps them to appropriate values for the musical coordinator to work with.

8.33.2 Member Function Documentation

8.33.2.1 mouthpiece_to_MIDI_note()

```
int PitchMapper::mouthpiece_to_MIDI_note (
            int8_t delta )
```

Maps capacitative mouthpiece readings to discrete MIDI notes.

Parameters

delta	Raw value supplied by the cap1188

8.33.2.2 tof_to_MIDI_bend()

```
int PitchMapper::tof_to_MIDI_bend (
            uint16_t tof_distance )
```

Maps raw LiDAR time-of-flight values to MIDI pitch bend scaled and clamped to range of 0-8192.

Parameters

tof_distance	Raw value returned from time-of-flight sensor.

Returns

Integer 0-8192

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/PitchMapper.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/PitchMapper.cpp

8.34 PressureSensor Class Reference

Wrapper around ADS1115rpi library (https://github.com/berndporr/rpi_ads1115) by Bernd Porr that provides an interface for pressure sensing.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/pressure↩
_sensor.hpp>
```

Public Types

- using **PressureCallback** = std::function< void(float)>

Public Member Functions

- **PressureSensor** (const PressureSensor &)=delete
- PressureSensor & **operator=** (const PressureSensor &)=delete
- void **registerCallback** (PressureCallback cb)
- void **start** (ADS1115settings settings=getDefaultSettings())
- void **stop** ()
- ADS1115rpi & **getADC** ()

Static Public Member Functions

- static ADS1115settings **getDefaultSettings** ()

8.34.1 Detailed Description

Wrapper around ADS1115rpi library (https://github.com/berndporr/rpi_ads1115) by Bernd Porr that provides an interface for pressure sensing.

The documentation for this class was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/pressure_sensor.hpp

8.35 RawInputEvent Struct Reference

Public Types

- enum class **Type** { **ToFDistance** , **PressureReading** , **MouthpieceReading** , **Keycontrol** }

Public Attributes

- Type **type**
- uint16_t **tofDistance**
- float **pressureReading**
- int8_t **mouthpieceReading**
- char **keycontrol**

The documentation for this struct was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/eventHandler.hpp

8.36 RtMidiSink Class Reference

Public Member Functions

- void **send** (const MidiMessage &message)
- void **DetectUSBMidiOutput** ()
- bool **GetDeviceStatus** ()

The documentation for this class was generated from the following files:

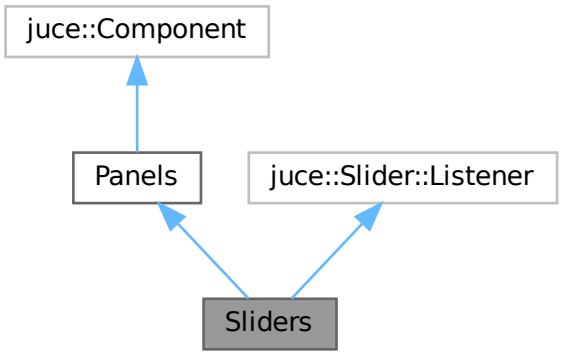
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/USBMidi.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/drivers/USBMidi.cpp

8.37 Sliders Class Reference

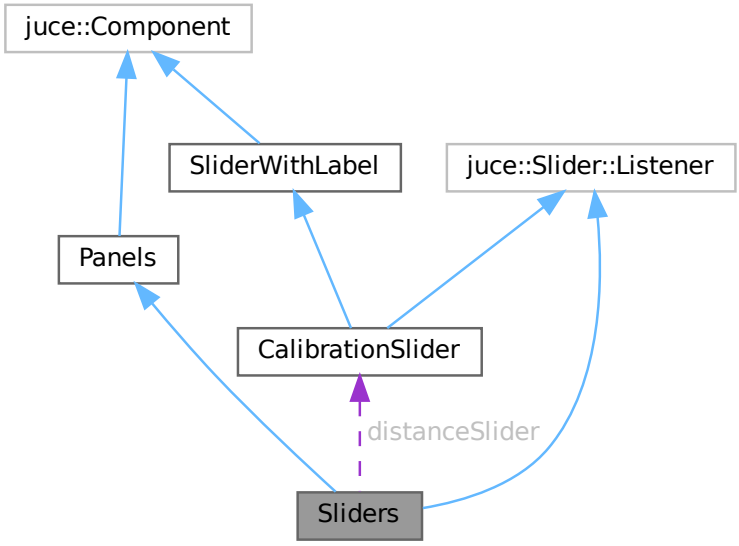
A class which adds all the linear sliders being used within the calibration section to be within the same panel.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp>
```

Inheritance diagram for Sliders:



Collaboration diagram for Sliders:



Public Member Functions

- **Sliders** ()

  The constructor function for the `Sliders` class which sets the maximum and minimum distance of the slider as well as its knobs' initial positions and other such initialisation steps.

- ~**Sliders** ()

  The destructor function for the `Sliders` class which removes the look and feel of the sliders from the sliders as well as their listeners.

- void resized () override

  Adding a function that will allow for the `Sliders` class to be resized when the window size is adjusted to fit best.
- void sliderValueChanged (juce::Slider *sliderChanged) override

  A callback function which is automatically called upon when one of the sliders present has the value of one of it's knobs changed.
- void setMinimumDistance (float distance)

  A function to set the minimum distance to which the slider may be set to.
- float getMinimumDistance (void)

  A function to get the current minimum distance to which the slider may be set to.
- void setMaximumDistance (float distance)

  A function to set the maximum distance to which the slider may be set to.
- float getMaximumDistance (void)

  A function to get the current maximum distance to which the slider may be set to.
- void setStepDistance (float step)

  A function to set the minimum step to which the slider may be moved by the user.
- float getStepDistance (void)

  A function to get the current minimum distance to which the slider may be moved by the user (step size or interval).
- void setDistanceRange (float range)

  A function to set the minimum range between the maximum and minimum fingers on the slider.
- float getDistanceRange (void)

  A function to get the current minimum distance to which the sliders lower finger may be away from the higher finger.

Public Member Functions inherited from Panels

- **Panels** ()=default

  The constructor function for the `Panels` class which currently provides no other function than creating an instance of the `Panels` class. I.e. no initialisation functions or settings.
- ~**Panels** ()=default

  The destructor function for the `Panels` class which currently does nothing extra as there is no initialisation within the class.
- void paint (juce::Graphics &g) override

  This function draws onto the parent function the default shape and colour of the panels designed.
- void setBackgroundColour (juce::Colour colour)

  A function to set the background colour of the panels for the main window.
- juce::Colour getBackgroundColour (void)

  A function to get the current background colour being used for the panels within the main window.

Public Attributes

- CalibrationSlider distanceSlider

8.37.1 Detailed Description

A class which adds all the linear sliders being used within the calibration section to be within the same panel.

8.37.2 Member Function Documentation

8.37.2.1 getDistanceRange()

```
float Sliders::getDistanceRange (
            void  )
```

A function to get the current minimum distance to which the sliders lower finger may be away from the higher finger.

Return values

The	floating point representaion of the minimum distance the between the slider fingers.

8.37.2.2 getMaximumDistance()

```
float Sliders::getMaximumDistance (
            void  )
```

A function to get the current maximum distance to which the slider may be set to.

Return values

The	floating point representaion of the maximum distance the slider is currently set to.

8.37.2.3 getMinimumDistance()

```
float Sliders::getMinimumDistance (
            void  )
```

A function to get the current minimum distance to which the slider may be set to.

Return values

The	floating point representaion of the minimum distance the slider is currently set to.

8.37.2.4 getStepDistance()

```
float Sliders::getStepDistance (
            void  )
```

A function to get the current minimum distance to which the slider may be moved by the user (step size or interval).

Return values

The	floating point representaion of the minimum distance the slider can currently be moved by.

8.37.2.5 resized()

```
void Sliders::resized ( )  [override]
```

Adding a function that will allow for the `Sliders` class to be resized when the window size is adjusted to fit best.

Note

For this function, this sets the size of the labels to be a maximum size and scales them downwards if the relative size of the window is too small to fit the phrase. This scales the length of the trombone to be able to fit in a window with a width of any size.

8.37.2.6 setDistanceRange()

```
void Sliders::setDistanceRange (
            float range )
```

A function to set the minimum range between the maximum and minimum fingers on the slider.

Parameters

range	The floating point representation of the minimum range.

8.37.2.7 setMaximumDistance()

```
void Sliders::setMaximumDistance (
            float distance )
```

A function to set the maximum distance to which the slider may be set to.

Parameters

distance	The floating point representation of the maximum distance.

8.37.2.8 setMinimumDistance()

```
void Sliders::setMinimumDistance (
            float distance )
```

A function to set the minimum distance to which the slider may be set to.

Parameters

distance	The floating point representation of the minimum distance.

8.37.2.9 setStepDistance()

```
void Sliders::setStepDistance (
            float step )
```

A function to set the minimum step to which the slider may be moved by the user.

Parameters

step	The floating point representation of the minimum step.

8.37.2.10 sliderValueChanged()

```
void Sliders::sliderValueChanged (
            juce::Slider * sliderChanged )  [override]
```

A callback function which is automatically called upon when one of the sliders present has the value of one of it's knobs changed.

Parameters

sliderChanged	A pointer to the slider which has been modified. This allows for the comparison between the different sliders within the class to detect which one has been affected.

Note

This is where the interaction with the Core side of the Tromboneless takes place by referring to the specific variable which stores all the calibration data for the Tromboneless.

8.37.3 Member Data Documentation

8.37.3.1 distanceSlider

```
CalibrationSlider Sliders::distanceSlider
```

Creating an instance of the CalibrationSlider class to be used for calibrating the maximum and minimum distance which the trombonist wishes to have their hand for the pitch bend.

The documentation for this class was generated from the following files:

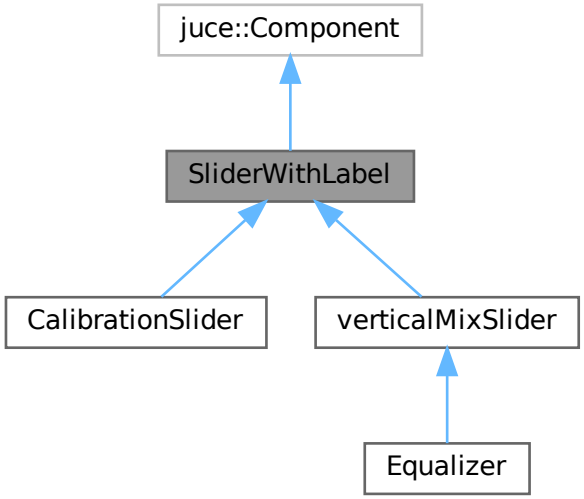
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Panels.cpp

8.38 SliderWithLabel Class Reference

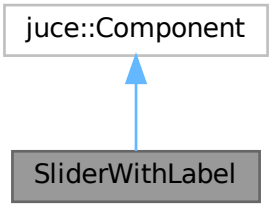
A class which inherits attributes from the juce::Component class to create a new component containing a 8 possible label locations around the slider.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for SliderWithLabel:



Collaboration diagram for SliderWithLabel:



Public Types

- enum LabelPositions_t {
  UpperLeft , **UpperCentre** , **UpperRight** , **MiddleLeft** ,
  MiddleRight , **LowerLeft** , **LowerCentre** , **LowerRight** }

  Enumerated list of possible positions for the label to be attached relative to the slider.
- enum InitialiseLabelErrors_t { **LabelOK** , **PositonNotAccepted** }

  Enumerated list of possible errors when initialising the labels within this component.

Public Member Functions

- **SliderWithLabel** (juce::Slider::SliderStyle style)

  Constructor which sets the style of the slider being used and removes the text entry option.
- InitialiseLabelErrors_t CreateLabel (LabelPositions_t position, juce::String phrase, juce::NotificationType NotificationEnable=juce::dontSendNotification)

  A function which may be called to create or edit a specified label around the slider.
- void resized () override

  A function which is automatically called whenever the window size is modified. This automatically adjusts the bounds of each of the labels as well as the slider to ensure the size is correct with respect to the screen size for the component.
- void setTopLabelBounds (int bound)

  A function to set the bounds for which the top labels may be situated in.
- int getTopLabelBounds (void)

  A function to get the bounds for which the top labels are situated in.
- void setLeftLabelBounds (int bound)

  A function to set the bounds for which the labels on the left may be situated in.
- int getLeftLabelBounds (void)

  A function to get the bounds for which the labels on the left are situated in.
- void setRightLabelBounds (int bound)

  A function to set the bounds for which the labels on the right may be situated in.
- int getRightLabelBounds (void)

  A function to get the bounds for which the labels on the right are situated in.
- void setBottomLabelBounds (int bound)

  A function to set the bounds for which the bottom labels may be situated in.
- int getBottomLabelBounds (void)

  A function to get the bounds for which the bottom labels are situated in.

Public Attributes

- juce::Slider **slider**

  The slider which is having the labels added to it.

8.38.1 Detailed Description

A class which inherits attributes from the juce::Component class to create a new component containing a 8 possible label locations around the slider.

8.38.2 Member Function Documentation

8.38.2.1 CreateLabel()

```
SliderWithLabel::InitialiseLabelErrors_t SliderWithLabel::CreateLabel (
            LabelPositions_t position,
            juce::String phrase,
            juce::NotificationType NotificationEnable = juce::dontSendNotification )
```

A function which may be called to create or edit a specified label around the slider.

Parameters

position	The position index of the label which should be modified. These are saved as an enum in `Widgets.h`.
phrase	An array of type juce::String which contains the phrase which should be displayed where the label is.
NotificationEnable	The type of notification which should be attached to the label. This is set to `dontSendNotification` by default but may be modified.

Note

This function may only address one label at a time, but may be called upon for each label needed. There may only be one label in each position.

8.38.2.2 getBottomLabelBounds()

```
int SliderWithLabel::getBottomLabelBounds (
            void  )
```

A function to get the bounds for which the bottom labels are situated in.

Return values

The	integer representation of the bottom label bounds.

8.38.2.3 getLeftLabelBounds()

```
int SliderWithLabel::getLeftLabelBounds (
            void  )
```

A function to get the bounds for which the labels on the left are situated in.

Return values

The	integer representation of the left label bounds.

8.38.2.4 getRightLabelBounds()

```
int SliderWithLabel::getRightLabelBounds (
            void  )
```

A function to get the bounds for which the labels on the right are situated in.

Return values

The	integer representation of the right label bounds.

8.38.2.5 getTopLabelBounds()

```
int SliderWithLabel::getTopLabelBounds (
            void  )
```

A function to get the bounds for which the top labels are situated in.

Return values

The	integer representation of the top label bounds.

8.38.2.6 resized()

```
void SliderWithLabel::resized ( )  [override]
```

A function which is automatically called whenever the window size is modified. This automatically adjusts the bounds of each of the labels as well as the slider to ensure the size is correct with respect to the screen size for the component.

Note

The bounds are preset to be 20 for the top and bottom labels and 40 for the left and right labels. These values may be overwritten by writting to the ____`LabelBounds` where ___ may be `top`, `right`, `left` or `bottom`.

8.38.2.7 setBottomLabelBounds()

```
void SliderWithLabel::setBottomLabelBounds (
            int bound )
```

A function to set the bounds for which the bottom labels may be situated in.

Parameters

bound	The integer representation of the bounds which are to be set.

8.38.2.8 setLeftLabelBounds()

```
void SliderWithLabel::setLeftLabelBounds (
            int bound )
```

A function to set the bounds for which the labels on the left may be situated in.

Parameters

bound	The integer representation of the bounds which are to be set.

8.38.2.9 setRightLabelBounds()

```
void SliderWithLabel::setRightLabelBounds (
            int bound )
```

A function to set the bounds for which the labels on the right may be situated in.

Parameters

bound	The integer representation of the bounds which are to be set.

8.38.2.10 setTopLabelBounds()

```
void SliderWithLabel::setTopLabelBounds (
            int bound )
```

A function to set the bounds for which the top labels may be situated in.

Parameters

bound	The integer representation of the bounds which are to be set.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.39 SynthesiserParameters_t Struct Reference

A struct to store any of the synthesier parameters that may be changed within the app. A data changed flag variable should be added to this struct to tell the core to read the new values.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Shared/tromboneless_↩
data.hpp>
```

Public Attributes

- bool **synthEnable**

  The variable used to determine when the internal synth and equalizer is used.
- float **gains** [10] = {1, 1, 1, 1, 1, 1, 1, 1, 1, 1}

  The chosen gains for the various frequency bands in the equalizer.

8.39.1 Detailed Description

A struct to store any of the synthesier parameters that may be changed within the app. A data changed flag variable should be added to this struct to tell the core to read the new values.

The documentation for this struct was generated from the following file:

- /home/ryan-mcbride/Documents/Tromboneless/Shared/tromboneless_data.hpp

8.40 ToFSensor Class Reference

Public Types

- using **DistanceCallback** = std::function< void(uint16_t)>

Public Member Functions

- **ToFSensor** (I2CBus &bus, uint8_t i2cAddress7Bit=0x29, const std::string &gpioChipPath="/dev/gpiochip0", unsigned int gpioLine=4)
- bool **initialise** ()
- void **registerCallback** (DistanceCallback cb)
- void **start** ()
- void **stop** ()

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/Core/include/drivers/tof_sensor.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/drivers/tof_sensor.cpp

8.41 TrombonelessParameters_t Struct Reference

A struct which stores all the necessary parameters that may be changed on the app side effecting the core functionality. This struct should add a flag which is triggered whenever a parameter has been changed.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Shared/tromboneless_↩
data.hpp>
```

Public Attributes

- ShiftKeyingOptions_t **shiftKeyingOption**

  The chosen shiftkey option by the user corresponding to the type of trombone.
- double **nearDistance**

  The nearest distance the user wishes to have their hand from the sensor for pitch bend.
- double **farDistance**

  The furthest distance the user wishes to have their hand from the sensor for pitch bend.
- double **lowPressure**

  The lower pressure gain which the user has chosen for the mouthpiece corresponding to the peak input.
- double **highPressure**

  The upper pressure gain which the user has chosen for the mouthpiece corresponding to the peak input.
- EmbachureOptions_t **embouchureOption**

  The chosen embouchure option which the user wishes to calibrate.
- bool **triggerEmbouchureCalibrate**

  High/true when the user has pressed the button to calibrate the Embouchure.

8.41.1 Detailed Description

A struct which stores all the necessary parameters that may be changed on the app side effecting the core functionality. This struct should add a flag which is triggered whenever a parameter has been changed.

The documentation for this struct was generated from the following file:

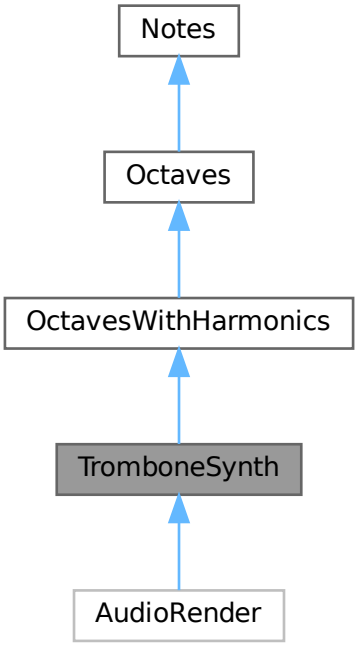
- /home/ryan-mcbride/Documents/Tromboneless/Shared/tromboneless_data.hpp

8.42 TromboneSynth Class Reference

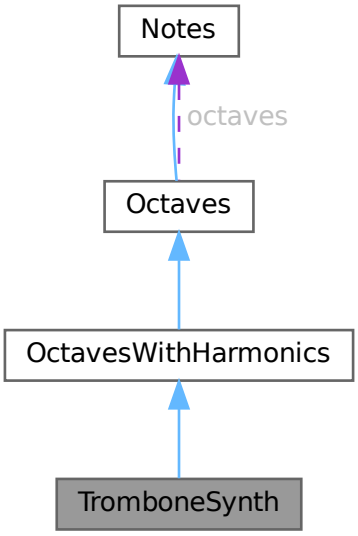
A class which may be used to create a custom synth with an envelope attached to it.

```
#include </home/ryan-mcbride/Documents/Tromboneless/Core/include/trombone↩
Synth.hpp>
```

Inheritance diagram for TromboneSynth:



Collaboration diagram for TromboneSynth:



Public Member Functions

- TromboneSynth (int sampleRate=44100, float attack=10.0f, float decay=50.0f, float sustain=1.0f, float rest=10.0f)

  The constructor function for the `TromboneSynth` class.
- void StartTromboneNote (Notes::Notes_t note_in, int octave_in)

  A function which may be called to set the note and octave of a new sound by the user. If there is already a note playing, this will stop the previous not and play this one without phase shifting (initially).
- void ChangeTromboneNote (Notes::Notes_t note_in, int octave_in)

  A function that can change the current note without needed to change the stage that the envelope is in.
- float ReadTromboneAudio (void)

  A function which may be called to read the current amplitude of the audio signal being generated.
- void **StopTromboneNote** (void)

  A function which sends the synthesiser into its rest state.
- void NewTromboneNoteMIDI (int MIDINote)

  A function which can take in the MIDI equivalent note and map it to the Notes_t options to match the rest of the synth interface.
- void **HandleMIDINoteOn** ()

  A method to handle NoteOn messages as passed from the MIDI Coordinator.
- void **HandleMIDINoteOff** ()
- void **HandleMIDIPitchBend** (int midipitchbend)
- void **HandleMIDINoteChange** (int new_note)
- void setAttackMS (float attack_time)

  A function to set the amount of time in milliseconds that the attack stage takes.
- float getAttackMS (void)

  A function to get the current attack time in milliseconds.
- void setDecayMS (float decay_time)

A function to set the amount of time in milliseconds that the decay stage takes.

- float getDecayMS (void)

  A function to get the current decay time in milliseconds.
- void setRestMS (float rest_time)

  A function to set the amount of time in milliseconds that the rest stage takes.
- float getRestMS (void)

  A function to get the current rest time in milliseconds.
- void setSustain (float sustain_in)

  A function to set the relative amplitude of the sustain stage.
- float getSustain (void)

  A function to get the current sustain value.
- void setNHarmonics (int n)

  A function to set the number of harmonics which are being generated for every note being played.
- int getNHarmonics (void)

  A function to get the number of harmonics which are being generated for every note being played.
- int getOctave (void)

  A function to get the current octave which the current note is in.
- Notes::Notes_t getNote (void)

  A function to get the current note which is being used.
- void setPitchBend (int bend)

  A function to set the pitch bend of the device.
- int getPitchBend (void)

  A function to get the pitch bend of the device.
- float getAdjustedFrequency (void)

  A function which determines the new frequency which should be being played depending upon the current note and octave as well as the pitch bend being input.
- Envelope **getEnvelope** (void)

Public Member Functions inherited from OctavesWithHarmonics

- **OctavesWithHarmonics** ()

  Creates an instance of the `OctavesWithHarmonics`.
- float PlayingNoteWithHarmonics (int n, int octave, Notes_t note, float phase)

  A function which may be called upon to stop playing a range of harmonic notes when a specific note is called upon to be played.
- float PlayingFrequencyWithHarmonics (int n, float phase)

  A function which may be called upon to stop playing a range of harmonic frequencies when a specific note is called upon to be played.
- float HarmonicDecay (int n)

  A function which calculates how much the amplitude of the harmonic should decay in comparison to the first harmonic.
- float getHarmonicDecayMax (int n)

  A function which returns the maximum amplitude possible for a set number of harmonics if they are all constructively interfering.
- void setDecayConstant (float decayC)

  A function that can change the value of the decay constant used to set the maxmimum amplitude of each successive harmonic.
- float getDecayConstant (void)

  A function which returns the decay constant being used to determine the relative ampltiude of each successive harmonic.

Public Member Functions inherited from Octaves

- float PlayingNote (int octave, Notes_t note, float phase)

  A function which may be called to determine the amplitude of a specified note at a specified phase.
- float PlayingFrequency (float phase)

  A function which may be called to determine the amplitude of a specified frequency at a specified time.

Public Member Functions inherited from Notes

- Notes (float C_in=16.35, float Db_in=17.32, float D_in=18.35, float Eb_in=19.45, float E_in=20.60, float F↩_in=21.83, float Gb_in=23.12, float G_in=24.50, float Ab_in=25.96, float A_in=27.50, float Bb_in=29.14, float B_in=30.87)

  Constructor for the `Notes` class to create a new set of notes and assign specified frequencies to each. Each frequency is set to `Octave 0` by default.
- void setC (float frequency)

  A function that can change the value of the frequency that the C note plays.
- void setC_sharp (float frequency)

  A function that can change the value of the frequency that the C# note plays.
- void setDb (float frequency)

  A function that can change the value of the frequency that the Db note plays.
- void setD (float frequency)

  A function that can change the value of the frequency that the D note plays.
- void setD_sharp (float frequency)

  A function that can change the value of the frequency that the D# note plays.
- void setEb (float frequency)

  A function that can change the value of the frequency that the Eb note plays.
- void setE (float frequency)

  A function that can change the value of the frequency that the E note plays.
- void setF (float frequency)

  A function that can change the value of the frequency that the F note plays.
- void setF_sharp (float frequency)

  A function that can change the value of the frequency that the F# note plays.
- void setGb (float frequency)

  A function that can change the value of the frequency that the Gb note plays.
- void setG (float frequency)

  A function that can change the value of the frequency that the G note plays.
- void setG_sharp (float frequency)

  A function that can change the value of the frequency that the G# note plays.
- void setAb (float frequency)

  A function that can change the value of the frequency that the Ab note plays.
- void setA (float frequency)

  A function that can change the value of the frequency that the A note plays.
- void setA_sharp (float frequency)

  A function that can change the value of the frequency that the A# note plays.
- void setBb (float frequency)

  A function that can change the value of the frequency that the Bb note plays.
- void setB (float frequency)

  A function that can change the value of the frequency that the B note plays.
- float setNote (Notes_t note, float freq)

  An alternative function which can allow for the user to alter the frequency of a specific note based upon the Notes_t enumerated values rather than specifying a function call.

- float getC (void)

  A function that returns the frequency of the C note for the specified object.
- float getC_sharp (void)

  A function that returns the frequency of the C# note for the specified object.
- float getDb (void)

  A function that returns the frequency of the Db note for the specified object.
- float getD (void)

  A function that returns the frequency of the D note for the specified object.
- float getD_sharp (void)

  A function that returns the frequency of the D# note for the specified object.
- float getEb (void)

  A function that returns the frequency of the Eb note for the specified object.
- float getE (void)

  A function that returns the frequency of the E note for the specified object.
- float getF (void)

  A function that returns the frequency of the F note for the specified object.
- float getF_sharp (void)

  A function that returns the frequency of the F# note for the specified object.
- float getGb (void)

  A function that returns the frequency of the Gb note for the specified object.
- float getG (void)

  A function that returns the frequency of the G note for the specified object.
- float getG_sharp (void)

  A function that returns the frequency of the G# note for the specified object.
- float getAb (void)

  A function that returns the frequency of the Ab note for the specified object.
- float getA (void)

  A function that returns the frequency of the A note for the specified object.
- float getA_sharp (void)

  A function that returns the frequency of the A# note for the specified object.
- float getBb (void)

  A function that returns the frequency of the Bb note for the specified object.
- float getB (void)

  A function that returns the frequency of the B note for the specified object.
- float getNote (Notes_t note)

  An alternative function which can allow for the user to request a specific note based upon the Notes_t enumerated values rather than specifying a function call.

Additional Inherited Members

Public Types inherited from Notes

- enum Notes_t {
  note_C , **note_Db** , **note_D** , **note_Eb** ,
  note_E , **note_F** , **note_Gb** , **note_G** ,
  note_Ab , **note_A** , **note_Bb** , **note_B** ,
  note_C_sharp = note_Db , **note_D_sharp** = note_Eb , **note_F_sharp** = note_Gb , **note_G_sharp** = note↩_Ab ,
  note_A_sharp = note_Bb }

  Creation of the enumerated list of possible notes.

Public Attributes inherited from Octaves

- Notes **octaves** [nOctaves]

  Initialising an array of the octaves and their notes.

Static Public Attributes inherited from Octaves

- static const char **nOctaves** = 9

  Stores the number of octaves being used.

Protected Member Functions inherited from OctavesWithHarmonics

- void **updateHarmonicCache** (int n)

Protected Attributes inherited from OctavesWithHarmonics

- float **decayConstant** = 0.1f
- std::array< float, maxHarmonics > **harmonicWeights** {}
- int **cachedNHarmonics** = 0
- bool **harmonicCacheDirty** = true

Static Protected Attributes inherited from OctavesWithHarmonics

- static constexpr int **maxHarmonics** = 64

8.42.1 Detailed Description

A class which may be used to create a custom synth with an envelope attached to it.

8.42.2 Constructor & Destructor Documentation

8.42.2.1 TromboneSynth()

```
TromboneSynth::TromboneSynth (
            int sampleRate = 44100,
            float attack = 10.0f,
            float decay = 50.0f,
            float sustain = 1.0f,
            float rest = 10.0f )
```

The constructor function for the `TromboneSynth` class.

Parameters

sampleRate	The sample rate which will be used to measure the amplitude of the sound wave being synthesised. This should be passed through in Hz.
attack	The amount of time in ms that the attack should last for.
decay	The amount of time in ms that the decay should last for.
sustain	The relative amplitude that is being output at the steady state.
rest	The amount of time in milliseconds that the rest stage should last for.

8.42.3 Member Function Documentation

8.42.3.1 ChangeTromboneNote()

```
void TromboneSynth::ChangeTromboneNote (
            Notes::Notes_t note_in,
            int octave_in )
```

A function that can change the current note without needed to change the stage that the envelope is in.

Parameters

note_in	The note which the synthesiser should start to play.
octave↩ _in	The octave in which the note is in.

8.42.3.2 getAdjustedFrequency()

```
float TromboneSynth::getAdjustedFrequency (
            void )
```

A function which determines the new frequency which should be being played depending upon the current note and octave as well as the pitch bend being input.

Return values

The	floating point representation of the frequency.

8.42.3.3 getAttackMS()

```
float TromboneSynth::getAttackMS (
            void )
```

A function to get the current attack time in milliseconds.

Return values

The	time in milliseconds as a floating point value.

8.42.3.4 getDecayMS()

```
float TromboneSynth::getDecayMS (
            void )
```

A function to get the current decay time in milliseconds.

Return values

The	time in milliseconds as a floating point value.

8.42.3.5 getNHarmonics()

```
int TromboneSynth::getNHarmonics (
            void  )
```

A function to get the number of harmonics which are being generated for every note being played.

Return values

The	integer representation of the number of harmonics.

8.42.3.6 getNote()

```
Notes::Notes_t TromboneSynth::getNote (
            void  )
```

A function to get the current note which is being used.

Return values

The	note of type Notes::Notes_t being played.

8.42.3.7 getOctave()

```
int TromboneSynth::getOctave (
            void  )
```

A function to get the current octave which the current note is in.

Return values

The	octave as an integer representation.

8.42.3.8 getPitchBend()

```
int TromboneSynth::getPitchBend (
            void  )
```

A function to get the pitch bend of the device.

Return values

The	integer representation between 8192 and 0 of how much of the pitch has been bent.

8.42.3.9 getRestMS()

```
float TromboneSynth::getRestMS (
            void  )
```

A function to get the current rest time in milliseconds.

Return values

The	time in milliseconds as a floating point value.

8.42.3.10 getSustain()

```
float TromboneSynth::getSustain (
            void  )
```

A function to get the current sustain value.

Return values

The	floating point relative amplitude

8.42.3.11 NewTromboneNoteMIDI()

```
void TromboneSynth::NewTromboneNoteMIDI (
            int MIDINote )
```

A function which can take in the MIDI equivalent note and map it to the Notes_t options to match the rest of the synth interface.

Parameters

MIDINote	The integer value between 0 and 127 representing the note which is being played.

8.42.3.12 ReadTromboneAudio()

```
float TromboneSynth::ReadTromboneAudio (
            void  )
```

A function which may be called to read the current amplitude of the audio signal being generated.

Return values

Returns	a floating point representation of the amplitude normalised between -1 and 1.

8.42.3.13 setAttackMS()

```
void TromboneSynth::setAttackMS (
            float attack_time )
```

A function to set the amount of time in milliseconds that the attack stage takes.

Parameters

attack_time	The time in milliseconds.

8.42.3.14 setDecayMS()

```
void TromboneSynth::setDecayMS (
            float decay_time )
```

A function to set the amount of time in milliseconds that the decay stage takes.

Parameters

decay_time	The time in milliseconds.

8.42.3.15 setNHarmonics()

```
void TromboneSynth::setNHarmonics (
            int n )
```

A function to set the number of harmonics which are being generated for every note being played.

Parameters

n	The number of harmonics which should be added to each note.

8.42.3.16 setPitchBend()

```
void TromboneSynth::setPitchBend (
            int bend )
```

A function to set the pitch bend of the device.

Parameters

bend	The integer representation between 8192 and 0 of how much of the pitch has been bent.

8.42.3.17 setRestMS()

```
void TromboneSynth::setRestMS (
            float rest_time )
```

A function to set the amount of time in milliseconds that the rest stage takes.

Parameters

rest_time	The time in milliseconds.

8.42.3.18 setSustain()

```
void TromboneSynth::setSustain (
            float sustain_in )
```

A function to set the relative amplitude of the sustain stage.

Parameters

sustain↩ _in	The floating point relative amplitde between 0 and 1.

8.42.3.19 StartTromboneNote()

```
void TromboneSynth::StartTromboneNote (
            Notes::Notes_t note_in,
            int octave_in )
```

A function which may be called to set the note and octave of a new sound by the user. If there is already a note playing, this will stop the previous not and play this one without phase shifting (initially).

Parameters

note_in	The note which should start to play.
octave↩ _in	The octave which the note is in.

The documentation for this class was generated from the following files:

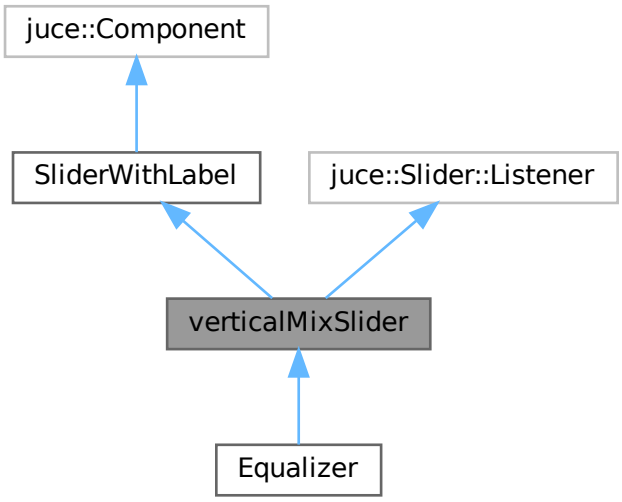
- /home/ryan-mcbride/Documents/Tromboneless/Core/include/tromboneSynth.hpp
- /home/ryan-mcbride/Documents/Tromboneless/Core/src/tromboneSynth.cpp

8.43 verticalMixSlider Class Reference

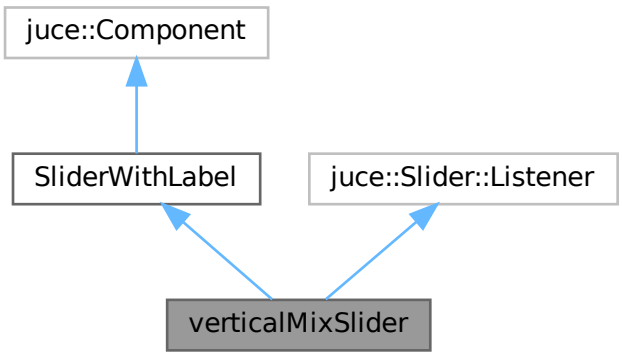
A class that adds more functionality to the slider class from Juce.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp>
```

Inheritance diagram for verticalMixSlider:



Collaboration diagram for verticalMixSlider:



Public Member Functions

- **verticalMixSlider** ()

  Constructor function for the vericalMixSlider which creates an instance of the class and sets the parameters to match it best such as setting the range to be between -9 dB and 9 dB with a step size of 3 dB as well as label size.
- ∼**verticalMixSlider** ()

  Destructor function for the verticalMixSlider class which removes the listener from the slider before killing the object.
- void sliderValueChanged (juce::Slider ∗sliderChanged) override

  A callback function for when the slider used has been adjusted and its value may have been changed.

Public Member Functions inherited from SliderWithLabel

- **SliderWithLabel** (juce::Slider::SliderStyle style)

  Constructor which sets the style of the slider being used and removes the text entry option.
- InitialiseLabelErrors_t CreateLabel (LabelPositions_t position, juce::String phrase, juce::NotificationType NotificationEnable=juce::dontSendNotification)

  A function which may be called to create or edit a specified label around the slider.
- void resized () override

  A function which is automatically called whenever the window size is modified. This automatically adjusts the bounds of each of the labels as well as the slider to ensure the size is correct with respect to the screen size for the component.
- void setTopLabelBounds (int bound)

  A function to set the bounds for which the top labels may be situated in.
- int getTopLabelBounds (void)

  A function to get the bounds for which the top labels are situated in.
- void setLeftLabelBounds (int bound)

  A function to set the bounds for which the labels on the left may be situated in.
- int getLeftLabelBounds (void)

  A function to get the bounds for which the labels on the left are situated in.
- void setRightLabelBounds (int bound)

  A function to set the bounds for which the labels on the right may be situated in.
- int getRightLabelBounds (void)

  A function to get the bounds for which the labels on the right are situated in.
- void setBottomLabelBounds (int bound)

  A function to set the bounds for which the bottom labels may be situated in.
- int getBottomLabelBounds (void)

  A function to get the bounds for which the bottom labels are situated in.

Additional Inherited Members

Public Types inherited from SliderWithLabel

- enum LabelPositions_t {
  UpperLeft , **UpperCentre** , **UpperRight** , **MiddleLeft** ,
  MiddleRight , **LowerLeft** , **LowerCentre** , **LowerRight** }

  Enumerated list of possible positions for the label to be attached relative to the slider.
- enum InitialiseLabelErrors_t { **LabelOK** , **PositonNotAccepted** }

  Enumerated list of possible errors when initialising the labels within this component.

Public Attributes inherited from SliderWithLabel

- juce::Slider **slider**

  The slider which is having the labels added to it.

8.43.1 Detailed Description

A class that adds more functionality to the slider class from Juce.

8.43.2 Member Function Documentation

8.43.2.1 sliderValueChanged()

```
void verticalMixSlider::sliderValueChanged (
            juce::Slider * sliderChanged )  [override]
```

A callback function for when the slider used has been adjusted and its value may have been changed.

Parameters

sliderChanged	This is a pointer to the slider which has been changed.

Note

As this class only has one slider that is in it, there is no need to use it but the JUCE function passes it anyway.

The documentation for this class was generated from the following files:

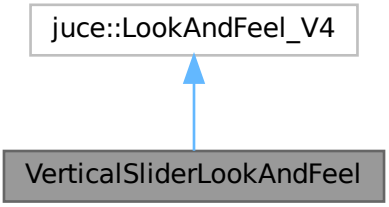
- /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/Widgets.cpp

8.44 VerticalSliderLookAndFeel Class Reference

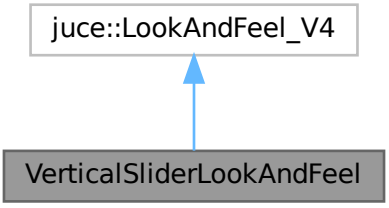
Creation of a class which can be used to change the look of the sliders used within the equalizer for the Tromboneless project.

```
#include </home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.↩
hpp>
```

Inheritance diagram for VerticalSliderLookAndFeel:



Collaboration diagram for VerticalSliderLookAndFeel:



Public Member Functions

- VerticalSliderLookAndFeel ()

  The constructor function for the `VerticalSliderLookAndFeel` class that creates an instance of the look and feel style.
- virtual ~**VerticalSliderLookAndFeel** ()=default

  The destructor function for the `VerticalSliderLookAndFeel` class which is currently set to default as there is no need for a deconstructor for this class.
- virtual void drawLinearSlider (juce::Graphics &g, int x, int y, int width, int height, float sliderPos, float min↩ SliderPos, float maxSliderPos, const juce::Slider::SliderStyle style, juce::Slider &slider) override

  When this look and feel is attached to an object which has a vertical slider in it, this function sets the look and feel of the slider being used. This makes it look like an average equalizer slider.
- void setDialColour (const juce::Colour colour)

  A function to set the colour of the finger/dial of the slider.
- juce::Colour getDialColour (void)

  A function to get the current colour of the finger/dial of the slider.
- void setDialEdgeColour (const juce::Colour &colour)

  A function to set the colour of the edge of the finger/dial on the slider.
- juce::Colour getDialEdgeColour (void)

  A function to get the current colour of the edge of the finger /dial of the slider.

- void setTickColour (juce::Colour colour)

  A function to set the colour of the ticks on the slider.
- juce::Colour getTickColour (void)

  A function to get the current colour of the ticks on the slider.
- void setTrackColour (juce::Colour colour)

  A function to set the colour of the track of the slider.
- juce::Colour getTrackColour (void)

  A function to get the current colour of the track of the slider.
- void setBackgroundColour (juce::Colour colour)

  A function to set the colour of the background of the slider.
- juce::Colour getBackgroundColour (void)

  A function to get the current colour of the background of the slider.
- void setTextColour (juce::Colour colour)

  A function to set the colour of the text around the slider.
- juce::Colour getTextColour (void)

  A function to get the current colour of the text around the slider.
- void setTrackWidth (float width)

  A function to set the size of the track width of the slider.
- float getTrackWidth (void)

  A function to get the track width of the slider.
- void setTickThickness (float thickness)

  A function to set the thickness of the ticks to indicate what step the slider is positioned in.
- float getTickThickness (void)

  A function to get what the current tick thickness is for the object.
- void setTickWidth (float width)

  A function to set the width of the ticks to indicate what step the slider is positioned in.
- float getTickWidth (void)

  A function which may be used to find the current tick width of the object.
- void setFingerWidth (float width)

  A function to set the width of the slider's finger/knob when the slider is set to have a rectangular shape.
- float getFingerWidth (void)

  A function to get the current width of the slider's finger/knob within the object when the slider is set to have a rectangular shape.
- void setFingerHeight (float height)

  A function to set the height of the slider's finger/knob when the slider is set to have a rectangular shape.
- float getFingerHeight (void)

  A function to get the current height of the slider's finger/knob within the object when the slider is set to have a rectangular shape.

8.44.1 Detailed Description

Creation of a class which can be used to change the look of the sliders used within the equalizer for the Tromboneless project.

8.44.2 Constructor & Destructor Documentation

8.44.2.1 VerticalSliderLookAndFeel()

```
VerticalSliderLookAndFeel::VerticalSliderLookAndFeel ( )
```

The constructor function for the `VerticalSliderLookAndFeel` class that creates an instance of the look and feel style.

Note

Currently there is not extra functionality attached to the contructor as the style of the vertical slider is defined in the `drawLinearSlider()` function and there are no other components being modified by this style currently.

8.44.3 Member Function Documentation

8.44.3.1 drawLinearSlider()

```
void VerticalSliderLookAndFeel::drawLinearSlider (
            juce::Graphics & g,
            int x,
            int y,
            int width,
            int height,
            float sliderPos,
            float minSliderPos,
            float maxSliderPos,
            const juce::Slider::SliderStyle style,
            juce::Slider & slider )  [override], [virtual]
```

When this look and feel is attached to an object which has a vertical slider in it, this function sets the look and feel of the slider being used. This makes it look like an average equalizer slider.

Parameters

g	A pointer to the `juce::Graphics` instance which is being drawn to. This is likely the main window or else the parent component.
x	The integer representation of the slider's left corner in the global window.
y	The integer representation of the sliders top corner in the global window.
width	The integer representation of the width of the slider. i.e. the difference between the left most and right most part of the slider.
height	The integer representation of the width of the slider. i.e. the difference between the top most and bottom most parts of the slider's bounds.
sliderPos	The position of the sliders finger. The position represents x if the slider is horizontal and y if the slider is vertical.
minSliderPos	The minimum slider position.
maxSliderPos	The maximum slider position.
style	The style of the slider whose look and feel is getting modified. For example, LinearVertical or Rotary.
slider	A pointer to the slider which is being modified. This can allow for extraction of more data from the slider.

8.44.3.2 getBackgroundColour()

```
juce::Colour VerticalSliderLookAndFeel::getBackgroundColour (
            void  )
```

A function to get the current colour of the background of the slider.

Return values

Returns	the colour as a juce::Colour type.

8.44.3.3 getDialColour()

```
juce::Colour VerticalSliderLookAndFeel::getDialColour (
            void  )
```

A function to get the current colour of the finger/dial of the slider.

Return values

Returns	the colour as a juce::Colour type.

8.44.3.4 getDialEdgeColour()

```
juce::Colour VerticalSliderLookAndFeel::getDialEdgeColour (
            void  )
```

A function to get the current colour of the edge of the finger /dial of the slider.

Return values

Returns	the colour as a juce::Colour type.

8.44.3.5 getFingerHeight()

```
float VerticalSliderLookAndFeel::getFingerHeight (
            void  )
```

A function to get the current height of the slider's finger/knob within the object when the slider is set to have a rectangular shape.

Return values

The	floating point representation of the height of the object's finger/knob.

8.44.3.6 getFingerWidth()

```
float VerticalSliderLookAndFeel::getFingerWidth (
            void  )
```

A function to get the current width of the slider's finger/knob within the object when the slider is set to have a rectangular shape.

Return values

The	floating point representation of the width of the object's finger/knob.

8.44.3.7 getTextColour()

```
juce::Colour VerticalSliderLookAndFeel::getTextColour (
            void  )
```

A function to get the current colour of the text around the slider.

Return values

Returns	the colour as a juce::Colour type.

8.44.3.8 getTickColour()

```
juce::Colour VerticalSliderLookAndFeel::getTickColour (
            void  )
```

A function to get the current colour of the ticks on the slider.

Return values

Returns	the colour as a juce::Colour type.

8.44.3.9 getTickThickness()

```
float VerticalSliderLookAndFeel::getTickThickness (
            void  )
```

A function to get what the current tick thickness is for the object.

Return values

Returns	the floating point representation of the tick thickness of the object.

8.44.3.10 getTickWidth()

```
float VerticalSliderLookAndFeel::getTickWidth (
            void  )
```

A function which may be used to find the current tick width of the object.

Return values

Returns	the floating point representation of the tick's width.

8.44.3.11 getTrackColour()

```
juce::Colour VerticalSliderLookAndFeel::getTrackColour (
            void  )
```

A function to get the current colour of the track of the slider.

Return values

Returns	the colour as a juce::Colour type.

8.44.3.12 getTrackWidth()

```
float VerticalSliderLookAndFeel::getTrackWidth (
            void  )
```

A function to get the track width of the slider.

Return values

The	floating point representation of the width of the track in terms of the resoltion.

8.44.3.13 setBackgroundColour()

```
void VerticalSliderLookAndFeel::setBackgroundColour (
            juce::Colour colour )
```

A function to set the colour of the background of the slider.

Parameters

colour	The colour to which the background should be set to.

8.44.3.14 setDialColour()

```
void VerticalSliderLookAndFeel::setDialColour (
            const juce::Colour colour )
```

A function to set the colour of the finger/dial of the slider.

Parameters

colour	The colour to which the dial should be set to.

8.44.3.15 setDialEdgeColour()

```
void VerticalSliderLookAndFeel::setDialEdgeColour (
            const juce::Colour & colour )
```

A function to set the colour of the edge of the finger/dial on the slider.

Parameters

colour	The colour to which the edge of the dial should be set to.

8.44.3.16 setFingerHeight()

```
void VerticalSliderLookAndFeel::setFingerHeight (
            float height )
```

A function to set the height of the slider's finger/knob when the slider is set to have a rectangular shape.

Parameters

height	The floating point representation of the height of the finger/knob in terms of the resolution.

8.44.3.17 setFingerWidth()

```
void VerticalSliderLookAndFeel::setFingerWidth (
            float width )
```

A function to set the width of the slider's finger/knob when the slider is set to have a rectangular shape.

Parameters

width	The floating point representation of the width of the finger/knob in terms of the resolution.

8.44.3.18 setTextColour()

```
void VerticalSliderLookAndFeel::setTextColour (
            juce::Colour colour )
```

A function to set the colour of the text around the slider.

Parameters

colour	The colour to which the text should be set to.

8.44.3.19 setTickColour()

```
void VerticalSliderLookAndFeel::setTickColour (
            juce::Colour colour )
```

A function to set the colour of the ticks on the slider.

Parameters

colour	The colour to which the ticks should be set to.

8.44.3.20 setTickThickness()

```
void VerticalSliderLookAndFeel::setTickThickness (
            float thickness )
```

A function to set the thickness of the ticks to indicate what step the slider is positioned in.

Parameters

thickness	The thickess of the ticks (vertical height) in terms of the resolution.

8.44.3.21 setTickWidth()

```
void VerticalSliderLookAndFeel::setTickWidth (
            float width )
```

A function to set the width of the ticks to indicate what step the slider is positioned in.

Parameters

width	The width of the ticks (horizontal) in terms of the resolution.

8.44.3.22 setTrackColour()

```
void VerticalSliderLookAndFeel::setTrackColour (
            juce::Colour colour )
```

A function to set the colour of the track of the slider.

Parameters

colour	The colour to which the track should be set to.

8.44.3.23 setTrackWidth()

```
void VerticalSliderLookAndFeel::setTrackWidth (
            float width )
```

A function to set the size of the track width of the slider.

Parameters

width	The floating point representation of the width of the track in terms of the resolution.

The documentation for this class was generated from the following files:

- /home/ryan-mcbride/Documents/Tromboneless/App/inc/CustomStyles.hpp
- /home/ryan-mcbride/Documents/Tromboneless/App/src/CustomStyles.cpp

Chapter 9

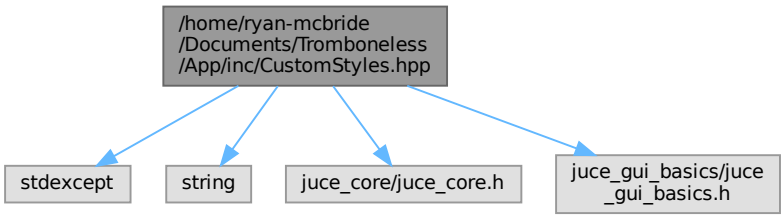
File Documentation

9.1 /home/ryan-mcbride/Documents/Tromboneless/App/inc/Custom↩ Styles.hpp File Reference

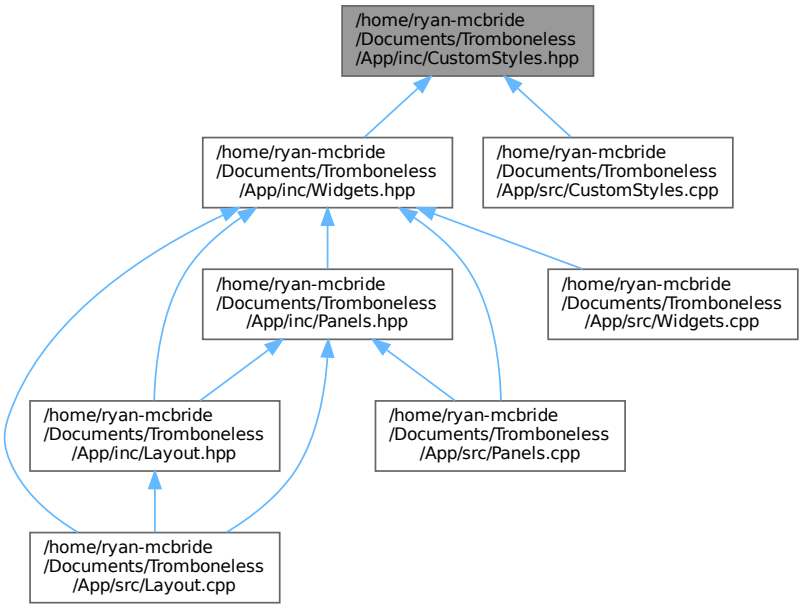
This file holds the declarations of all functions and variables needed by the Tromboneless project to change the looks and feel of various different widgets which are being displayed to the end user.

```
#include <stdexcept>
#include <string>
#include <juce_core/juce_core.h>
#include <juce_gui_basics/juce_gui_basics.h>
```

Include dependency graph for CustomStyles.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class VerticalSliderLookAndFeel

  Creation of a class which can be used to change the look of the sliders used within the equalizer for the Tromboneless project.
- class CalibrationSliderLookAndFeel

  A class to set the look and feel of a two finger linear slider to look like a trombone where the left most finger is the main body and the right most is the end of the trombones slide.
- class NeedleLookAndFeel

  A class which may be attached to a rotary slider from the juce framework to change its look and feel. This changes the look of the slider to have an arrow rotate about the centre of the slider pointing to the different values rather than a knob moving in an arc as the basic rotary slider.

Functions

- juce::Path PaintTrombone (float x=300, float y=300, float width=50, float height=20, bool incPipes=true)

  A function which returns a path matching the perimeter of a trombone.
- juce::Path PaintArc (float x, float y, float thickness, float gap, bool direction)

  A function that traces out the shape of an arc based upon the thickness of the arc and the gap between its inner sides. i.e the inner circle diameter that would remove an area from the larger outer circle.
- juce::Path RotaryArc (float x=100, float y=100, float thickness=5, float width=50, float height=50, float angularRange=5, float rotation=0)

  A function which can draw an arc with a specified thickness and nominal radius.
- juce::Path drawCircle (float x, float y, float xRadius, float yRadius)

  A simpified function for drawing a circle using the juce framework reducing the number of parameters compared to the ellipse function in juce.

9.1.1 Detailed Description

This file holds the declarations of all functions and variables needed by the Tromboneless project to change the looks and feel of various different widgets which are being displayed to the end user.

Author

Ryan McBride

9.1.2 Function Documentation

9.1.2.1 drawCircle()

```
juce::Path drawCircle (
            float x,
            float y,
            float xRadius,
            float yRadius )
```

A simpified function for drawing a circle using the juce framework reducing the number of parameters compared to the ellipse function in juce.

Parameters

x	The centre x position of the circle.
y	The centre y position of the circle.
xRadius	The radius of the circle.
yRadius	The radius of the circle.

Note

The reason there is an x and y radius is to compare which is smaller and set the circles radius to that size to stop the circle from being out of bounds or being distorted.

9.1.2.2 PaintArc()

```
juce::Path PaintArc (
            float x,
            float y,
            float thickness,
            float gap,
            bool direction )
```

A function that traces out the shape of an arc based upon the thickness of the arc and the gap between its inner sides. i.e the inner circle diameter that would remove an area from the larger outer circle.

Parameters

x	The x positon of the path, where this is the left most pixel of the area allowed for the path.
y	The y position of the path, where thes is the top most pixel of the area allowed by the path.

Parameters

thickness	The thickness of the arc itself.
gap	The inner diameter of the arc itself. The arcs center diameter would be the equivalent of gap + (thickness/2).
direction	A boolean option where 0 represents the arc having its outer face facing right, and 1 represents the arc having its outer face facing left.

Return values

Returns	a juce::Path which traces the perimeter of an arc.

9.1.2.3 PaintTrombone()

```
juce::Path PaintTrombone (
            float x = 300,
            float y = 300,
            float width = 50,
            float height = 20,
            bool incPipes = true )
```

A function which returns a path matching the perimeter of a trombone.

Parameters

x	The x coordinate from which the paths minimum x starts. The default value for this is 300.
y	The y coordinate from which the paths minimum y starts. The default value for this is 300.
width	The maximum width which the path is allowed to enclose. The default value for this is 50.
height	The maximum height which the path is allowed to enclose. The default value for this is 20.
incPipes	A boolean value determining if the trombones pipes should be included within the path or whether only the end of the pipe should be returned. The pipes are included by default.

Return values

Returns	a juce::Path which traces the perimeter of a trombone.

9.1.2.4 RotaryArc()

```
juce::Path RotaryArc (
            float x = 100,
            float y = 100,
            float thickness = 5,
            float width = 50,
            float height = 50,
            float angularRange = 5,
            float rotation = 0 )
```

A function which can draw an arc with a specified thickness and nominal radius.

Parameters

x	The x position of the centre point of the arc. This is set to be 100 by default
y	The y position of the centre point of the arc. This is set to be 100 by default.
thickness	The thickness of the arc which is being drawn. This is set to be 5 by default.
width	The radius in the x axis of the arc. This is set to be 50 by default.
height	the radius in the y axis of the arc. This is set to be 50 by default.
angularRange	This sets the difference in minimum versus maximum angle of the arc which is being drawn. This is measured in radians and is set to be 5 by default.
rotation	This parameter rotates the arc to be facing the orientated the preferred way by the user. This function splits the `angularRange` in half from the up direction and has the left and right arcs of equal length. By default, this parameter is set to 0.

Note

This is built upon using the juce::centredArc function and line functions to create a path. If the user chooses to draw the shape outline as opposed to fill, this will be visible.

9.2 CustomStyles.hpp

Go to the documentation of this file.

```
00001
00009 #pragma once
00010
00011 // #include <juce_gui_extra/juce_gui_extra.h>
00012 #include <stdexcept>
00013 #include <string>
00014 #include <juce_core/juce_core.h>
00015 #include <juce_gui_basics/juce_gui_basics.h>
00016
00020 class VerticalSliderLookAndFeel : public juce::LookAndFeel_V4
00021 {
00022     public:
00030         VerticalSliderLookAndFeel();
00031
00036         virtual ~VerticalSliderLookAndFeel() = default;
00037
00064         virtual void drawLinearSlider(juce::Graphics& g, int x, int y, int width,
00065             int height, float sliderPos, float minSliderPos, float maxSliderPos,
00066             const juce::Slider::SliderStyle style, juce::Slider& slider) override;
00067
00071         void setDialColour(const juce::Colour colour);
00072
00077         juce::Colour getDialColour(void);
00078
00083         void setDialEdgeColour(const juce::Colour& colour);
00084
00089         juce::Colour getDialEdgeColour(void);
00090
00094         void setTickColour(juce::Colour colour);
00095
00100         juce::Colour getTickColour(void);
00101
00105         void setTrackColour(juce::Colour colour);
00106
00111         juce::Colour getTrackColour(void);
00112
00116         void setBackgroundColour(juce::Colour colour);
00117
00122         juce::Colour getBackgroundColour(void);
00123
00127         void setTextColour(juce::Colour colour);
00128
00133         juce::Colour getTextColour(void);
00134
00139         void setTrackWidth(float width);
00140
00145         float getTrackWidth(void);
00146
```

```
        void setTickThickness(float thickness);

        float getTickThickness(void);

        void setTickWidth(float width);

        float getTickWidth(void);

#ifdef USE_CIRCLE_FINGER
        void setFingerRadius(float radius);

        float getFingerRadius(void);
#else
        void setFingerWidth(float width);

        float getFingerWidth(void);

        void setFingerHeight(float height);

        float getFingerHeight(void);

#endif

private:
    /*  Setting the main colour of the knob (dial) of the slider. */
    juce::Colour dialColour           = juce::Colours::white;

    /*  Setting the secondary colour of the knob (dial) of the slider.
        currently, this is for the lines within the knob. */
    juce::Colour dialEdgeColour       = juce::Colours::red;

    /*  Setting the colour of the ticks along the slider. */
    juce::Colour tickColour           = juce::Colours::darkgrey;

    /*  Setting the colour of the track which the knob will be moving
        across. */
    juce::Colour trackColour          = juce::Colours::darkgrey;

    /*  Setting the colour which will fill the bounds of the slider. */
    juce::Colour backgroundColour     = juce::Colours::black;

    /*  Setting the colour of the text which will be beside the slider. */
    juce::Colour textColour           = juce::Colours::white;

    /*          Sizes.       */
    /*  This sets the track width of the slider. */
    float trackWidth          = 5;

    /*  This sets the height of the ticks along the slider. */
    float tickThickness       = 3.0;

    /*  This sets the width of the ticks along the slider. */
    float tickWidth           = 10;

/*  Checking the he slider should use a circular finger or rectangular. */
#ifdef USE_CIRCLE_FINGER
    /*  This sets the radius of the finger of the dial position. */
    float fingerRadius        = 12;
#else
    /*  This sets the width of the finger of the slider. */
    float fingerWidth         = 25;

    /*  This sets the height of the finger of the slider. */
    float fingerHeight        = 10;
#endif
};

class CalibrationSliderLookAndFeel  :  public   juce::LookAndFeel_V4{
    public:
        CalibrationSliderLookAndFeel();

        virtual ~CalibrationSliderLookAndFeel() = default;

        virtual void drawLinearSlider(juce::Graphics& g, int x, int y,
            int width, int height, float sliderPos, float minSliderPos,
            float maxSliderPos, const juce::Slider::SliderStyle style,
            juce::Slider& slider);

        void setEmptyTrackColour(juce::Colour colour);

        juce::Colour getEmptyTrackColour(void);

        void setFullTrackColour(juce::Colour colour);

        juce::Colour getFullTrackColour(void);

        void setMinThumbColour(juce::Colour colour);
```

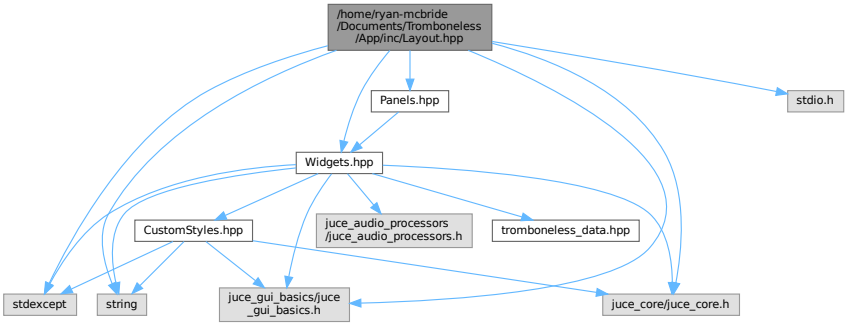
```
00339
00344         juce::Colour getMinThumbColour(void);
00345
00349         void setMaxThumbColour(juce::Colour colour);
00350
00355         juce::Colour getMaxThumbColour(void);
00356
00361         void setTrackWidth(float width);
00362
00368         float getTrackWidth(void);
00369
00370     private:
00371         juce::Colour  emptyTrackColour= juce::Colours::darkgrey;
00372         juce::Colour  fullTrackColour = juce::Colours::gold;
00373         juce::Colour  thumbOneColour  = juce::Colours::yellow;
00374         juce::Colour  thumbTwoColour  = juce::Colours::pink;
00375
00376         float trackWidth        =   5;
00377
00378 };
00379
00393 juce::Path PaintTrombone(float x=300, float y=300, float width=50, float height = 20, bool incPipes =
      true);
00394
00395
00411 juce::Path PaintArc(float x, float y, float thickness, float gap, bool direction);
00412
00418 class NeedleLookAndFeel :   public juce::LookAndFeel_V4
00419 {
00420     public:
00424         NeedleLookAndFeel() = default;
00425
00429         virtual ~NeedleLookAndFeel() = default;
00430
00453         virtual void drawRotarySlider(juce::Graphics& g, int x, int y, int width,
00454             int height, float sliderPos, const float rotaryStartAngle,
00455                 const float rotaryEndAngle, juce::Slider& slider) override;
00456
00461         void setNeedleColour(juce::Colour colour);
00462
00467         juce::Colour getNeedleColour(void);
00468
00469     private:
00470         /*  Setting the colour of the needle/arrow which is being drawn. */
00471         juce::Colour needleColour =   juce::Colours::black;
00472
00473
00474 };
00475
00501 juce::Path RotaryArc(float x = 100, float y = 100, float thickness = 5, float width = 50,
00502     float height = 50, float angularRange = 5, float rotation = 0);
00503
00504
00516 juce::Path drawCircle(float x, float y, float xRadius, float yRadius);
```

9.3 /home/ryan-mcbride/Documents/Tromboneless/App/inc/Layout.hpp File Reference

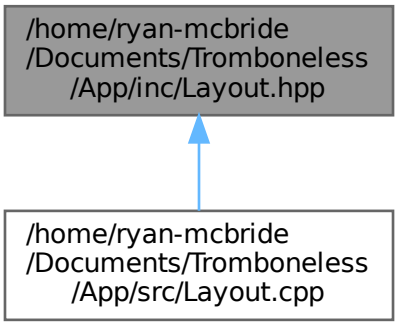
This file holds the declarations of all functions and variables needed by the Tromboneless project to define the layout of the window which is being displayed to the end user.

```
#include <stdexcept>
#include <string>
#include <juce_core/juce_core.h>
#include <juce_gui_basics/juce_gui_basics.h>
#include "Widgets.hpp"
#include "Panels.hpp"
#include <stdio.h>
```

Include dependency graph for Layout.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class Layout

  Main class that creates the window.

9.3.1 Detailed Description

This file holds the declarations of all functions and variables needed by the Tromboneless project to define the layout of the window which is being displayed to the end user.

Author

Ryan McBride

9.4 Layout.hpp

Go to the documentation of this file.

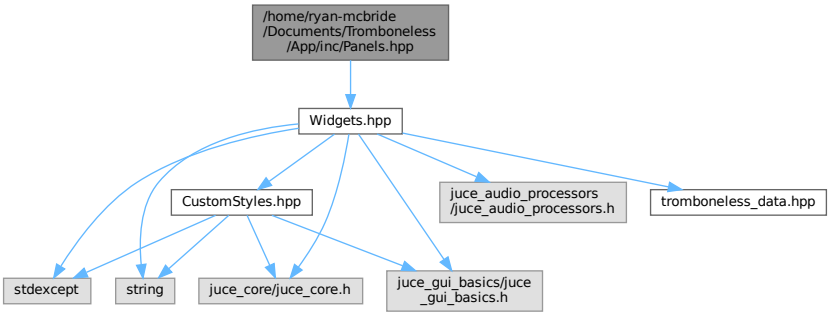
```
/*  Preventing recurssive definitons. */
#pragma once

/*  Adding all the necessary header files to be included in the file */
// #include <juce_gui_extra/juce_gui_extra.h>
#include <stdexcept>
#include <string>
#include <juce_core/juce_core.h>
#include <juce_gui_basics/juce_gui_basics.h>
#include "Widgets.hpp"
#include "Panels.hpp"
#include <stdio.h>

class Layout final : public juce::Component
{
    public:

        // Constructor
        Layout();

        // Deconstructor
        ~Layout() = default;

        void paint (juce::Graphics& g) override;


        void resized() override;



    private:
        /* ========== This is where the buttons and sliders should be initialised and attached to the profiler. ========== */

        /* Shift keying drop down menu and label */
        DropDownMenus   dropDownMenus;

        /* Sliders and labels for calibrating the range. */
        Sliders sliders;

        /* Sliders and labels for calibrating the pressure maximum and minimum. */
        Barometer pressureSlider;

        /* Adding an option to calibrate the embouchure. */
        CalibrateEmbouchure calibrateEmbouchure;


        /* Creation of the equalizer. */
        EqualizerPanel equalizer;

        void showMessageBox (const juce::String& title, const juce::String& message)
        {
            juce::AlertWindow::showMessageBoxAsync(
                juce::MessageBoxIconType::InfoIcon,
                title,
                message
            );
        }

        JUCE_DECLARE_NON_COPYABLE_WITH_LEAK_DETECTOR (Layout)
};
```

9.5 /home/ryan-mcbride/Documents/Tromboneless/App/inc/Panels.hpp File Reference

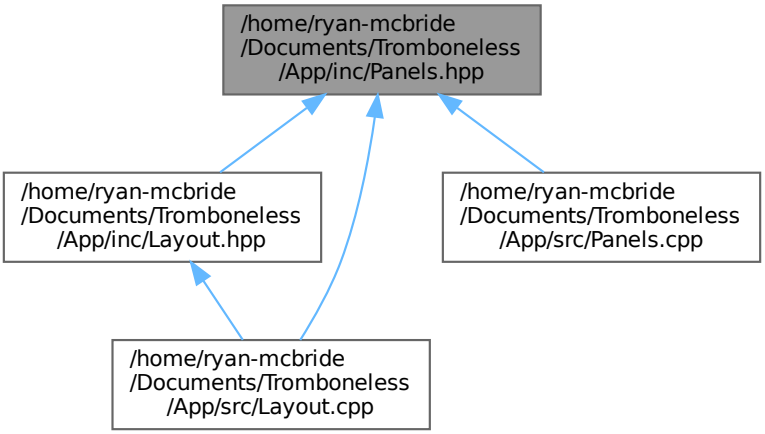
A header file to contain the panels used within the Tromboneless user interface window. This will allow for the encapsulation of different areas as opposed to having large amounts of repeat calls in the `Layout.hpp` file.

```
#include "Widgets.hpp"
```

Include dependency graph for Panels.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class Panels

  A class which is inherited from Juce:Component reponsible for setting the general style of a panel within the main window.
- class DropDownMenus

  A class which is created to display the drop down menus required by the Tromboneless UI to choose the chosen option for calibration settings.
- class Sliders

  A class which adds all the linear sliders being used within the calibration section to be within the same panel.
- class EqualizerPanel

  A class which may be called to add a panel to the window contains a series of sliders for the equalizer allowing for live editting of the synthesisers sound to ensure it sounds how the Trombonist wishes it to.

9.5.1 Detailed Description

A header file to contain the panels used within the Tromboneless user interface window. This will allow for the encapsulation of different areas as opposed to having large amounts of repeat calls in the `Layout.hpp` file.

Author

Ryan McBride

9.6 Panels.hpp

Go to the documentation of this file.

```
00001
00010 /* Preventing recurssion. */
00011 #pragma once
00012
00013 /* Adding any necessary include files. */
00014 #include   "Widgets.hpp"
00015
00016 /* Class definitions. */
00017
00025 class Panels   :   public juce::Component
00026 {
00027    public:
00033        Panels() = default;
00034
00039        ~Panels() = default;
00040
00050        void paint(juce::Graphics& g) override;
00051
00057        void setBackgroundColour(juce::Colour colour);
00058
00063        juce::Colour getBackgroundColour(void);
00064
00065    private:
00066        /*  Definition of the colour which the panel should be. Currently a dark
00067            grey colour with a slight transparency.*/
00068        juce::Colour backgroundColour         = juce::Colour((unsigned int) (0x43080810));
00069 };
00070
00075 class DropDownMenus :   public Panels
00076 {
00077    public:
00084        DropDownMenus();
00085
00090        ~DropDownMenus() = default;
00091
00095        void resized() override;
00096
00097    private:
00098        /*  Shift keying drop down menu and label */
00099        DropDownMenu shiftKeyChoice;
00100
00101        /*  Creating an instance of the CalibrateEmbouchure class. */
00102        CalibrateEmbouchure calibrateEmbouchure;
00103 };
00104
00109 class Sliders : public Panels,
00110                 public juce::Slider::Listener
00111 {
00112    public:
00118        CalibrationSlider distanceSlider;
00119
00125        Sliders();
00126
00131        ~Sliders();
00132
00141        void resized() override;
00142
00154        void sliderValueChanged(juce::Slider* sliderChanged) override;
00155
00160        void setMinimumDistance(float distance);
00161
00167        float getMinimumDistance(void);
00168
00173        void setMaximumDistance(float distance);
00174
```

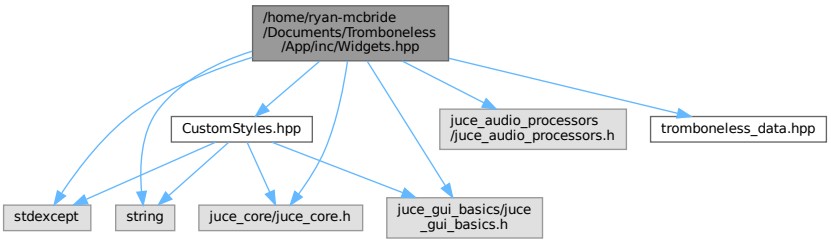
```
        float getMaximumDistance(void);

        void setStepDistance(float step);

        float getStepDistance(void);

        void setDistanceRange(float range);

        float getDistanceRange(void);

    private:
        /*  Creating an instance of the CalibrationSliderLookAndFeel
            class which will later be applied to the slider to look
            like a trombone being actuated to its full length. */
        CalibrationSliderLookAndFeel LandF;


        /*  Sets the minimum distance the users hand may be from the Tromboneless for it to
            be detected and counted as a position on the pitch bend. */
        float minimumDistance = 5;
        /*  Sets the maxmimum distance the users hand may be from the Tromboneless for it to
            be detected and counted as a position on the pitch bend. */
        float maximumDistance = 60;
        /*  Sets the minimum change in distance the user can have when adjusting the slider.
            This is essentially the resolution of the slider. */
        float stepDistance = 0.1;
        /*  Setting the minimum distance that the maximum and minimum hand positions must be
            from each other. This reduces the chances of having ambiguity in the pitch bend. */
        float distanceRange = 5;
};

class EqualizerPanel   :   public Panels,
                           public juce::Button::Listener
{
    public:
        EqualizerPanel();

        ~EqualizerPanel();

        void resized() override;

        void buttonClicked(juce::Button* buttonClicked) override;

    private:
        /*  Initialising the button which will be used to show or hide the equalizer. */
        juce::ToggleButton  button;
        /*  Initialising the equalizer which will allow the user to interact with the
            gains of the frequency bands for the synthesiser. */
        Equalizer equalizer;
        /*  Initialising the label which will appear beside the enable/disable button
            which will indicate what pressing the button will do. */
        juce::Label buttonLabel;

};
```

9.7 /home/ryan-mcbride/Documents/Tromboneless/App/inc/Widgets.hpp File Reference

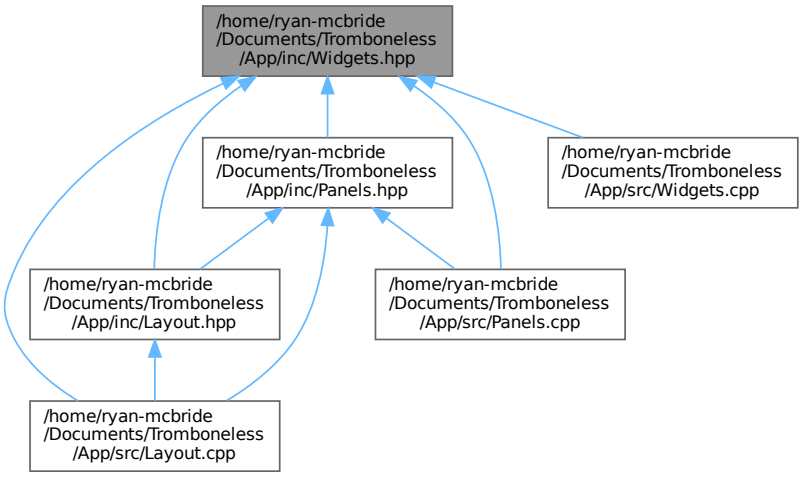
A file which sets the declarations of any custom widgets which have been built off of the juce framework. A large portion of these widgets have been created specifically for the Tromboneless project.

```
#include <stdexcept>
#include <string>
#include <juce_core/juce_core.h>
#include <juce_gui_basics/juce_gui_basics.h>
#include <juce_audio_processors/juce_audio_processors.h>
#include "CustomStyles.hpp"
#include <tromboneless_data.hpp>
```

Include dependency graph for Widgets.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class SliderWithLabel

  A class which inherits attributes from the juce::Component class to create a new component containing a 8 possible label locations around the slider.
- class CalibrationSlider

  A class that adds more functionality to the slider class from Juce.
- class DualRotarySlider

  A class that adds more functionality to the slider class from Juce.
- class Barometer

  A class which can allow for the look and feel of a set of rotary sliders be modified to look like a barometer (pressure gauge).
- class verticalMixSlider

  A class that adds more functionality to the slider class from Juce.
- class Equalizer

  A class which inherits its sliders from `verticalMixSlider` made to act as an equalizer.
- class DropDownMenu

A class that creates a drop down menu or 'comboBox' with a label attached to it. This takes up 95% of the available local area. The label is set to have a maxmimum width of 200, but scales as the window size is reduced.

- class CalibrationOnClick

  A class which allows for the additon of a button to the screen without inheriting the button directly. This allows for other widgets to be added to it later without compatibility issues.

- class CalibrateEmbouchure

  A class which is inherited from the CalibrationOnClick class that adds a dropdown menu next to the button for the user to select which option they wish to calibrate.

9.7.1 Detailed Description

A file which sets the declarations of any custom widgets which have been built off of the juce framework. A large portion of these widgets have been created specifically for the Tromboneless project.

Author

Ryan McBride

9.8 Widgets.hpp

Go to the documentation of this file.

```
00001
00010 #pragma once
00011
00012 // #include <juce_gui_extra/juce_gui_extra.h>
00013 #include <stdexcept>
00014 #include <string>
00015 #include <juce_core/juce_core.h>
00016 #include <juce_gui_basics/juce_gui_basics.h>
00017 #include <juce_audio_processors/juce_audio_processors.h>
00018 #include "CustomStyles.hpp"
00019 #include <tromboneless_data.hpp>
00020
00021 /* A header file containing custom widgets to appear on the screen to reduce and repetitive
      definitions within the main Layout.cpp file. */
00022
00026 class SliderWithLabel : public juce::Component{
00027     public:
00028
00030         juce::Slider slider;
00031
00033         SliderWithLabel(juce::Slider::SliderStyle style){
00034             slider.setSliderStyle(style);
00035             slider.setTextBoxStyle(juce::Slider::TextEntryBoxPosition::NoTextBox, true, 0, 0);
00036             addAndMakeVisible(slider);
00037         }
00038
00040         typedef enum{
00041             UpperLeft, UpperCentre, UpperRight,
00042             MiddleLeft,              MiddleRight,
00043             LowerLeft, LowerCentre, LowerRight,
00044         } LabelPositions_t;
00045
00047         typedef enum{
00048             LabelOK,
00049             PositonNotAccepted,
00050         } InitialiseLabelErrors_t;
00051
00065         InitialiseLabelErrors_t CreateLabel(LabelPositions_t position,
00066             juce::String phrase,
00067             juce::NotificationType NotificationEnable = juce::dontSendNotification);
00068
00080         void resized() override;
00081
00087         void setTopLabelBounds(int bound);
00088
00093         int getTopLabelBounds(void);
00094
00100         void setLeftLabelBounds(int bound);
00101
```

```
        int getLeftLabelBounds(void);

        void setRightLabelBounds(int bound);

        int getRightLabelBounds(void);

        void setBottomLabelBounds(int bound);

        int getBottomLabelBounds(void);

    private:

        void labelSetup(juce::Label* Label, juce::String phrase,
                        juce::NotificationType NotificationEnable);

        /* Top label bound parameter (maximum height of the above the slider). */
        int topLabelBounds = 50;
        /* Left label bound parameter (maximum width of the label to the left of the slider). */
        int leftLabelBounds = 200;
        /* Right label bound parameter (maximum width of the label to the right of the slider). */
        int rightLabelBounds = 200;
        /* Bottom label bound parameter (maximum height of the label under the slider). */
        int bottomLabelBounds = 50;

        /* Creation of labels that may be used with the slider. */
        juce::Label UpperLeftLabel;
        juce::Label UpperCentreLabel;
        juce::Label UpperRightLabel;
        juce::Label MiddleLeftLabel;
        juce::Label MiddleRightLabel;
        juce::Label LowerLeftLabel;
        juce::Label LowerCentreLabel;
        juce::Label LowerRightLabel;
};


class CalibrationSlider :   public SliderWithLabel,
                            public juce::Slider::Listener
{
public:
    CalibrationSlider();

    ~CalibrationSlider();

    void setMinDifference(float difference);

    float getMinDifference(void);


    void sliderValueChanged(juce::Slider* sliderChanged) override;

    private:
        float minDistance = 1.0;
};

class DualRotarySlider : public juce::Component,
                         public juce::Slider::Listener
{
public:

    juce::Slider    maxSlider;
    juce::Slider    minSlider;
    juce::Label     rotaryLabel;

    DualRotarySlider();

    ~DualRotarySlider();

    void sliderValueChanged(juce::Slider* sliderChanged) override;

    void resized() override;

    void setMinDifference(float difference);

    float getMinDifference(void);

    void setMinSliderRadius(float radius);

    float getMinSliderRadius(void);

    void setMaxSliderRadius(float radius);

    float getMaxSliderRadius(void);

    void setLabelHeight(float height);

    float getLabelHeight(void);
```

```
    void setMinLimit(float limit);

    float getMinLimit(void);

    void setMaxLimit(float limit);

    float getMaxLimit(void);

    void setInterval(float step);

    float getInterval(void);

    void setMinAngle(float angle);

    float getMinAngle(void);

    void setMaxAngle(float angle);

    float getMaxAngle(void);

private:

    /* Setting the relative radius of the minimum slider. */
    float minSliderRadius =   0.6;

    /* Setting the relative radius of the maximum slider. */
    float maxSliderRadius =   0.9;

    /*  Setting the minimum limit of the slider. */
    float minLimit = 0.1;
    /*  Setting the maximum limit of the slider. */
    float maxLimit = 2.0;

    /*  Setting the minimum step to which the slider may be moved. */
    float interval = 0.1;

    /* This is the minimum difference between the min and max sliders with respect ot the non normalised values. */
    float minDifference = 0.2;

    /* Setting the relative height of the label. */
    float labelHeight;

    /*  These values should be moved to be private and set/get functions added. */
    /*  Setting the minimum angle which the slider can reach. */
    float minAngle      =   M_PI + 0.5;
    /*  Setting the maximum angle which the slider can reach. */
    float maxAngle      =   3*M_PI - 0.5;

    /* Finding the minimum difference in angle for the finger differences by
       normalising the minDifference with respect to the max and min limits
       and then mulitplying it by the difference between the max and min
       angles. */
    float minAngleDifference = (maxAngle - minAngle)*(minDifference/(maxLimit - minLimit));

    void updateMinAngleDifference(void);
};

class Barometer :   public DualRotarySlider
{
    public:
        Barometer();

        ~Barometer();

        void sliderValueChanged(juce::Slider* sliderChanged) override;

        void paint (juce::Graphics& g) override;

        void setBackgroundColour(juce::Colour colour);

        juce::Colour getBackgroundColour(void);

        void setTextColour(juce::Colour colour);

        juce::Colour getTextColour(void);

        void setBoarderColour(juce::Colour colour);

        juce::Colour getBoarderColour(void);

        void setEdgeColour(juce::Colour colour);

        juce::Colour getEdgeColour(void);

        void setOuterRadius(float radius);
```

```
        float getOuterRadius(void);

        void setInnerRadius(float radius);

        float getInnerRadius(void);

        void setLabelRadius(float radius);

        float getLabelRadius(void);

    private:
        /*  Creation of an instance of the NeedleLookAndFeel class to be
            added to the slider knobs to have the correct aesthetic. */
        NeedleLookAndFeel arrow;

        /* Defining the colours used within the barometer. */
        juce::Colour backgroundColour   = juce::Colours::white;
        juce::Colour textColour         = juce::Colours::black;
        juce::Colour boarderColour      = juce::Colours::gold;
        juce::Colour edgeColour         = juce::Colour((unsigned int) (0x43080810));

        /* Dimensions */
        /* Setting the size of the outer arc on the barometer. */
        float relativeOuterRadius = 0.4;

        /* Setting the size of the inner arc on the barometer.*/
        float relativeInnerRadius = 0.35;

        /* Setting the distance of the gain labels relative to the centre point. */
        float relativeLabelRadius = 0.3;
};

class verticalMixSlider : public SliderWithLabel,
                          public juce::Slider::Listener
{
public:
    verticalMixSlider();

    ~verticalMixSlider();

    void sliderValueChanged(juce::Slider* sliderChanged) override;

};


class Equalizer : public verticalMixSlider
{
    public:

        juce::OwnedArray <verticalMixSlider> eqSliders;

        Equalizer(char numberOfSliders = 10);

        ~Equalizer();

        void resized() override;

        void sliderValueChanged(juce::Slider* sliderChanged) override;

    private:
        /*  Creation of an array which stores the different labels which will be set
            as the frequency ranges of the sliders. */
        juce::Array <juce::String> frequencyLabels;

        /*  Stores the previous label which is being written to the bottom of the
            slider. i.e. the minimum frequency in the frequency band which is being
            adjusted by the slider. */
        float prevLabel;

        /*  Stores the current label which is being written to the top of the slider.
            i.e the maximum frequency in the frequency band which is being adjusted
            by the slider. */
        float newLabel;

        /*  Creating an instance of the VerticalSliderLookAndFeel class to apply a new
            look and feel to the sliders within this class to match that of a standard
            equalizer. */
        VerticalSliderLookAndFeel customLook;

        /* variable to store the number of sliders which are being used. */
        char nSliders;

};


 class DropDownMenu :   public juce::Component
```

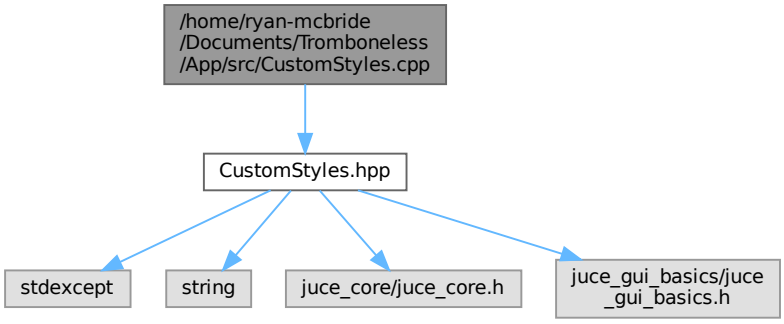
```
{
    public:
        DropDownMenu(juce::String labelPhrase = "ComboBox");

        ~DropDownMenu() = default;

        void resized() override;

        void AddItem(juce::String itemName, int itemValue);

        template <typename T> void OnChange(T* updateVariable){
            dropDownChoice.onChange = [updateVariable, this] {*updateVariable = static_cast<T>
(dropDownChoice.getSelectedId());};
        }

        void ChangeLabelText(juce::String text, juce::NotificationType notification =
juce::dontSendNotification);

        juce::String getLabelText(void);

    private:
        /* Adding the label which will be next to the drop down menu. */
        juce::Label dropDownLabel;

        /* Adding the drop down menu to the class. */
        juce::ComboBox dropDownChoice;
};

class CalibrationOnClick   :   public juce::Component,
                               public juce::Button::Listener
{
    public:

        CalibrationOnClick();

        ~CalibrationOnClick();

        void resized() override;

        juce::TextButton button;

};

class CalibrateEmbouchure    :   public CalibrationOnClick
{
    public:
        CalibrateEmbouchure();

        ~CalibrateEmbouchure() = default;

        void resized() override;

        void buttonClicked(juce::Button* thisButtonClicked) override;

    private:
        /*  Adding a dropdown menu next to the calibration button to select
            the embachure option which is being calibrated. */
        DropDownMenu embouchureChoice;

        /*  Adding a tootltipWindow to the class which may be displayed when
            the user runs their cursor over the button for at least 300ms. */
        juce::TooltipWindow tooltipWindow;

};
```

9.9 /home/ryan-mcbride/Documents/Tromboneless/App/src/CustomStyles.cpp File Reference

This file holds the defintions of all functions and variables needed by the Tromboneless project to change the looks and feel of various different widgets which are being displayed to the end user.

```
#include "CustomStyles.hpp"
```

Include dependency graph for CustomStyles.cpp:



Functions

- juce::Path PaintTrombone (float x, float y, float width, float height, bool incPipes)

  A function which returns a path matching the perimeter of a trombone.
- juce::Path PaintArc (float x, float y, float thickness, float gap, bool direction)

  A function that traces out the shape of an arc based upon the thickness of the arc and the gap between its inner sides. i.e the inner circle diameter that would remove an area from the larger outer circle.
- juce::Path RotaryArc (float x, float y, float thickness, float width, float height, float angularRange, float rotation)

  A function which can draw an arc with a specified thickness and nominal radius.
- juce::Path drawCircle (float x, float y, float xRadius, float yRadius)

  A simpified function for drawing a circle using the juce framework reducing the number of parameters compared to the ellipse function in juce.

9.9.1 Detailed Description

This file holds the defintions of all functions and variables needed by the Tromboneless project to change the looks and feel of various different widgets which are being displayed to the end user.

Author

Ryan McBride

9.9.2 Function Documentation

9.9.2.1 drawCircle()

```
juce::Path drawCircle (
            float x,
            float y,
            float xRadius,
            float yRadius )
```

A simpified function for drawing a circle using the juce framework reducing the number of parameters compared to the ellipse function in juce.

Parameters

x	The centre x position of the circle.
y	The centre y position of the circle.
xRadius	The radius of the circle.
yRadius	The radius of the circle.

Note

The reason there is an x and y radius is to compare which is smaller and set the circles radius to that size to stop the circle from being out of bounds or being distorted.

9.9.2.2 PaintArc()

```
juce::Path PaintArc (
            float x,
            float y,
            float thickness,
            float gap,
            bool direction )
```

A function that traces out the shape of an arc based upon the thickness of the arc and the gap between its inner sides. i.e the inner circle diameter that would remove an area from the larger outer circle.

Parameters

x	The x positon of the path, where this is the left most pixel of the area allowed for the path.
y	The y position of the path, where thes is the top most pixel of the area allowed by the path.
thickness	The thickness of the arc itself.
gap	The inner diameter of the arc itself. The arcs center diameter would be the equivalent of gap + (thickness/2).
direction	A boolean option where 0 represents the arc having its outer face facing right, and 1 represents the arc having its outer face facing left.

Return values

Returns	a juce::Path which traces the perimeter of an arc.

9.9.2.3 PaintTrombone()

```
juce::Path PaintTrombone (
            float x = 300,
            float y = 300,
            float width = 50,
            float height = 20,
            bool incPipes = true )
```

A function which returns a path matching the perimeter of a trombone.

Parameters

x	The x coordinate from which the paths minimum x starts. The default value for this is 300.
y	The y coordinate from which the paths minimum y starts. The default value for this is 300.
width	The maximum width which the path is allowed to enclose. The default value for this is 50.
height	The maximum height which the path is allowed to enclose. The default value for this is 20.
incPipes	A boolean value determining if the trombones pipes should be included within the path or whether only the end of the pipe should be returned. The pipes are included by default.

Return values

Returns	a juce::Path which traces the perimeter of a trombone.

9.9.2.4 RotaryArc()

```
juce::Path RotaryArc (
            float x = 100,
            float y = 100,
            float thickness = 5,
            float width = 50,
            float height = 50,
            float angularRange = 5,
            float rotation = 0 )
```

A function which can draw an arc with a specified thickness and nominal radius.

Parameters

x	The x position of the centre point of the arc. This is set to be 100 by default
y	The y position of the centre point of the arc. This is set to be 100 by default.
thickness	The thickness of the arc which is being drawn. This is set to be 5 by default.
width	The radius in the x axis of the arc. This is set to be 50 by default.
height	the radius in the y axis of the arc. This is set to be 50 by default.
angularRange	This sets the difference in minimum versus maximum angle of the arc which is being drawn. This is measured in radians and is set to be 5 by default.
rotation	This parameter rotates the arc to be facing the orientated the preferred way by the user. This function splits the `angularRange` in half from the up direction and has the left and right arcs of equal length. By default, this parameter is set to 0.

Note

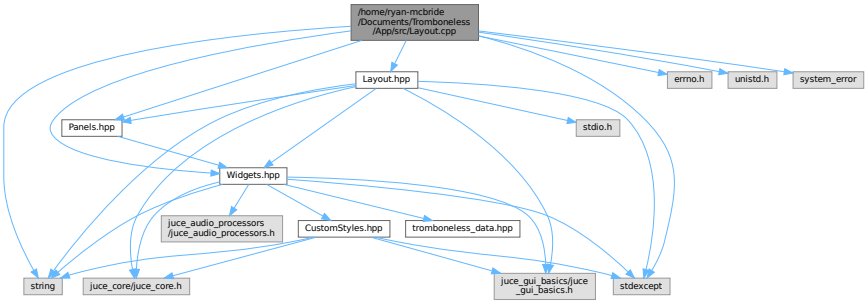
This is built upon using the juce::centredArc function and line functions to create a path. If the user chooses to draw the shape outline as opposed to fill, this will be visible.

9.10 /home/ryan-mcbride/Documents/Tromboneless/App/src/Layout.cpp File Reference

This file holds the defintions of all functions and variables needed by the Tromboneless project to define the layout of the window which is being displayed to the end user.

```
#include "Layout.hpp"
#include "Widgets.hpp"
#include "Panels.hpp"
#include <errno.h>
#include <unistd.h>
#include <stdexcept>
#include <string>
#include <system_error>
```

Include dependency graph for Layout.cpp:



9.10.1 Detailed Description

This file holds the defintions of all functions and variables needed by the Tromboneless project to define the layout of the window which is being displayed to the end user.

Author

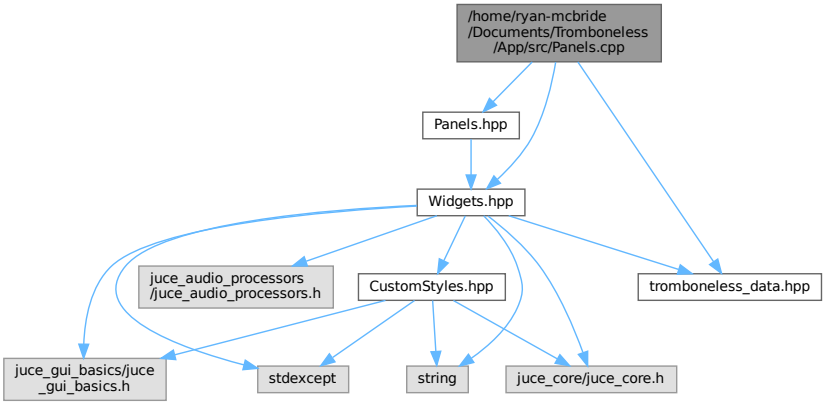
Ryan McBride

9.11 /home/ryan-mcbride/Documents/Tromboneless/App/src/Panels.cpp File Reference

This file contains the function defintions for to create the necessary panels that may be used within the Tromboneless user interface.

```
#include "Panels.hpp"
#include "Widgets.hpp"
#include <tromboneless_data.hpp>
```

Include dependency graph for Panels.cpp:



9.11.1 Detailed Description

This file contains the function defintions for to create the necessary panels that may be used within the Tromboneless user interface.

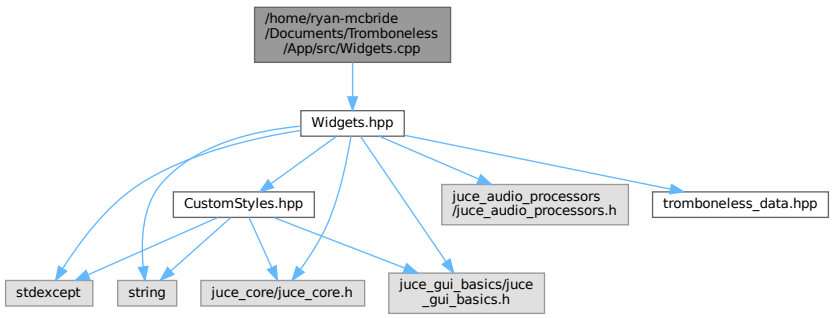
Author

Ryan McBride

9.12 /home/ryan-mcbride/Documents/Tromboneless/App/src/↩ Widgets.cpp File Reference

A file which defines the functions which have been declared within the `Widgets.hpp` file.

`#include "Widgets.hpp"`
Include dependency graph for Widgets.cpp:



9.12.1 Detailed Description

A file which defines the functions which have been declared within the `Widgets.hpp` file.

Author

Ryan McBride

9.13 AmplitudeMapper.hpp

```
#pragma once
#include <cstdio>
#include <vector>
#include <chrono>

// This code defines a class that takes a voltage level sample from
// ADS1115 representing breath pressure
// and compares it against a baseline to determine whether it
// sufficiently high enough to switch note on.

class AmplitudeMapper {

    private:
        // Attributes
        float pressureSample = 0.0f;
        float pressureBaseline = 0.0f;
        float pressureThreshold;
        bool pressureState = false;
        std::vector<float> buffer;
        std::chrono::steady_clock::time_point startTime;

    public:
        AmplitudeMapper();
        bool calculateBaseline(float pressureSample);
        float getBaseline();
        bool noteEdge(float pressureSample);


    };
```

9.14 audioRender.hpp

```
#pragma once

#include <atomic>
#include <thread>
#include <cstdint>
#include <mutex>

#include "tromboneSynth.hpp"
#include "drivers/audio_output.hpp"

class AudioRender : public TromboneSynth {
public:
    static constexpr int frames = 128;
    static constexpr int sample_rate = 44100;

    AudioRender();
    ~AudioRender();

    AudioRender(const AudioRender&) = delete;
    AudioRender& operator=(const AudioRender&) = delete;

    void start();
    void stop();

    bool isRunning() const { return running.load(); }

    // Thread-safe wrappers for synth control from other threads
    void noteOnFromMidi(int midiNote);
    void noteOffFromMidi();
    void noteChangeFromMidi(int midiNote);
    void pitchBendFromMidi(int bend);

    // Optional debug mode to prove audio path works
```

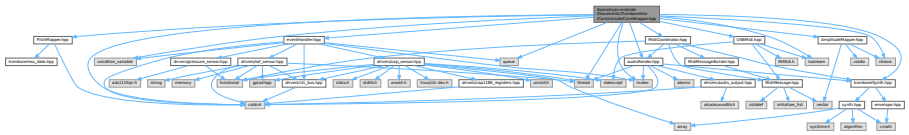
```
    void setDebugTone(bool enabled);

private:
    void renderLoop();
    void renderReleaseTail();

    AudioOutput output;

    std::thread renderThread;
    std::atomic<bool> running{false};
    std::atomic<bool> stopRequested{false};
    std::atomic<bool> debugToneEnabled{false};

    std::mutex synthMutex;

    int16_t buffer[frames]{0};
    float debugPhase{0.0f};
};
```

9.15 /home/ryan-mcbride/Documents/Tromboneless/Core/include/Core↩ Wrapper.hpp File Reference

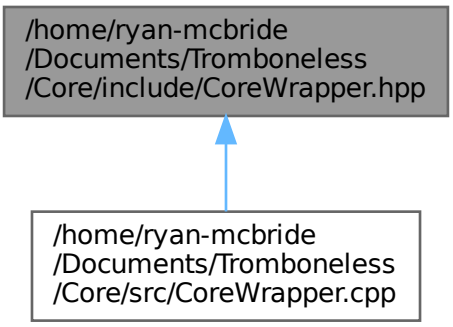
A file to allow for the app to view methods attached to the main functioanlity of the core.

```
#include "eventHandler.hpp"
#include <condition_variable>
#include <cstdint>
#include <iostream>
#include <thread>
#include <mutex>
#include <queue>
#include <chrono>
#include "MidiCoordinator.hpp"
#include "tromboneSynth.hpp"
#include "USBMidi.hpp"
#include "audioRender.hpp"
#include "PitchMapper.hpp"
#include "AmplitudeMapper.hpp"
```

Include dependency graph for CoreWrapper.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class CoreWrapper

9.15.1 Detailed Description

A file to allow for the app to view methods attached to the main functioanlity of the core.

Author

Ryan McBride Aidan McIntosh

9.16 CoreWrapper.hpp

Go to the documentation of this file.

```
#pragma once

#include "eventHandler.hpp"
#include <condition_variable>
#include <cstdint>
#include <iostream>
#include <thread>
#include <mutex>
#include <queue>
#include <chrono>
#include "MidiCoordinator.hpp"
#include "tromboneSynth.hpp"
#include "USBMidi.hpp"
#include "audioRender.hpp"
#include "PitchMapper.hpp"
#include "AmplitudeMapper.hpp"

class CoreWrapper{
    public:
        /*  ==========================================  */
        /*  Adding the constructor and destructor.      */
        /*  ==========================================  */

        CoreWrapper();
        ~CoreWrapper();
```

```

        /*  ========================================  */
        /*  Adding functions to start the threads.      */
        /*  ========================================  */
        void start();
        void stop();

        /*  ========================================  */
        /* Adding the getters for the classes' instances */
        /*  ========================================  */

    private:
        /*  Setting up queue for the threading. */
        std::queue<RawInputEvent> eventQueue;
        std::mutex eventQueueMutex;
        std::condition_variable eventQueueCv;

        /*  Creating the instances. */
        EventHandler eventHandler = EventHandler(eventQueue, eventQueueMutex, eventQueueCv);
        RtMidiSink midiSink;
        AudioRender render;
        AmplitudeMapper amplitudemapper;
        PitchMapper pitchmapper;
        MidiCoordinator coordinator = MidiCoordinator(render);


        /*  Core loop. */
        std::thread eventThread;
        bool running = false;
        void eventLoop();

};
```

9.17 audio_output.hpp

```
#pragma once

#include <alsa/asoundlib.h>
#include <cstdint>

class AudioOutput {
public:
    AudioOutput();
    ~AudioOutput();

    void writeSamples(const int16_t* buffer, int frames);

private:
    snd_pcm_t* handle = nullptr;
};
```

9.18 cap1188_registers.hpp

```
#pragma once

#include <cstdint>

// ============================================================
//  CAP1188 - 8-Channel Capacitive Touch Sensor
//  Raspberry Pi Linux I2C driver (i2c-dev / SMBus)
//
//  Register set sourced from Microchip datasheet DS00001620B
// ============================================================

// ------------------------------------------------------------
//  I2C address (set by AD pin wiring)
//  AD floating / 3V3    -> 0x29 (default)
//  AD -> 390K to GND    -> 0x28
//  AD -> 100K to GND    -> 0x2A
//  AD -> 680K to GND    -> 0x2B
//  AD -> GND            -> SPI mode only, not used here
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_DEFAULT_ADDR   = 0x29;
static constexpr uint8_t CAP1188_390K_ADDR      = 0x28;         // Used for Tromboneless

// ------------------------------------------------------------
//  General Registers
// ------------------------------------------------------------
```

```
static constexpr uint8_t CAP1188_REG_MAIN_CTRL          = 0x00; // R/W  Main Control             - Controls general power states and power dissipation        Default: 0x00
static constexpr uint8_t CAP1188_REG_GENERAL_STATUS     = 0x02; // R    General Status           - Stores general status bits                                 Default: 0x00
static constexpr uint8_t CAP1188_REG_SENSOR_STATUS      = 0x03; // R    Sensor Input Status       - State of sampled capacitive touch sensor inputs            Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_STATUS         = 0x04; // R    LED Status                - Stores status bits for LEDs                                Default: 0x00
static constexpr uint8_t CAP1188_REG_NOISE_FLAG_STATUS  = 0x0A; // R    Noise Flag Status         - Stores the noise flags for sensor inputs                   Default: 0x00

// ------------------------------------------------------------
//  Delta Count Registers (read-only, one per channel)
//  Stores the difference between current count and base count.
//  A rising value indicates a touch. Compared against threshold
//  to determine if a touch event is flagged.
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS1    = 0x10; // R    Sensor Input 1 Delta Count - Delta count for CS1                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS2    = 0x11; // R    Sensor Input 2 Delta Count - Delta count for CS2                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS3    = 0x12; // R    Sensor Input 3 Delta Count - Delta count for CS3                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS4    = 0x13; // R    Sensor Input 4 Delta Count - Delta count for CS4                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS5    = 0x14; // R    Sensor Input 5 Delta Count - Delta count for CS5                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS6    = 0x15; // R    Sensor Input 6 Delta Count - Delta count for CS6                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS7    = 0x16; // R    Sensor Input 7 Delta Count - Delta count for CS7                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_DELTA_COUNT_CS8    = 0x17; // R    Sensor Input 8 Delta Count - Delta count for CS8                                          Default: 0x00

// ------------------------------------------------------------
//  Sensitivity & Configuration Registers
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_SENSITIVITY        = 0x1F; // R/W  Sensitivity Control          - Controls sensitivity of threshold/delta counts and base count scaling  Default: 0x2F
static constexpr uint8_t CAP1188_REG_CONFIG             = 0x20; // R/W  Configuration                - Controls general functionality                                         Default: 0x20
static constexpr uint8_t CAP1188_REG_SENSOR_ENABLE      = 0x21; // R/W  Sensor Input Enable          - Controls which capacitive touch sensor inputs are sampled              Default: 0xFF
static constexpr uint8_t CAP1188_REG_SENSOR_CFG1        = 0x22; // R/W  Sensor Input Configuration   - Controls max duration and auto-repeat delay (full power state)         Default: 0xA4
static constexpr uint8_t CAP1188_REG_SENSOR_CFG2        = 0x23; // R/W  Sensor Input Configuration 2 - Controls MPRESS controls for all sensor inputs                         Default: 0x07
static constexpr uint8_t CAP1188_REG_AVG_SAMP_RATE      = 0x24; // R/W  Averaging and Sampling Config - Controls averaging and sampling window                                Default: 0x39
static constexpr uint8_t CAP1188_REG_CALIB_ACTIVATE     = 0x26; // R/W  Calibration Activate         - Forces re-calibration for capacitive touch sensor inputs               Default: 0x00
static constexpr uint8_t CAP1188_REG_INTERRUPT_ENABLE   = 0x27; // R/W  Interrupt Enable             - Enables interrupts associated with capacitive touch sensor inputs      Default: 0xFF
static constexpr uint8_t CAP1188_REG_REPEAT_RATE_ENABLE = 0x28; // R/W  Repeat Rate Enable           - Enables repeat rate for all sensor inputs                              Default: 0xFF
static constexpr uint8_t CAP1188_REG_MULTI_TOUCH_CFG    = 0x2A; // R/W  Multiple Touch Configuration - Determines number of simultaneous touches to flag                      Default: 0x80
static constexpr uint8_t CAP1188_REG_MULTI_TOUCH_PAT_CFG= 0x2B; // R/W  Multiple Touch Pattern Config- Determines MTP configuration                                          Default: 0x00
static constexpr uint8_t CAP1188_REG_MULTI_TOUCH_PAT    = 0x2D; // R/W  Multiple Touch Pattern       - Determines pattern/number of sensor inputs used by MTP                 Default: 0xFF
static constexpr uint8_t CAP1188_REG_RECALIB_CFG        = 0x2F; // R/W  Recalibration Configuration  - Determines re-calibration timing and sampling window                   Default: 0x8A

// ------------------------------------------------------------
//  Threshold Registers (one per channel)
//  Delta count must exceed this value to register a touch.
//  Default value 0x40 = 64 counts.
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_THRESH_CS1         = 0x30; // R/W  Sensor Input 1 Threshold - Delta count threshold for touch on CS1                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS2         = 0x31; // R/W  Sensor Input 2 Threshold - Delta count threshold for touch on CS2                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS3         = 0x32; // R/W  Sensor Input 3 Threshold - Delta count threshold for touch on CS3                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS4         = 0x33; // R/W  Sensor Input 4 Threshold - Delta count threshold for touch on CS4                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS5         = 0x34; // R/W  Sensor Input 5 Threshold - Delta count threshold for touch on CS5                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS6         = 0x35; // R/W  Sensor Input 6 Threshold - Delta count threshold for touch on CS6                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS7         = 0x36; // R/W  Sensor Input 7 Threshold - Delta count threshold for touch on CS7                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_THRESH_CS8         = 0x37; // R/W  Sensor Input 8 Threshold - Delta count threshold for touch on CS8                                  Default: 0x40
static constexpr uint8_t CAP1188_REG_NOISE_THRESHOLD    = 0x38; // R/W  Sensor Input Noise Threshold - Controls noise threshold for all sensor inputs                        Default: 0x01
```

```

// ------------------------------------------------------------
//  Standby Configuration Registers
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_STANDBY_CHANNEL    = 0x40; // R/W  Standby Channel        - Controls which sensor inputs are enabled in standby          Default: 0x00
static constexpr uint8_t CAP1188_REG_STANDBY_CFG        = 0x41; // R/W  Standby Configuration  - Controls averaging and cycle time in standby                Default: 0x39
static constexpr uint8_t CAP1188_REG_STANDBY_SENS       = 0x42; // R/W  Standby Sensitivity    - Controls sensitivity settings used in standby                Default: 0x02
static constexpr uint8_t CAP1188_REG_STANDBY_THRESH     = 0x43; // R/W  Standby Threshold      - Touch detection threshold for active inputs in standby       Default: 0x40
static constexpr uint8_t CAP1188_REG_CONFIG2            = 0x44; // R/W  Configuration 2        - Additional configuration controls for the device             Default: 0x40

// ------------------------------------------------------------
//  Base Count Registers (read-only, one per channel)
//  Stores the reference (baseline) capacitance for each pad,
//  established during calibration. Delta count = current - base.
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS1     = 0x50; // R    Sensor Input 1 Base Count - Reference count value for sensor input 1                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS2     = 0x51; // R    Sensor Input 2 Base Count - Reference count value for sensor input 2                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS3     = 0x52; // R    Sensor Input 3 Base Count - Reference count value for sensor input 3                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS4     = 0x53; // R    Sensor Input 4 Base Count - Reference count value for sensor input 4                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS5     = 0x54; // R    Sensor Input 5 Base Count - Reference count value for sensor input 5                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS6     = 0x55; // R    Sensor Input 6 Base Count - Reference count value for sensor input 6                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS7     = 0x56; // R    Sensor Input 7 Base Count - Reference count value for sensor input 7                   Default: 0xC8
static constexpr uint8_t CAP1188_REG_BASE_COUNT_CS8     = 0x57; // R    Sensor Input 8 Base Count - Reference count value for sensor input 8                   Default: 0xC8

// ------------------------------------------------------------
//  LED Control Registers
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_LED_OUTPUT_TYPE    = 0x71; // R/W  LED Output Type              - Controls output type for LED outputs                   Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_LINKING        = 0x72; // R/W  Sensor Input LED Linking     - Controls linking of sensor inputs to LED channels      Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_POLARITY       = 0x73; // R/W  LED Polarity                 - Controls output polarity of LEDs                       Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_OUTPUT_CTRL    = 0x74; // R/W  LED Output Control           - Controls the output state of the LEDs                  Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_LINKED_TRANS   = 0x77; // R/W  LED Linked Transition Control - Controls transition when LEDs are linked to CS channels Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_MIRROR_CTRL    = 0x79; // R/W  LED Mirror Control           - Controls mirroring of duty cycles for the LEDs         Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_BEHAVIOUR1     = 0x81; // R/W  LED Behavior 1               - Controls behaviour and response of LEDs 1-4            Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_BEHAVIOUR2     = 0x82; // R/W  LED Behavior 2               - Controls behaviour and response of LEDs 5-8            Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_PULSE1_PERIOD  = 0x84; // R/W  LED Pulse 1 Period           - Controls the period of each breathe during a pulse     Default: 0x20
static constexpr uint8_t CAP1188_REG_LED_PULSE2_PERIOD  = 0x85; // R/W  LED Pulse 2 Period           - Controls the period of breathing during breathe/pulse  Default: 0x14
static constexpr uint8_t CAP1188_REG_LED_BREATHE_PERIOD = 0x86; // R/W  LED Breathe Period           - Controls the period of an LED breathe operation        Default: 0x5D
static constexpr uint8_t CAP1188_REG_LED_CONFIG         = 0x88; // R/W  LED Config                   - Controls LED configuration                             Default: 0x04
static constexpr uint8_t CAP1188_REG_LED_PULSE1_DUTY    = 0x90; // R/W  LED Pulse 1 Duty Cycle       - Min and max duty cycle for pulse operation             Default: 0xF0
static constexpr uint8_t CAP1188_REG_LED_PULSE2_DUTY    = 0x91; // R/W  LED Pulse 2 Duty Cycle       - Min and max duty cycle for breathe and pulse operation Default: 0xF0
static constexpr uint8_t CAP1188_REG_LED_BREATHE_DUTY   = 0x92; // R/W  LED Breathe Duty Cycle       - Min and max duty cycle for breathe operation           Default: 0xF0
static constexpr uint8_t CAP1188_REG_LED_DIRECT_DUTY    = 0x93; // R/W  LED Direct Duty Cycle        - Min and max duty cycle for Direct mode LED operation   Default: 0xF0
static constexpr uint8_t CAP1188_REG_LED_DIRECT_RAMP    = 0x94; // R/W  LED Direct Ramp Rates        - Rising and falling edge ramp rates of the LEDs         Default: 0x00
static constexpr uint8_t CAP1188_REG_LED_OFF_DELAY      = 0x95; // R/W  LED Off Delay                - Determines the off delay for all LED behaviours        Default: 0x00

// ------------------------------------------------------------
//  Calibration Registers (read-only, one per channel)
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_CALIB_CS1          = 0xB1; // R    Sensor Input 1 Calibration - Upper 8-bit calibration value for sensor input 1    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS2          = 0xB2; // R    Sensor Input 2 Calibration - Upper 8-bit calibration value for sensor input 2    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS3          = 0xB3; // R    Sensor Input 3 Calibration - Upper 8-bit calibration value for sensor input 3    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS4          = 0xB4; // R    Sensor Input 4 Calibration
```

```
- Upper 8-bit calibration value for sensor input 4    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS5          = 0xB5; // R    Sensor Input 5 Calibration
- Upper 8-bit calibration value for sensor input 5    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS6          = 0xB6; // R    Sensor Input 6 Calibration
- Upper 8-bit calibration value for sensor input 6    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS7          = 0xB7; // R    Sensor Input 7 Calibration
- Upper 8-bit calibration value for sensor input 7    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_CS8          = 0xB8; // R    Sensor Input 8 Calibration
- Upper 8-bit calibration value for sensor input 8    Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_LSB1         = 0xB9; // R    Sensor Input Calibration LSB 1
- 2 LSBs of calibration value for sensor inputs 1-4  Default: 0x00
static constexpr uint8_t CAP1188_REG_CALIB_LSB2         = 0xBA; // R    Sensor Input Calibration LSB 2
- 2 LSBs of calibration value for sensor inputs 5-8  Default: 0x00

// ------------------------------------------------------------
//  Identification Registers (read-only)
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_REG_PRODUCT_ID         = 0xFD; // R    Product ID      - Fixed value
identifying this product                                Default: 0x50
static constexpr uint8_t CAP1188_REG_MANUFACTURER_ID    = 0xFE; // R    Manufacturer ID - Fixed value
identifying Microchip                                   Default: 0x5D
static constexpr uint8_t CAP1188_REG_REVISION           = 0xFF; // R    Revision        - Fixed value
representing the chip revision                          Default: 0x83

// ------------------------------------------------------------
//  Known identification values
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_PRODUCT_ID_VALUE       = 0x50;
static constexpr uint8_t CAP1188_MANUFACTURER_ID_VALUE  = 0x5D;
static constexpr uint8_t CAP1188_REVISION_VALUE         = 0x83;

// ------------------------------------------------------------
//  Main Control register bit masks (REG_MAIN_CTRL 0x00)
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_MAIN_INT_BIT           = (1 « 0); // Interrupt flag - clear to
acknowledge a touch interrupt
static constexpr uint8_t CAP1188_MAIN_DSLEEP_BIT        = (1 « 4); // Deep sleep enable
static constexpr uint8_t CAP1188_MAIN_STBY_BIT          = (1 « 5); // Standby enable
static constexpr uint8_t CAP1188_MAIN_GAIN1_BIT         = (1 « 6); // Gain bit 1 (GAIN[1:0] at bits
7:6)
static constexpr uint8_t CAP1188_MAIN_GAIN2_BIT         = (1 « 7); // Gain bit 2

// ------------------------------------------------------------
//  Sensitivity control values (bits 6:4 of REG_SENSITIVITY 0x1F)
//  Lower value = more sensitive (smaller delta required)
// ------------------------------------------------------------
static constexpr uint8_t CAP1188_SENSITIVITY_128X       = 0x00;
static constexpr uint8_t CAP1188_SENSITIVITY_64X        = 0x01;
static constexpr uint8_t CAP1188_SENSITIVITY_32X        = 0x02; // default
static constexpr uint8_t CAP1188_SENSITIVITY_16X        = 0x03;
static constexpr uint8_t CAP1188_SENSITIVITY_8X         = 0x04;
static constexpr uint8_t CAP1188_SENSITIVITY_4X         = 0x05;
static constexpr uint8_t CAP1188_SENSITIVITY_2X         = 0x06;
static constexpr uint8_t CAP1188_SENSITIVITY_1X         = 0x07;

static constexpr std::array<uint8_t, 8> CAP1188_SENSITIVITY_OPTIONS = {128, 64, 32, 16, 8, 4, 2, 1};
static constexpr std::array<uint8_t, 8> CAP1188_AVERAGING_OPTIONS = {1, 2, 4, 8, 16, 32, 64, 128};
static std::array<std::string, 4> CAP1188_SAMP_TIME_OPTIONS = {"320us", "640us", "1.28ms", "2.56ms"};
static std::array<std::string, 4> CAP1188_CYCLE_TIME_OPTIONS = {"35ms", "70ms", "105ms", "140ms"};
```

9.19 cap_sensor.hpp

```
#pragma once

#include <array>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
#include <assert.h>
#include <linux/i2c-dev.h>
#include <thread>
#include <gpiod.hpp>
#include <functional>
#include <stdexcept>
#include "drivers/i2c_bus.hpp"
#include "drivers/cap1188_registers.hpp"
```

```
class CAP1188;

class CAP1188Channel{
    public:
        bool isPinTouched();
        int8_t pinDelta();
        int getThreshold();
        void setThreshold(int threshold);
        void pinRecalibrate();
    private:
        friend class CAP1188;
        CAP1188Channel();
        ~CAP1188Channel();
        void initialise(CAP1188* cap1188, int pin);
        CAP1188* cap1188_;
        int pin_;
};

class CAP1188
{
    public:
        using CAP1188CallbackInterface = std::function<void(uint8_t)>;

        // Class constructor, includes setting
        CAP1188(I2CBus& bus,
                uint8_t address                 = 0x28,
                const std::string& gpioChipPath = "/dev/gpiochip0",
                unsigned int gpioLine = 27,
                unsigned int gpioReset = 22,
                uint8_t enabledChannels = 0x01,
                uint8_t interruptEnabledChannels = 0x01);

        // Destructor
        ~CAP1188();

        void start();
        void stop();

        // Initialise sensor
        bool initialise();

        // Sensitivity Getter & Setter
        uint8_t sensitivityGetter();
        void sensitivitySetter(uint8_t sensitivityVal);

        uint8_t getAveraging();
        void setAveraging(uint8_t averageingVal);

        std::string getSampleTime();
        void setSampleTime(std::string newSampleTime);

        std::string getCycleTime();
        void setCycleTime(std::string newCycleTime);

        std::array<bool, 8> touched_pins();
        int touched();
        int8_t deltaCount(int pin);
        void recalibratePins(uint8_t pins);
        std::array<uint8_t, 8> getThresholds();
        void setThresholds(std::array<uint8_t, 8> newThresholds);

        // REALTIME FUNCTIONALITY
        // Worker thread: blocks on GPIO edge events
        void worker();

        // Called for data = ready
        void handleDataReady();

        void disableStandby();
        void clearInterrupt();


        // Register subscriber callback
        void registerCallback(CAP1188CallbackInterface ci);

        void tester();

    private:

        I2CBus& bus_;
        uint8_t address_;
        std::string gpioChipPath_;
        unsigned int gpioLine_;
        unsigned int gpioReset_;
        uint8_t enabledChannels_;
        uint8_t interruptEnabledChannels_;
        CAP1188Channel channels[8];
```

```

        uint8_t interruptPolarity_ = 0;
        bool running_ = false;

        // Stored callback
        CAP1188CallbackInterface cap1188CallbackInterface_;

        std::shared_ptr<gpiod::chip> chip_;
        std::shared_ptr<gpiod::line_request> request_;
        std::thread thread_;

};
```

9.20 i2c_bus.hpp

```
#pragma once

#include <cstdint>

/*
 * Minimal Linux I2C bus wrapper
 *
 * Responsibility:
 * - Open/close I2C device (e.g. /dev/i2c-1)
 * - Perform register-based read/write transactions
 *
 * Does NOT:
 * - Know anything about specific sensors
 * - Perform any higher-level logic
 */

class I2CBus
{
public:
    // Construct with device path (default = Raspberry Pi I2C bus 1)
    explicit I2CBus(const char* devicePath = "/dev/i2c-1");

    // Destructor ensures bus is closed
    ~I2CBus();

    // Open the I2C device file
    bool openBus();

    // Close the I2C device file
    void closeBus();

    /*
     * Write a block of data to a device register
     *
     * address : 7-bit I2C device address
     * reg     : 16-bit register address
     * data    : pointer to data buffer
     * length  : number of bytes to write
     *
     * Returns 0 on success, -1 on failure
     */
    int writeBlock16(uint8_t address, uint16_t reg, const uint8_t* data, uint8_t length);

    /*
     * Read a block of data from a device register
     *
     * address : 7-bit I2C device address
     * reg     : 16-bit register address
     * data    : buffer to store result
     * length  : number of bytes to read
     *
     * Returns 0 on success, -1 on failure
     */
    int readBlock16(uint8_t address, uint16_t reg, uint8_t* data, uint8_t length);

        /*
```

```
     * Write a block of data to a device register
     *
     * address : 7-bit I2C device address
     * reg     : 8-bit register address
     * data    : pointer to data buffer
     * length  : number of bytes to write
     *
     * Returns 0 on success, -1 on failure
     */
    int writeBlock8(uint8_t address, uint8_t reg, const uint8_t* data, uint8_t length);

        /*
     * Read a block of data to a device register
     *
     * address : 7-bit I2C device address
     * reg     : 8-bit register address
     * data    : pointer to data buffer
     * length  : number of bytes to write
     *
     * Returns 0 on success, -1 on failure
     */
    int readBlock8(uint8_t address, uint8_t reg, uint8_t* data, uint8_t length);

private:
    const char* devicePath_;   // e.g. "/dev/i2c-1"
    int fd_;                   // file descriptor for the I2C device
};
```

9.21 Pin_Definitions.h

```
/* Defining main.h to have so there is no recursion*/

#ifndef __PIN_DEF_
#define __PIN_DEF_
#ifdef C_PLUS_PLUS
C{
#endif



/* Pin definitions*/
#define LED 2



#ifdef C_PLUS_PLUS
}
#endif
#endif
```

9.22 pressure_sensor.hpp

```
#pragma once

#include <functional>
#include "ads1115rpi.h"  // From ADS1115 submodule

class PressureSensor {
public:
    using PressureCallback = std::function<void(float)>;

    PressureSensor() = default;
    ~PressureSensor() { stop(); }

    PressureSensor(const PressureSensor&) = delete;
    PressureSensor& operator=(const PressureSensor&) = delete;


    void registerCallback(PressureCallback cb) {
        callback = cb;
        adc.registerCallback([this](float voltage) {
            if (callback) {
                callback(voltage);
            }
        });
    }


```

```
    void start(ADS1115settings settings = getDefaultSettings()) {
        adc.start(settings);
    }


    void stop() {
        adc.stop();
    }

    static ADS1115settings getDefaultSettings() {
        ADS1115settings s;
        s.samplingRate = ADS1115settings::FS128HZ;
        s.channel = ADS1115settings::AIN0;
        s.pgaGain = ADS1115settings::FSR0_512;
        return s;
    }

    ADS1115rpi& getADC() { return adc; }

private:
    ADS1115rpi adc;
    PressureCallback callback;
};
```

9.23 tof_sensor.hpp

```
#pragma once

#include <cstdint>
#include <functional>
#include <memory>
#include <string>
#include <thread>

#include <gpiod.hpp>

#include "drivers/i2c_bus.hpp"

class ToFSensor
{
public:
    using DistanceCallback = std::function<void(uint16_t)>;

    ToFSensor(I2CBus& bus,
              uint8_t i2cAddress7Bit = 0x29,
              const std::string& gpioChipPath = "/dev/gpiochip0",
              unsigned int gpioLine = 4);

    ~ToFSensor();

    bool initialise();
    void registerCallback(DistanceCallback cb);
    void start();
    void stop();

private:
    void softReset();

    void startRanging();
    void stopRanging();
    uint16_t readDistanceMm();

    void worker();
    void handleDataReady();

    I2CBus& bus_;

    // Linux I2C address form, e.g. 0x29
    uint8_t i2cAddress7Bit_;

    // ST ULD address form, e.g. 0x52
    uint16_t stDevAddress_;

    std::string gpioChipPath_;
    unsigned int gpioLine_;

    DistanceCallback distanceCallback_;

    bool running_ = false;
    std::thread thread_;

    std::shared_ptr<gpiod::chip> chip_;
    std::shared_ptr<gpiod::line_request> request_;
};
```

9.24 USBMidi.hpp

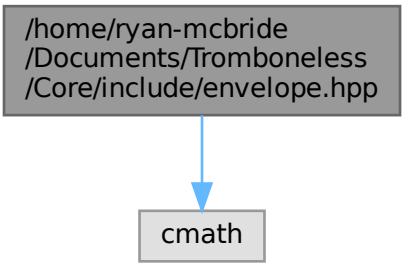
```
#pragma once

#include <RtMidi.h>
#include "MidiMessage.hpp"
#include <iostream>
#include <stdexcept>

class RtMidiSink {
    public:
        RtMidiSink();
        void send(const MidiMessage& message);

        void DetectUSBMidiOutput();

        bool GetDeviceStatus();

    private:
        RtMidiOut midiOut;

        bool EXTERNAL_DEVICE_PRESENT = false;
        int externalPortIndex = -1;
};
```

9.25 /home/ryan-mcbride/Documents/Tromboneless/↩ Core/include/envelope.hpp File Reference

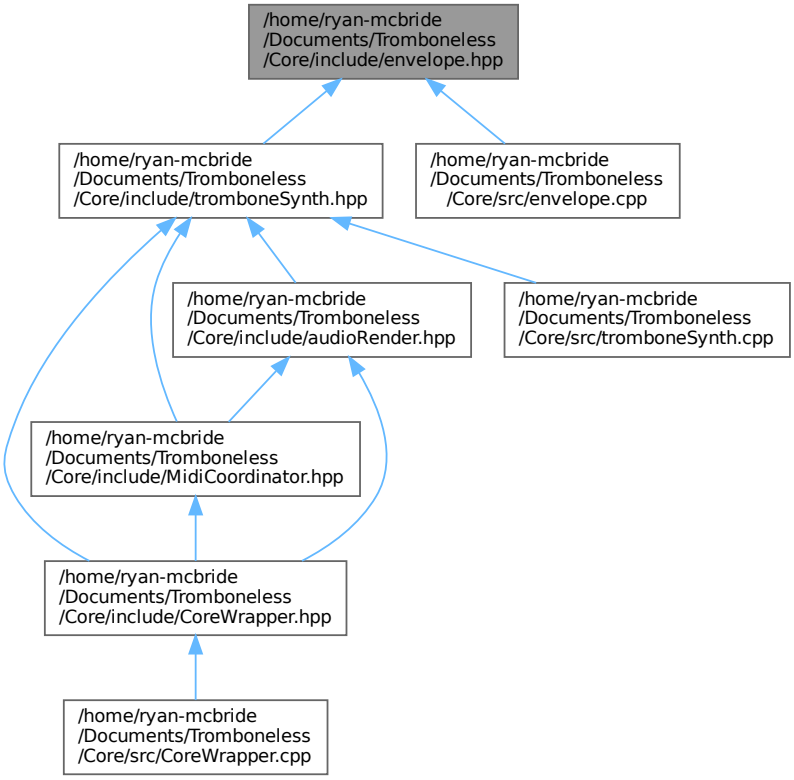
A header file containing all the necessary declarations and references to create a functional envelope class using the ADSR standard.

`#include <cmath>`

Include dependency graph for envelope.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class Envelope

  A class which can allow for the create of an envelope function which outputs a value between 0 and 1 depending on what stage the envelope is in. This uses the ADSR standard in music synthesis where A is the Attack, D is the decay, S is the saturation and R is the rest stage.

9.25.1 Detailed Description

A header file containing all the necessary declarations and references to create a functional envelope class using the ADSR standard.

Author

Ryan McBride

9.26 envelope.hpp

Go to the documentation of this file.

```
/* Preventing recursion. */
#pragma once

/* Addition of all necessary header files. */
#include <cmath>

class Envelope{
    public:

        typedef enum{
            no_stage,
            attack_stage,
            decay_stage,
            sustain_stage,
            rest_stage,
        }   envelope_stages_t;

        Envelope(int attack_in = 10, int decay_in = 20, float sustain_in = 0.98, int rest_in = 10);

        ~Envelope() = default;

        float getAmplitude(void);

        void startEnvelope(void);

        void endEnvelope(void);

        void setAttack(int attack_in);

        int getAttack(void);

        void setDecay(int decay_in);

        int getDecay(void);

        void setSustain(float sustain_in);

        float getSustain(void);

        void setRest(int rest_in);

        int getRest(void);

        float attackFunction(void);

        float decayFunction(void);

        float restFunction(void);

        envelope_stages_t getStage(void);

        void setStage(envelope_stages_t stage_in);

    private:
        /*  ADSR Variables */
        int attack;
        int decay;
        float sustain;
        int rest;

        float clamp01(float value);
        /*  Timing Variables */
        int counter = 0;
        envelope_stages_t stage = no_stage;
};
```

9.27 eventHandler.hpp

```
#pragma once

#include <cstdint>
#include <condition_variable>
#include <mutex>
#include <queue>

#include "drivers/i2c_bus.hpp"
#include "drivers/pressure_sensor.hpp"
```

```
#include "drivers/tof_sensor.hpp"
#include "drivers/cap_sensor.hpp"

struct RawInputEvent {
    enum class Type {
        ToFDistance,
        PressureReading,
        MouthpieceReading,
        Keycontrol
    };

    Type type;
    uint16_t tofDistance;
    float pressureReading;
    int8_t mouthpieceReading;
    char keycontrol;
};

class EventHandler {
public:
    using EventQueue = std::queue<RawInputEvent>;

    EventHandler(EventQueue& queue,
                 std::mutex& queueMutex,
                 std::condition_variable& queueCv);

    bool initialise();
    void start();

private:
    void handleToFDistance(uint16_t distance);

    void handleKeyControl(char key);

    void handleEmbouchure(int8_t embouchure);

    /* Takes pressure value as a float */
    void handlePressureReading(float pressure);

    EventQueue& eventQueue;
    std::mutex& eventQueueMutex;
    std::condition_variable& eventQueueCv;

    I2CBus bus;
    ToFSensor tofSensor;
    PressureSensor pressureSensor;
    CAP1188 cap1188;

};
```

9.28 main.hpp

```
/* Defining main.h to have so there is no recursion */

#ifndef __MAIN_H_
#define __MAIN_H_
#ifdef C_PLUS_PLUS
C{
#endif

/* Including header files */
// #include "Pin_Definitions.h"
// #include "MidiCoordinator.hpp"
// #include "USBMidi.hpp"
// #include "RtMidi.h"
// #include "eventHandler.hpp"
#include <iostream>
#include "tromboneless_data.hpp"
//extern PitchMapper pitchmapper;

#ifdef C_PLUS_PLUS
}
#endif
#endif
```

9.29 MidiCoordinator.hpp

```

#pragma once
```

```
#include "MidiMessage.hpp"
#include "MidiMessageBuilder.hpp"
#include "functional"
#include "tromboneSynth.hpp"
#include "audioRender.hpp"

class MidiCoordinator{
    private:
        MidiMessage message;
        MidiMessageBuilder builder;
        AudioRender& synthRef;

        using CallbackInterface = std::function<void(const MidiMessage&)>;
        CallbackInterface callback;

        int latestNote = -1; // Updated by mouthpiece. [0:127]
        int latestExpr = -1; // Updated by pressure. [0:127]
        int latestBend = -1; // Updated by slide. [-8192:8191]

        int currentNote = -1; // Playing note
        int lastSentExpr = -1; // Saves last expression sent to device
        int lastSentBend = -1; // Saves last bend sent to device

        const int velocity = 127; // Clamped to max
        enum OutputType{
            EXTERNAL,
            INTERNAL
        };

        enum OutputType current_output = INTERNAL;

        enum State{
            IDLE,
            PLAYING,
            ERROR
        };

        enum State current_state = IDLE;

    public:
        // Methods should be input events
        MidiCoordinator(AudioRender& synth);
        void RegisterCallback(CallbackInterface cb);

        void PressureEdge(bool on);
        void ChangeNote(int note);
        void setBend(int bend);
        void setExpr(int expr);

        void setState(State newstate);

        void setDevice(bool external_device_present);

        AudioRender& getSynth(void);
};
```

9.30 MidiMessage.hpp

```
#pragma once

#include <cstddef>
#include <initializer_list>
#include <vector>

class MidiMessage
{
public:
    using Byte = unsigned char;

    MidiMessage() = default;

    explicit MidiMessage(std::vector<Byte> bytes);

    MidiMessage(std::initializer_list<Byte> bytes);

    const std::vector<Byte>& bytes() const noexcept;

    std::size_t size() const noexcept;

    bool empty() const noexcept;

private:
    std::vector<Byte> data_;
};
```

9.31 MidiMessageBuilder.hpp

```
#pragma once

#include "MidiMessage.hpp"

class MidiMessageBuilder
{
public:
    using Byte = MidiMessage::Byte;

    MidiMessage noteOn(int channel,
                       Byte note,
                       Byte velocity) const;

    MidiMessage noteOff(int channel,
                        Byte note,
                        Byte velocity = 0) const;

    MidiMessage pitchBend(int channel,
                          int value) const;

    MidiMessage expr(int channel,
                     int value) const;

private:
    static Byte clampChannel(int channel);
};
```

9.32 PitchMapper.hpp

```
    /*** @file PitchMapper.hpp
     *   @author Aidan McIntosh
    */
#pragma once
#include <cstdint>
#include "tromboneless_data.hpp"

class PitchMapper{
    private:



        typedef enum{
            piccolo = 6,
            soprano = 4,
            alto = 2,
            tenor = 0,
            bass = -2,
            contrabass = -4
        } Trombone_type_t;


        int slide_max_limit_mm = 500; /* Max limit of virtual trombone slide in mm.*/
        int slide_min_limit_mm = 0; /* Min limit of virtual trombone slide in mm.*/
        int mouthpiece_MIDI_note = 34; /* MIDI note produced by mouthpiece.*/
        int partial = 0;


        int trombone_type = Trombone_type_t::piccolo;

    public:

        struct PartialDef
        {
            int minDelta;
            int hysteresisDelta;
            int partial;
            int midiNote;
        };

        PitchMapper();

        int tof_to_MIDI_bend(uint16_t tof_distance);

        int mouthpiece_to_MIDI_note(int8_t delta);


        void SetSlideMaxLimit(int new_slide_max_limit_mm);

        int GetSlideMaxLimit();

        void SetSlideMinLimit(int new_slide_min_limit_mm);
```

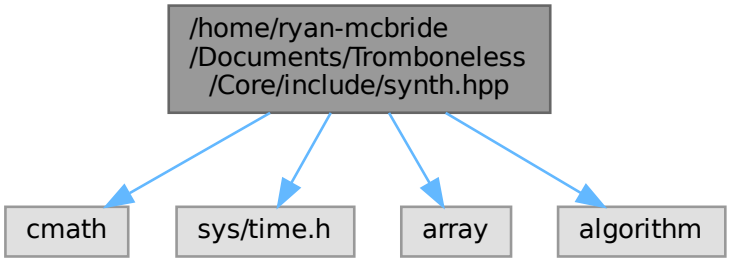
```
00070
00072         int GetSlideMinLimit();
00073
00075         void SetMouthMIDI(int new_MIDI_note);
00076
00078         int GetMouthMIDI();
00079
00080         void SetTromboneType(ShiftKeyingOptions_t trombone_choice);
00081 };
```

9.33 /home/ryan-mcbride/Documents/Tromboneless/↩ Core/include/synth.hpp File Reference

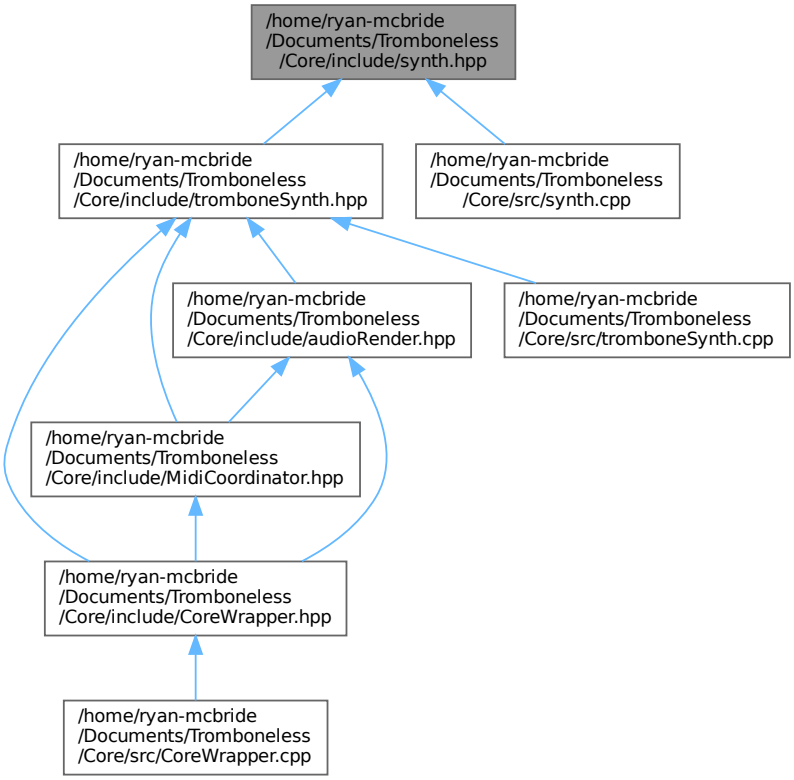
A header file containing all the necessary declarations and references to allow for the Tromboneless to synthesise the noise a trombone would make when playing the same notes.

```
#include <cmath>
#include <sys/time.h>
#include <array>
#include <algorithm>
```

Include dependency graph for synth.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class Notes

  A class to store the frequency of the notes which may be referenced. The notes are stored in an enum `Notes_t` which allow for the higher level functions to call upon the specific note as opposed to frequency.
- class Octaves

  Creation of a class to store the frequencies of notes at different octaves. Inherits from `Notes` class.

- class OctavesWithHarmonics

  A class which inherits functions and variables from the `Octaves` class allowing for the notes being output to have a set of harmonic frequencies attached to them too.

9.33.1 Detailed Description

A header file containing all the necessary declarations and references to allow for the Tromboneless to synthesise the noise a trombone would make when playing the same notes.

Author

Ryan McBride

Note

The envelope class lacks set/get functionality for its initialised parameters such as the attack, decay and rest times. This should be added in the next revision to allow for the sound to be modified without needing to create a new object each time.

9.34 synth.hpp

Go to the documentation of this file.

```
/* Preventing recursion. */
#pragma once

/* Addition of all necessary header files. */
#include <cmath>
#include <sys/time.h>
#include <array>
#include <algorithm>

/* ========================================================================================== */
/*                                    Class definitions                                       */
/* ========================================================================================== */

class Notes{
    public:

        typedef enum{
            note_C,
            note_Db,
            note_D,
            note_Eb,
            note_E,
            note_F,
            note_Gb,
            note_G,
            note_Ab,
            note_A,
            note_Bb,
            note_B,

            /* Addition of sharp notes. */
            note_C_sharp = note_Db,
            note_D_sharp = note_Eb,
            note_F_sharp = note_Gb,
            note_G_sharp = note_Ab,
            note_A_sharp = note_Bb,
        }   Notes_t;

        Notes(float C_in     =   16.35, float Db_in    =   17.32,
            float D_in       =   18.35, float Eb_in    =   19.45,
            float E_in       =   20.60, float F_in     =   21.83,
            float Gb_in      =   23.12, float G_in     =   24.50,
            float Ab_in      =   25.96, float A_in     =   27.50,
            float Bb_in      =   29.14, float B_in     =   30.87);

        ~Notes() = default;

        /* ========================================================================================== */
        /*                                  Note setters                                              */
        /* ========================================================================================== */

        void setC(float frequency);

        void setC_sharp(float frequency);

        void setDb(float frequency);

        void setD(float frequency);

        void setD_sharp(float frequency);

        void setEb(float frequency);

        void setE(float frequency);

        void setF(float frequency);

        void setF_sharp(float frequency);

        void setGb(float frequency);

        void setG(float frequency);

        void setG_sharp(float frequency);

        void setAb(float frequency);

        void setA(float frequency);
```

```
        void setA_sharp(float frequency);

        void setBb(float frequency);

        void setB(float frequency);

        float setNote(Notes_t note, float freq);

        /* ==========================================================================================
        /*                                Note getters
        /* ==========================================================================================

        float getC(void);

        float getC_sharp(void);

        float getDb(void);

        float getD(void);

        float getD_sharp(void);

        float getEb(void);

        float getE(void);

        float getF(void);

        float getF_sharp(void);

        float getGb(void);

        float getG(void);

        float getG_sharp(void);

        float getAb(void);

        float getA(void);

        float getA_sharp(void);

        float getBb(void);

        float getB(void);

        float getNote(Notes_t note);

    private:
        /* Initialisation of the possible note frequencies. */
        float   C, Db, D, Eb, E, F,
                Gb, G, Ab, A, Bb, B;

};


class Octaves   : public Notes
{
    public:

        static const char nOctaves = 9;

        Notes octaves[nOctaves];

        /* Constructor for the class*/
        Octaves();

        ~Octaves() = default;

        float PlayingNote(int octave, Notes_t note, float phase);

        float PlayingFrequency(float phase);


    private:
};

class OctavesWithHarmonics :    public Octaves
{
    public:

        OctavesWithHarmonics();

```

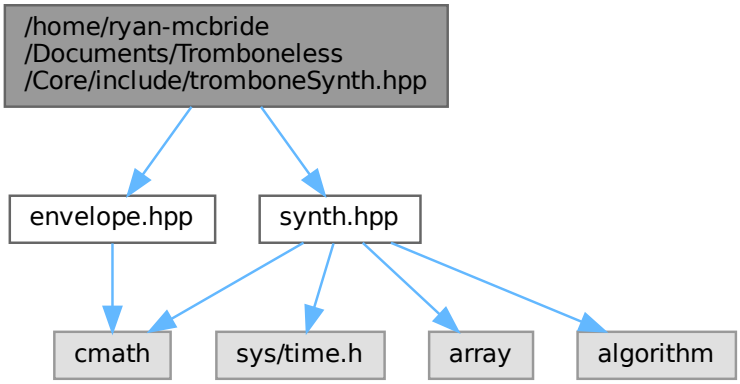
```
        ~OctavesWithHarmonics() = default;

        float PlayingNoteWithHarmonics(int n, int octave, Notes_t note, float phase);

        float PlayingFrequencyWithHarmonics(int n, float phase);

        float HarmonicDecay(int n);

        float getHarmonicDecayMax(int n);

        void setDecayConstant(float decayC);

        float getDecayConstant(void);

    protected:

        static constexpr int maxHarmonics = 64;

        float decayConstant = 0.1f;

        std::array<float, maxHarmonics> harmonicWeights{};
        int cachedNHarmonics = 0;
        bool harmonicCacheDirty = true;

        void updateHarmonicCache(int n);
};
```

9.35 /home/ryan-mcbride/Documents/Tromboneless/↩ Core/include/tromboneSynth.hpp File Reference

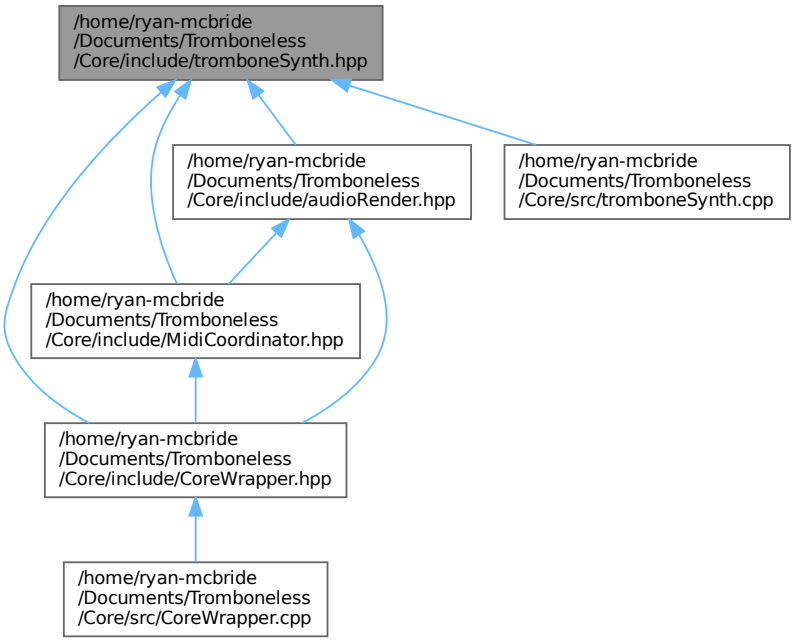
A file to declare the synthesiser parameters and functions for the trombone specifically using the classes and methods declared in the `synth.hpp` file.

```
#include "synth.hpp"
#include "envelope.hpp"
```

Include dependency graph for tromboneSynth.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class TromboneSynth

  A class which may be used to create a custom synth with an envelope attached to it.

9.35.1 Detailed Description

A file to declare the synthesiser parameters and functions for the trombone specifically using the classes and methods declared in the `synth.hpp` file.

Author

Ryan McBride

9.36 tromboneSynth.hpp

Go to the documentation of this file.

```
00001
00008     /* Preventing recurssion */
00009  #pragma once
00010
00011  /* Adding the necessary header files to be included. */
00012  #include "synth.hpp"
00013  #include "envelope.hpp"
00014
00018  class TromboneSynth    :   public OctavesWithHarmonics
00019  {
00020     public:
```

```
        TromboneSynth(int sampleRate = 44100, float attack = 10.0f, float decay = 50.0f,
             float sustain = 1.0f, float rest = 10.0f);

        void StartTromboneNote(Notes::Notes_t note_in, int octave_in);

        void ChangeTromboneNote(Notes::Notes_t note_in, int octave_in);

        float ReadTromboneAudio(void);

        void StopTromboneNote(void);

        void NewTromboneNoteMIDI(int MIDINote);

        void HandleMIDINoteOn();

        void HandleMIDINoteOff();

        void HandleMIDIPitchBend(int midipitchbend);

        void HandleMIDINoteChange(int new_note);

        void setAttackMS(float attack_time);

        float getAttackMS(void);

        void setDecayMS(float decay_time);

        float getDecayMS(void);

        void setRestMS(float rest_time);

        float getRestMS(void);

        void setSustain(float sustain_in);

        float getSustain(void);

        void setNHarmonics(int n);

        int getNHarmonics(void);

        int getOctave(void);

        Notes::Notes_t getNote(void);

        void setPitchBend(int bend);

        int getPitchBend(void);

        float getAdjustedFrequency(void);

        Envelope getEnvelope(void);

    private:
        /* Initialising the envelope class which will be used to create the various
        different noises for the trombone. This sets the sound of the trombone
        permanently. The ADSR values may need to be adjusted but this should sound
        brassy. */

        Envelope tromboneEnvelope = Envelope(10, 50, 1.0f, 10);

        float attack_ms = 10.0f;        /*  The attack time in milliseconds of the envelope. */
        float decay_ms = 50.0f;         /*  The decay time in milliseconds of the envelope. */
        float rest_ms = 10.0f;          /*  The rest time in milliseconds of the envelope. */
        float sustain = 1.0f;          /*  The sustain level of the envelope. */

        float cachedFrequency = 0.0f;
        float cachedPhaseIncrement = 0.0f;
        bool pitchCacheDirty = true;

        void updatePitchCache();


        float phase = 0.0f; /* Phase of oscillator wave */

        /*  synth   */
        int nHarmonics = 17;    /*  The number of harmonics which will be present in the final signal. */
        int octave = 4;         /*  The octave which the note will be present in. */
        Notes::Notes_t note = Notes::Notes::note_A;  /*  The note which will be played. */
        int pitchBend   =   8192;  /*  The MIDI message pitch bend dependent upon the position of the slider. */

        unsigned int samples = 0;
        unsigned int sampleRate = 44100;

        const unsigned int pitchBendMax = 8192;    /*  The maximum number which may be passed to the
```

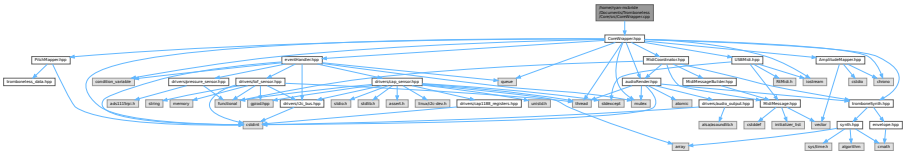
```
the pitch bending function by the MIDI input. */
        const unsigned int pitchBendSpread = 7;    /*  The maximum number of semitones that the pitch bend may change the playing note by*/
        const float semitoneSpread = pitchBendMax/pitchBendSpread;  /*   The number of data points between each adjacent semitone. */
        const int nNotes = 12;          /*  The number of notes per octave. */
};
```

9.37 /home/ryan-mcbride/Documents/Tromboneless/Core/src/Core↩ Wrapper.cpp File Reference

A file which encapsulations the main funcationality to allow for the app to act independently of it.

`#include "CoreWrapper.hpp"`

Include dependency graph for CoreWrapper.cpp:



9.37.1 Detailed Description

A file which encapsulations the main funcationality to allow for the app to act independently of it.

Author

Ryan McBride Aidan McIntosh

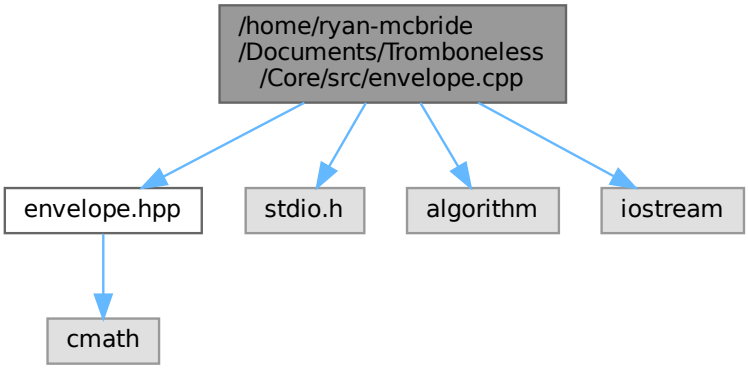
9.38 /home/ryan-mcbride/Documents/Tromboneless/↩ Core/src/envelope.cpp File Reference

Contains the defintions of the functions and variables needed to create an envelope.

```
#include "envelope.hpp"
#include <stdio.h>
#include <algorithm>
```

```
#include <iostream>
```

Include dependency graph for envelope.cpp:



9.38.1 Detailed Description

Contains the defintions of the functions and variables needed to create an envelope.

Author

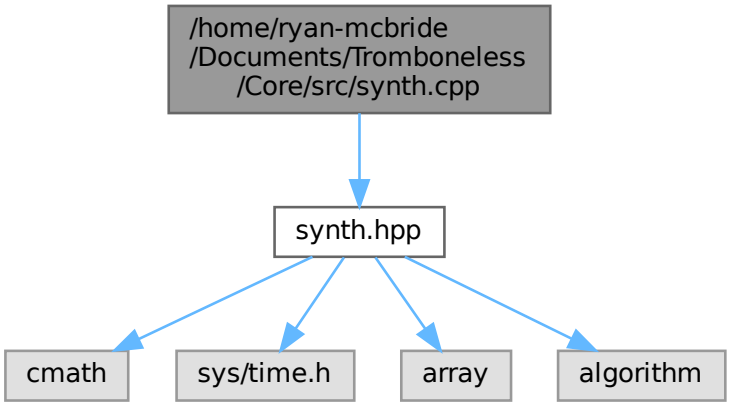
RyanMcBride

9.39 /home/ryan-mcbride/Documents/Tromboneless/Core/src/synth.cpp File Reference

C file which contains the defintions for the functions declared in the `synth.hpp` file.

```
#include "synth.hpp"
```

Include dependency graph for synth.cpp:



9.39.1 Detailed Description

C file which contains the defintions for the functions declared in the `synth.hpp` file.

Author

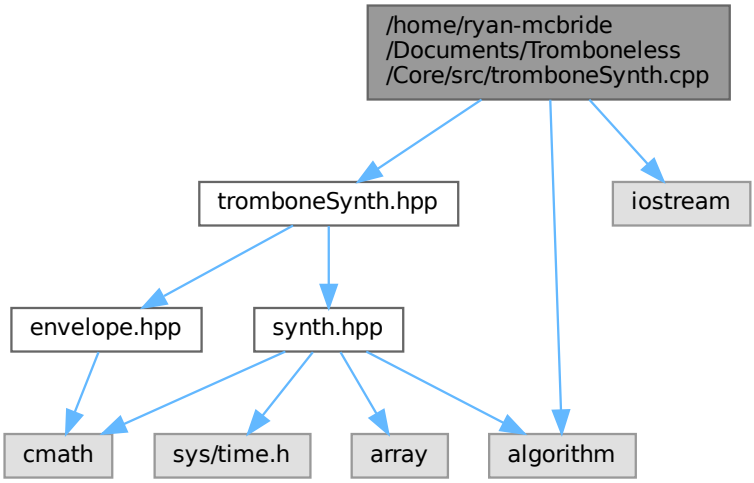
Ryan McBride

9.40 /home/ryan-mcbride/Documents/Tromboneless/Core/src/trombone↩ Synth.cpp File Reference

A file to define the synthesiser parameters and functions for the trombone specifically using the classes and methods declared in the `synth.hpp` file.

```
#include "tromboneSynth.hpp"
#include <algorithm>
#include <iostream>
```

Include dependency graph for tromboneSynth.cpp:



9.40.1 Detailed Description

A file to define the synthesiser parameters and functions for the trombone specifically using the classes and methods declared in the `synth.hpp` file.

Author

Ryan McBride

9.41 tromboneless_data.hpp

```
00001 /* Header file to pass functions which may have an effect on both the app and core functionality. */
00002 #ifndef _TROMBONELESS_DATA_H_
00003 #define _TROMBONELESS_DATA_H_
00004
00005 /* ============================= This is where the types are defined ============================
      */
00006
00011 typedef enum{
00013     SKOpt_EMPTY,
00015     SKOpt_BASS,
00017     SKOpt_PICCOLO,
00019     SKOpt_ALTO,
00021     SKOpt_TENOR,
00023     SKOpt_CONTRABASS,
00025     SKOpt_SOPRANO,
00026 } ShiftKeyingOptions_t;
00027
00032 typedef enum{
00034     EMBCR_EMPTY,
00036     EMBCR_LowPlacement,
00038     EMBCR_MediumPlacement,
00040     EMBCR_HighPlacement,
00041 } EmbachureOptions_t;
00042
00048 typedef struct{
00050     ShiftKeyingOptions_t shiftKeyingOption;
00051
00053     double nearDistance;
00054
```

```
    double farDistance;

    double lowPressure;
    double highPressure;
    EmbachureOptions_t embouchureOption;
    bool triggerEmbouchureCalibrate;
} Tromboneless Parameters_t;

typedef struct{
    bool synthEnable;
    float gains[10] = {1, 1, 1, 1, 1, 1, 1, 1, 1, 1};
} SynthesiserParameters_t;

/*  Creating an instance of the tromboneless Parameters_t struct to
    store the active parameter variables. */
extern Tromboneless Parameters_t tromboneless Parameters;
/*  Creating an instance of the SynthethiserParameters_t struct to
    store the active parameter variables. */
extern SynthesiserParameters_t synthesiserParameters;
#endif
```